

水泥烧成系统电除尘器的协同优化控制

摘 要

本文先分辨附件中哪些关系能由数据确定、哪些不能，再据此回答四问。

可由数据确定的是总功率： $P = 73.1 + 0.0850 \sum_i U_i^2 + 5.39 \times 10^4 \sum_i T_i^{-1}$ ，在未参与拟合的测试日上 $R^2 = 0.993$ 、RMSE=5.98 kW、平均偏差 -0.02 kW。第一项是电场功率，与电压平方成正比；第二项是振打功率，与振打频率 $60/T_i$ 成正比。由此得到一个易被忽略的事实：**振打环节占总功率的 38.8%**，其节电潜力与电压同属一级；入口温度、浓度、流量则对总功率无额外解释力。

问题一分两层作答。可确定的部分：上述功率律、历史操作规律（入口浓度升高时四个电场的振打周期一律缩短， $r = -0.21 \sim -0.28$ ，四个时段方向一致）以及两段持续异常工况的控制响应。不可确定的部分是出口浓度：10030 条有效记录全部落在 $48.74 \sim 50.00 \text{ mg/Nm}^3$ ，其中 5491 条恰为 50，与全部 12 个变量的相关系数绝对值均小于 0.017，属传感器量程截顶，故排放层只能按 Deutsch-Anderson 机理推演，并逐处标注其性质。振打峰值不再假定指数：由每周期积灰量 $\propto L \cdot T$ 可推出，单次峰值随周期近似线性放大，而其对小时均值的贡献 $\propto L \cdot T \cdot (1/T) = L$ ，**与周期无关**。

问题二：按入口三变量把运行状态分为 4 类（轮廓系数 0.4084）。以记录值 50 mg/Nm^3 为起点、不作任何缩放，在历史运行范围内求解，四类工况达到 10 mg/Nm^3 的最低功率为 $1851.1 \sim 2104.3 \text{ kW}$ ，加权 1973.0 kW ，比历史实际的 1783.2 kW **高 10.6%**（日电耗 $42.8 \rightarrow 47.4 \text{ MWh}$ ）。收紧排放必然增加电耗，同一排放水平下不存在额外的节能空间。

问题三：定义单位电功率的除尘收益为 $\alpha_i / (b_i U_i^{ref 2})$ 。第三电场参考电压最低，该收益在四类工况中均居首；原“前级优先”的结论实由排序约束 $U_1 \geq U_3 + 3$ 导出，而非权重 α_i 本身。

问题四：闭式解 $\Delta P = P_{field} \ln[(10 - B)/(5 - B)]/K_{10}$ 给出 $123.1 \sim 142.4 \text{ kW}$ ，与数值解相差不足 1%；加权功率由 1973.0 升至 2105.5 kW ，**增幅 6.72%**，日电耗多 3.2 MWh 。增幅与 $2/p$ 成正比（ $K \propto U^p$ ）， p 是影响功率增量与目标可达性的首要待标定参数，一次电压阶跃试验即可确定。更关键的约束不是电耗而是**电源容量**： 10 mg/Nm^3 仍在历史电压范围内， 5 mg/Nm^3 则需各场电压超出历史最大值最多 5%。若要求控制组合严格落在历史经验支持的范围内，八组解中七组仍可行，代价为多耗 1.5%~6.1% 电力。

关键词：电除尘器；可识别性审计；Deutsch-Anderson 灰箱；振打再飞扬；约束优化

1 问题重述与分析

1.1 问题背景

电除尘器通过高压电场使粉尘荷电并向收尘极迁移。提高电场电压通常能够增强荷电与迁移,但会提高运行功率并受到火花、绝缘和联锁条件限制;振打过于频繁会增加再飞扬次数,周期过长又可能导致积灰加重和单次脱落量增大。水泥工业颗粒物排放受到国家标准约束 [1], 题目进一步给出 10 mg/Nm^3 和 5 mg/Nm^3 两个控制目标, 要求同时考虑排放、总功率和振打峰值。

本文研究对象的四电场电除尘器结构及其可测量与可调量如图 1 所示: 含尘烟气依次流经四个电场, 每个电场由独立高压电源施加电压 U_i 并由独立振打装置按周期 T_i 清灰, 入口侧可测入口温度 H 、入口粉尘浓度 C_{in} 与烟气流量 Q , 出口侧可测出口粉尘浓度 C_{out} 与总功率 P 。

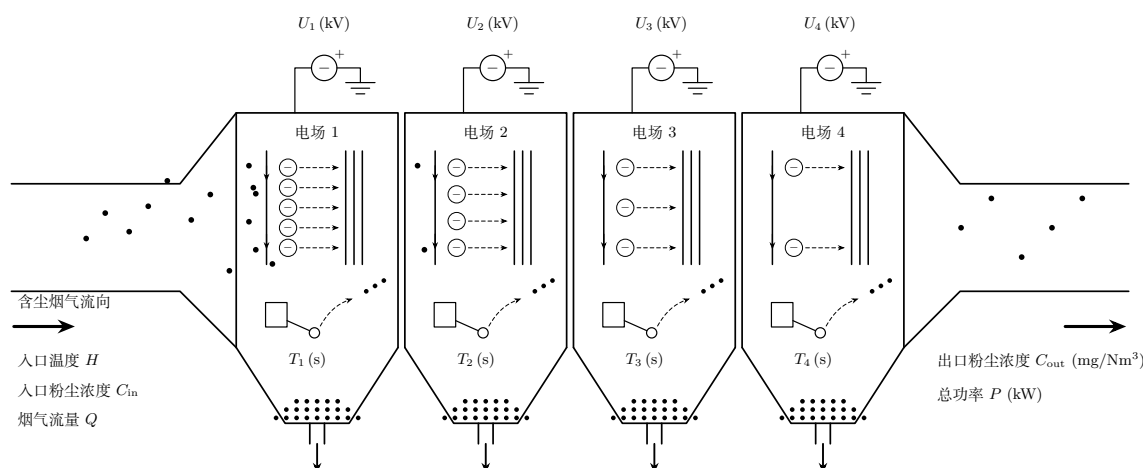


图 1 四电场电除尘器结构与变量定义

题目包含四项相互依赖的任务: 第一, 解释入口温度、入口浓度、烟气流量、四电场电压和振打周期与出口浓度之间的关系, 并分析振打峰值; 第二, 划分典型工况, 在 10 mg/Nm^3 约束下寻找最低功率控制参数; 第三, 选取差异显著的工况解释控制优先级; 第四, 将约束收紧至 5 mg/Nm^3 并估计功率代价。

1.2 总体思路

附件中有两列结果变量, 信息量却相差悬殊: 总功率几乎完全由控制量决定, 而出口浓度被压缩在量程上限附近, 几乎没有变化。若不加区分地将二者同时作为建模目标, 模型在出口浓度上学到的只会是传感器的截顶读数。本文因此采取先分层、后优化的路线:

1. 逐列检查哪些关系能被数据辨认出来, 把“可由附件验证的”与“只能靠机理推演的”分开;
2. 对可识别的总功率, 按物理形式 (电场功率 $\propto U^2$ 、振打功率 $\propto 1/T$) 建立并检验模型;
3. 对不可识别的排放层, 以记录值本身为锚点建立 Deutsch-Anderson 灰箱, 不作任何未经证实的量纲改写;
4. 在历史运行范围内求解优化问题, 并把“结果在哪些方面超出了历史经验”当作结论的一部分如实报告。

第 3 点是本文与常见做法的关键差别。附件记录的出口浓度约 50 mg/Nm^3 , 题目要求的却是 10 与 5 mg/Nm^3 , 相差 $5 \sim 10$ 倍。承认这个差距, 题目就自然变为: 控制参数应如何由现状调整至超低排放目标, 相应的电耗代价是多少。反过来, 若为了让数据靠近目标区间而假设记录值点错了小数点, 第二问就会变成“把排放放宽到 10 mg/Nm^3 能省多少电”——与题意正好相反。

2 数据审计与关系可识别性分析

2.1 数据完整性与派生变量

附件覆盖 2024 年 5 月 1 日 0 时至 5 月 7 日 23 时 59 分，共 10080 条记录，采样间隔 1 min，时间戳无重复、无缺口。原始字段包括入口温度 H_t 、入口浓度 $C_{in,t}$ 、烟气流量 Q_t 、四电场电压 $U_{i,t}$ 、四电场振打周期 $T_{i,t}$ 、出口浓度 $C_{out,t}$ 和总功率 P_t 。出口浓度缺失 50 条 (0.50%)，其余字段完整。

为表达过程负荷和振打风险，构造粉尘质量负荷

$$L_t = C_{in,t} Q_t / 1000 \quad (1)$$

其中 C_{in} 单位为 g/Nm^3 ， Q 单位为 Nm^3/h ，故 L 单位为 kg/h 。另构造电压平方和、振打频率 $60/T_i$ 与每周期积尘代理 $L_t T_i / 3600$ 。所有聚类与回归参数仅在训练期 (5 月 1—5 日) 估计，验证期 (5 月 6 日) 与回顾性测试期 (5 月 7 日) 不参与拟合，划分按时间顺序而非随机打乱。

2.2 逐列可识别性结论

对 14 列逐一审计的结果见表 1：只有出口浓度一列不可用，其余各列信息量充足。

表 1 附件逐列可识别性审计结论

数据块	审计证据	可用性
C_{out} 出口浓度	有效值 10030 条全部位于 48.74~50.00 mg/Nm^3 ，5491 条恰等于 50；与全部 12 个变量的同期相关系数绝对值均 < 0.017 ，1~120 min 滞后同样无稳定关系；封顶分类 AUC=0.5098，未封顶子样本回归验证 $R^2 = -0.019$	不可识别，不可修复
P 总功率	由控制量近乎确定，残差标准差约 6 kW (见 §5.2)	精确可辨识
H, C_{in}, Q 工况变量	连续波动并含两段持续异常事件 (§4.3)	工况分层与扰动分析
U_i, T_i 控制设定	与工况变量存在稳定相关结构 (§4.2)	可提取历史操作规律

出口浓度的分布与截顶情况见图 2。需要强调：以上结果只说明该记录不能提供可供学习的信号，不能据此反推物理变量对真实排放没有影响。已有研究曾用宽负荷下的机组负荷与各电场二次电流训练出口浓度模型 [8]；相比之下，本附件既没有二次电流，出口记录本身也几乎不变化，不具备同样的条件。附录 B 给出直接以 C_{out} 为目标建模的对照结果，可与其他方案交叉比较。

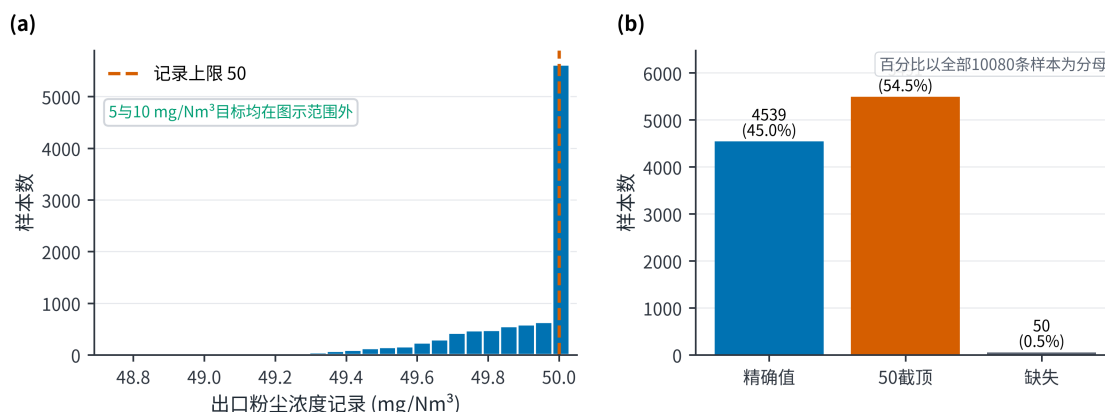


图 2 出口浓度数据诊断

首 24 h 过程量的变化如图 3 所示。入口变量和功率具有连续变化和工况迁移特征，而出口浓度记录几乎被压缩在单点附近。

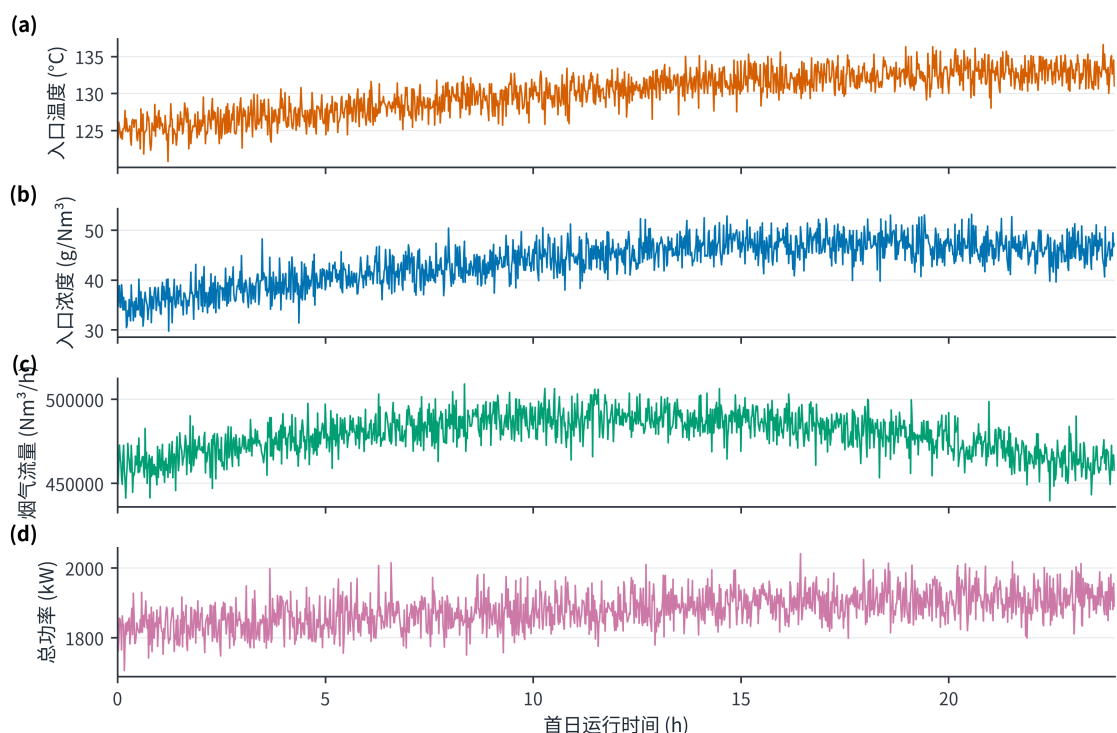


图3 首24 h过程量时序

2.3 排放锚点的取法

本文不对出口记录作任何缩放，直接把各工况出口浓度的记录中位数（都是 50 mg/Nm^3 ）当作机理模型的起算点（下称锚点）。由此确立的问题定位是：设备当前距超低排放水平尚远，本文求解的是由现状降至 10 与 5 mg/Nm^3 的控制路径及其代价。这一取法带来三个后果，后文均予明示：最优点必然落在历史电压范围的上端 (§5.5)； $10 \rightarrow 5 \text{ mg/Nm}^3$ 的功率增幅明显小于从更低起点出发时的估计 (§7)；排放层的数值仍属机理推演，须在补采未截顶的排放数据后重新标定（附录 C）。

3 模型假设、符号与总体框架

3.1 模型假设

本文将假设分为数据事实、情景假设、优化边界与运行约束四类，并逐条标注其证据状态，如表 2 所示。

表2 模型假设及证据状态

类别	假设	证据状态
数据事实	7 天记录按分钟连续，除 50 条出口浓度缺失外不填造目标值	附件可核验
数据事实	总功率按 U_i^2 与 $1/T_i$ 的物理形式建模，按时间滚动验证	可由验证与测试结果检查
数据事实	排放锚点取出口浓度记录中位数 50 mg/Nm^3 ，不作缩放	附件记录值
情景假设	去除指数按 $K = K_0 S(U)$ 随电压幂次 p 缩放，中心取 $p = 2$	Deutsch 迁移速度形式，未经现场标定
情景假设	电场贡献权重 α_i 取等权、弱前级与中心前级三种形状	工程先验与消融
情景假设	振打峰值指数 $\gamma = 1$ 由积灰量正比关系推导，扫描 $1.00/1.30/1.60$	本文推导+结构情景
优化边界	控制量位于各工况训练期实测最小—最大范围内	历史运行包络，不等同设备安全边界
优化边界	5 mg/Nm^3 档允许电压上界外推至历史最大值的 1.05 倍	显式声明的外推，须经容量核算
运行约束	前级电压高于后级至少 3 kV ，后级振打周期长于前级至少 100 s	工程排序约束

3.2 主要符号

表 3 主要符号与单位

符号	含义	单位
U_i	第 i 电场电压	kV
T_i	第 i 电场振打周期	s
C_{in}, C_{out}	入口、出口粉尘浓度	g/Nm ³ 、mg/Nm ³
Q	烟气流量	Nm ³ /h
L	粉尘质量负荷	kg/h
$P, \hat{P}$	实际、预测总功率	kW
b_i, c_i	电场功率系数、振打功率系数	kW/kV ² 、kW·s
K, K_0	去除指数及其历史锚点值	—
$S(U)$	电压强度因子, $U = U^{ref}$ 时为 1	—
p	去除指数对电压的幂次, $K \propto U^p$	—
$B(T)$	振打引起的峰值增量代理	mg/Nm ³
ρ, γ	振打相位叠加尺度、峰值—周期指数	—

框架上，工况与总功率由实测数据辨识，排放约束由显式先验建立，两者仅在支持域优化阶段合并；本文对每一个数值结论都标注其属于“实测辨识”还是“条件式推演”。

4 问题一：可识别的影响关系与振打峰值

第一问要求解释各变量与出口浓度的关系，并分析振打峰值。按 §2 的审计结论，本节分两层作答：先给出附件能够支持的三类实证结论，再给出只能靠机理推演的排放部分。

4.1 总功率的生成律（实证）

在训练期以物理形式拟合，得到

$$P = 73.09 + \sum_{i=1}^4 b_i U_i^2 + \sum_{i=1}^4 \frac{c_i}{T_i} \tag{2}$$

其中 $b = (0.0849, 0.0851, 0.0847, 0.0853)$ kW/kV²、 $c = (5.43, 5.37, 5.49, 5.26) \times 10^4$ kW·s。四个电场的 b_i 彼此相差不到 0.8%，说明四台高压电源的电气特性一致；写成合并形式即 $P \approx 73.0 + 0.0850 \sum U_i^2 + 5.39 \times 10^4 \sum T_i^{-1}$ 。这个式子的物理含义很直白：**电场功率与电压平方成正比，振打功率与振打频率 $60/T_i$ 成正比**。模型的检验结果见 §5.2。

从这个式子还能直接读出一个此前被忽略的事实。在训练期的中位设定下，电场功率为 1020.3 kW，振打功率为 693.7 kW，即**振打环节占总功率的 38.8%**。振打周期虽非题面强调的主要电气变量，但其对电耗的贡献与电压同一量级，不可忽略。相应地，把四个电场的周期从中位延长到历史 P95 可省下 56.1 kW，延长到历史最大值可省下 109.1 kW，代价是单次积灰量分别增加 8.4% 和 17.9%（图 5b）。§7.3 将说明，该可回收功率与排放限值收紧一档所增加的功率代价处于同一量级——这正是振打周期不宜为节电而无限延长的定量依据。

还有一点值得指出：把入口温度、浓度、流量也加进这个式子后，验证集与测试集的 R^2 变化都不到 10^{-3} （见表 8 末行）。**总功率只由控制量决定，与工况无关**。后文按工况分别优化所得的功率差异，全部来自各工况不同的排放锚点与取值范围，而非工况本身直接作用于功率。

4.2 历史操作规律（实证）

出口浓度不可识别，并不意味着附件不含操作策略方面的信息。图 4(c) 给出整周记录中工况变量与控制设定的相关系数，其中两类规律值得报告：

规律一（稳定）：入口浓度升高时，四个电场的振打周期一律缩短（ $r = -0.208$ 、 -0.211 、 -0.277 、 -0.278 ）。该方向在训练期、验证期与测试期三个时窗内符号一致，是附件中可复现的操作规律，与“高负荷下不宜延长周期”的工程直觉相符。

规律二（不稳定）：入口浓度升高时，前两个电场的电压随之升高（整周 $r = +0.172$ 、 $+0.166$ ），后两个电场基本不变（ $r = +0.019$ ）。但该规律并不贯穿整周： $r(C_{in}, U_1)$ 在前 5 天为 -0.052 ，第 6、7 天分别为 $+0.507$ 与 $+0.492$ 。即入口浓度升高时抬升前级电压这一做法，仅出现于最后两天。本文如实报告该非平稳性：前后级响应的不对称在整周与最后两天均成立，却不能视为贯穿全周的稳定规律。这也表明历史数据中至少发生过一次操作策略调整，故第三问只将其作为参照，而非定论。

表 4 工况变量与控制设定的历史相关系数（全记录期）

控制量	与入口温度 H	与入口浓度 C_{in}	与烟气流量 Q
U_1	-0.130	+0.172	+0.074
U_2	-0.127	+0.166	+0.070
U_3	-0.112	+0.019	-0.068
U_4	-0.113	+0.019	-0.063
T_1	-0.075	-0.208	-0.180
T_2	-0.064	-0.211	-0.175
T_3	-0.159	-0.277	-0.130
T_4	-0.151	-0.278	-0.135

4.3 两段持续异常工况（实证）

附件含两段持续数十分钟以上的异常工况，其控制端响应方向相反，构成理解设备行为的两个直接入口（图 4a、4b）。

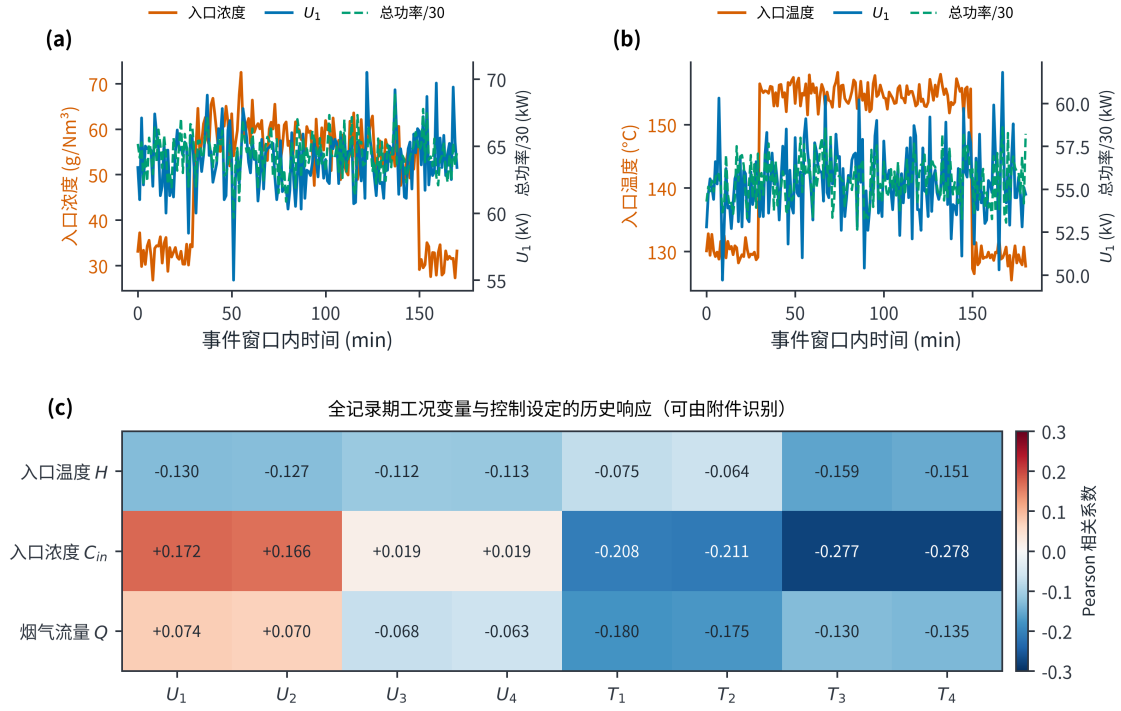
表 5 两段持续异常工况及控制端响应

事件	时间窗	特征	控制端响应
入口浓度尖峰	05-02 14:02—15:49，其中 41 min 超过 60 g/Nm ³	入口浓度峰值 72.47 g/Nm ³ （基线中位 37.41）	U_1 均值 63.7 kV（峰值 67.8），振打周期同步缩短，总功率均值 1924 kW、峰值 2020 kW
入口温度尖峰	05-06 13:00—14:59，持续 120 min	入口温度峰值 158.2 °C（基线中位 126.5），同期入口浓度降至均值 25.9 g/Nm ³	U_1 均值降至 55.5 kV，总功率均值降至 1668 kW

第二段事件在工程上更值得注意：**140~160 °C 正是水泥窑尾粉尘比电阻最高的温度区间**。比电阻过高会引发反电晕，使粉尘向极板迁移的速度下降、火花率上升，是电除尘最难运行的一段工况 [3-4]。附件在这段时间里的记录显示，操作端选择了降压而不是升压，这与“高比电阻区应限制电流密度以抑制反电晕”的常规处置一致；但附件没有二次电流和火花率记录，无法判断这是否为最优处置。由此可得两点：第一，第二问的工况划分把温度作为主轴之一是必要的；第二，本文四类工况的温度中心约为 120.7 与 130.4 °C，**都没有覆盖这段高比电阻区间**，因此本文结论不能外推到该区间，须单独补采数据重新标定 (§9.3)。

4.4 排放响应的灰箱骨架（条件式推演）

Deutsch-Anderson 关系用指数形式描述电除尘效率。它的理想化形式忽略了再飞扬、气流分布和粒径变化，因此只能当作**骨架**：形式取自机理，系数则须另行给定（本文称“灰箱”）[2-4]。White 的经典专著 [12] 与 Parker 的电除尘器电气运行专著 [13] 都强调，电气运行、粉尘性质与振打三者是相互耦合的。定义去除指数



注：振打周期规律（ $C_{in} \uparrow \Rightarrow T \downarrow$ ）在四个时窗内一致；电压规律并不平稳—— $r(C_{in}, U_1)$ 在前5天为 -0.05，在末日为 +0.49。

图 4 附件中可识别的运行证据

$$K = \ln \frac{1000C_{in}}{C_{out}}, \quad C_{out} = 1000C_{in}e^{-K} \quad (3)$$

K 越大表示除尘越彻底： K 每增加 $\ln 2$ ，出口浓度就减半。对工况 k ，以历史中位电压 $U_{i,k}^{ref}$ 和出口浓度记录中位数 $C_{0,k} = 50 \text{ mg/Nm}^3$ 为起算点，得 $K_{0,k} = \ln(1000C_{in,k}/C_{0,k})$ ，四类工况依次为 6.257、6.514、6.702、6.797。粉尘向极板迁移的速度与电场强度的平方近似成正比，故取

$$K_k(U, T) = K_{0,k} S_k(U) - \{G_k(T) - G_k(T^{ref})\}, \quad S_k(U) = \frac{\sum_i \alpha_i \left(U_i / U_{i,k}^{ref} \right)^p}{\sum_i \alpha_i} \quad (4)$$

$$G_k(T) = \sum_{i=1}^4 \beta_{i,k} \left(\frac{T_i}{T_{i,k}^*} + \frac{T_{i,k}^*}{T_i} - 2 \right), \quad \beta_{i,k} = \beta_i \left(\frac{L_k}{L_{med}} \right)^{1/2} \quad (5)$$

这样构造有一个好处：当 $U = U^{ref}$ 、 $T = T^{ref}$ 时式子严格还原为 $K = K_{0,k}$ ，即模型天然复现历史运行点，因此不再需要一个额外的整体比例参数 η ，全部电压敏感性都归结到一个幂次 p 上（中心取 $p = 2$ ）。 $G_k(T)$ 在 $T_i = T_{i,k}^*$ 处取最小值，表示周期过短或过长都不利：过短会频繁引起再飞扬，过长则积灰加重、单次脱落量变大。试验研究也确实观察到再飞扬与最大允许周期之间存在这种权衡 [6]。再加一点安全裕度 s ，连续排放的基础值为 $C_{base,k} = 1000C_{in,k} \exp[-K_k(U, T) + s]$ 。

本节全部排放系数均为工程先验，不是由附件出口浓度估计得到的参数。

4.5 振打峰值：由积灰量推导而非假定

附件只给出振打周期的设定值，没有实际振打时刻、相位和秒级浓度，所以逐次峰值本身无法辨认。但峰值随周期怎样变化可以由质量守恒直接推出，不必假定指数：设单位时间的积灰速率正

表 6 灰箱排放模型参数来源与证据状态

参数	中心取值	含义	来源	是否由附件识别
p	2.0	去除指数对电压的幂次	Deutsch 迁移速度形式	否
α_i	(1.15,1.10,0.95,0.80)	电场去除贡献权重	工程先验与消融	否
β_i	(0.30,0.27,0.20,0.17)	周期偏离损失	工程情景	否
b_0	1.2	峰值尺度	工程情景	否
γ	1.0	周期—峰值指数	本文由积灰量推导	否
ρ	1.0	相位叠加尺度	同相保守情景	否
s	0.08	安全裕度	工程情景	否

比于粉尘负荷 L ，那么一个周期内积下的灰饼质量就正比于 $L \cdot T$ ；一次振打抛落的粉尘量又正比于灰饼质量，于是

$$B_k(T; \rho, \gamma) = b_0 \rho \left(\frac{L_k}{L_{med}} \right) \sum_{i=1}^4 w_i \left(\frac{T_i}{T_{i,k}^*} \right)^\gamma, \quad \gamma = 1 \quad (6)$$

由此得到两条互补的结论，也正是第一问关于振打峰值的完整回答：

- **单次瞬时峰值随周期近似线性放大**：周期延长 10% 与 20% 时，单次峰值代理相应放大 10% 与 20%；
- **对小时均值的贡献与周期无关**：单位时间的总再飞扬量正比于 $L \cdot T \times (1/T) = L$ ，只取决于粉尘负荷。

换言之，延长振打周期并不改变再飞扬对小时均值的贡献，只是把等量粉尘集中于次数更少、单次幅度更高的峰值中释放。这条结论有两个直接用途：一是解释了为什么“延长周期省电”必须和瞬时峰值一起考虑 (§7.3)；二是由于现行标准按小时均值考核，本文把瞬时峰值直接卡到限值的做法**偏保守** (§7.4)。中心情景取 $\rho = 1$ ，即假设四个电场同时振打、峰值完全叠加，这是最不利的上界；若错开相位则取 $\rho = 0.80$ 或 0.65 。敏感性分析仍扫描 $\gamma \in \{1.00, 1.30, 1.60\}$ ，以检验推导值附近是否稳健。

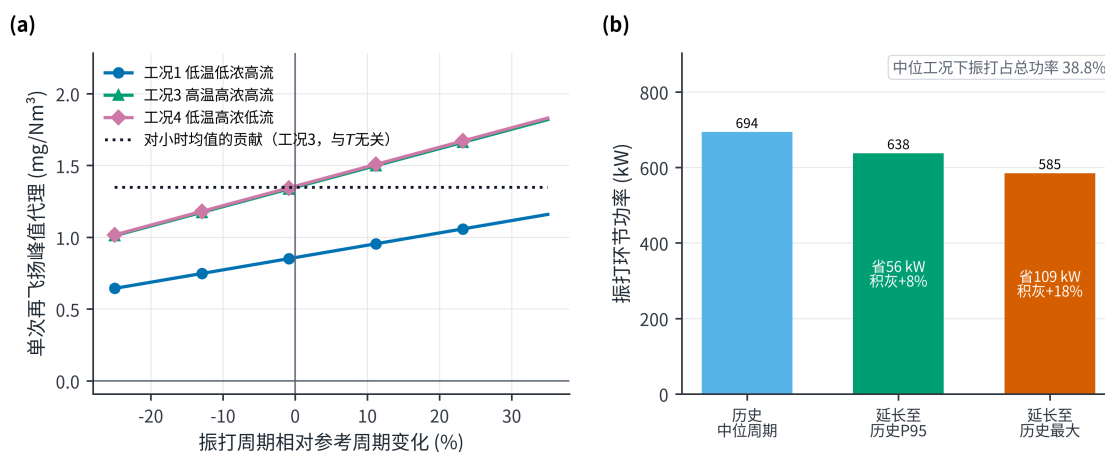


图 5 振打峰值机理与振打功率权衡

4.6 各因素方向性小结

在上述骨架中，其他条件不变时各因素的作用方向如下。提高 U_i 会增大 K 、降低连续排放，但总功率按 U^2 上升。 K 不变时，入口浓度升高会使出口浓度同比升高。流量增大会缩短烟气在电场中的停留时间、同时抬高粉尘负荷，因而需要更强的捕集能力。温度则通过气体性质、粉尘比电阻

和放电状态影响除尘效果，其中 140~160 °C 的高比电阻区间最为不利 (§4.3)。这些方向来自机理与文献 [2-4]，不是从附件回归出来的。

5 问题二：典型工况划分与 10 mg/Nm³ 最低功率

5.1 工况聚类

聚类只用入口温度、入口浓度和烟气流量三个变量——它们由上游工艺决定，不受控制策略影响，因此不会出现“用结果去定义工况”的问题。三者标准化后作 K-means 聚类 [9]。取 $k = 2, 3, 4, 5, 6$ 比较，轮廓系数 [10]（衡量类内紧凑程度与类间分离程度，取值越大越好）依次为 0.2963、0.3682、0.4084、0.3958、0.3934，在 $k = 4$ 最高。另一项 Calinski-Harabasz 指标依次为 3204.2、4399.0、4721.2、4835.5、4712.0，最大值却出现在 $k = 5$ ，两项判据并不一致。

本文仍取 $k = 4$ ，理由有二：轮廓系数在 $k = 4$ 最高；四类中心恰好构成温度两档（约 120.7 与 130.4 °C）与流量两档（约 44.0 与 48.1 万 Nm³/h）的 2×2 组合，物理含义清楚。但必须说明，CH 指标支持 $k = 5$ ，而 0.4084 的轮廓系数也只说明各类分得中等清楚。因此这四类工况是统计意义上的分层，而非唯一的物理状态划分；另外如 §4.3 所述，两段异常工况都落在四类中心之外。

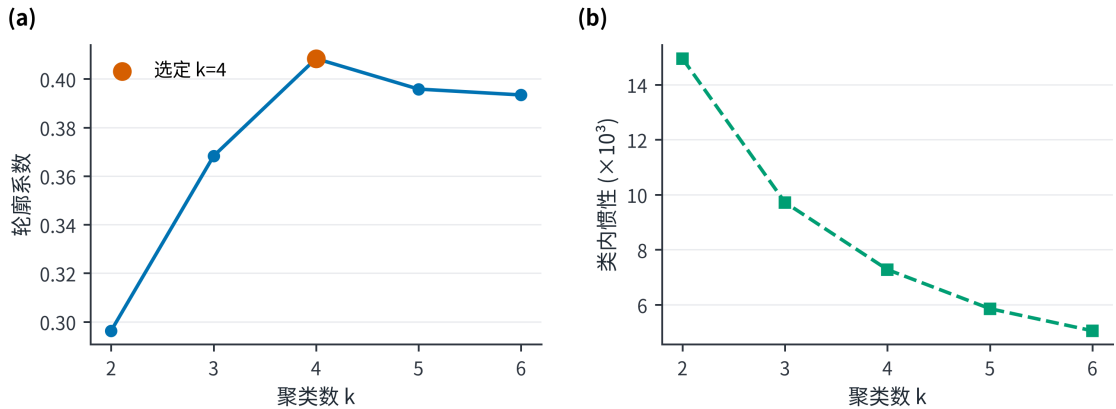


图 6 聚类数选择结果

表 7 四类典型工况画像

工况	占比	温度/°C	入口浓度/(g/Nm³)	流量/(Nm³/h)	粉尘负荷/(kg/h)	历史总功率/kW	锚点 K_0
1 低温 低浓度 高流量	23.35%	120.65	26.08	478257	12460	1818.35	6.257
2 高温 中浓度 低流量	20.06%	130.32	33.72	443591	14922	1870.89	6.514
3 高温 高浓度 高流量	34.93%	130.41	40.71	480851	19560	1763.08	6.702
4 低温 高浓度 低流量	21.67%	120.69	44.75	439990	19673	1696.55	6.797

5.2 功率模型的设定与检验

§4.1 采用的形式并不是唯一可选的。为说明“选什么形式”本身有多重要，本文在同一时间划分下比较四种设定：只用均值的基线、电压一次项+周期一次项的岭回归 [11]、电压二次项+周期

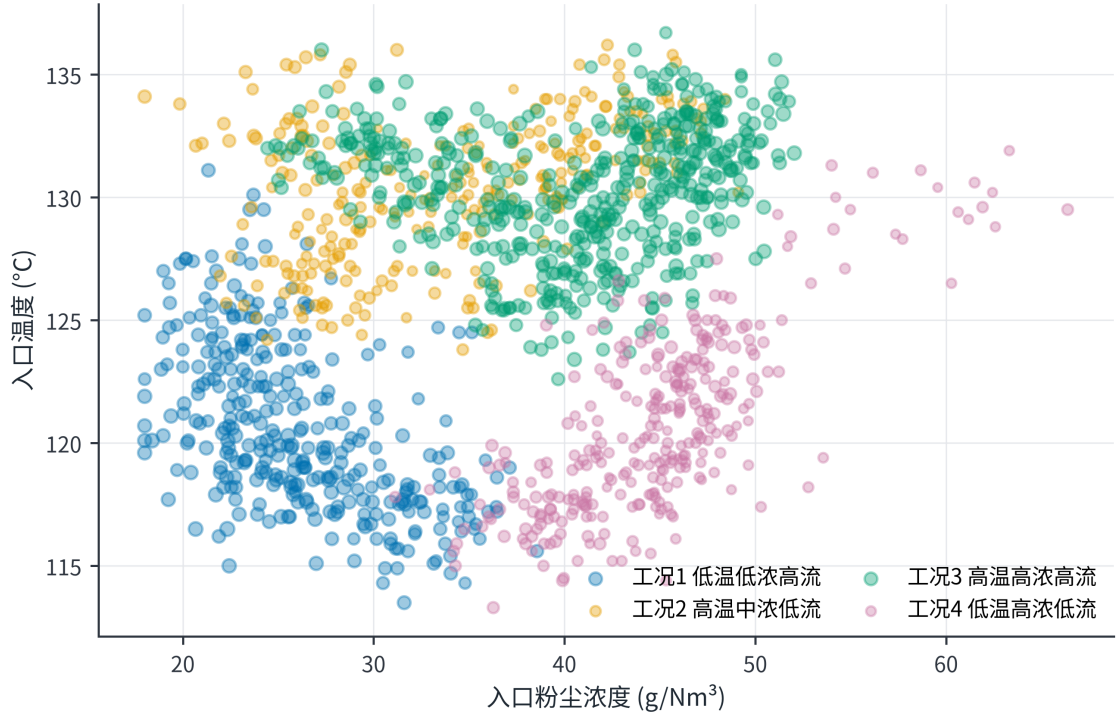


图7 训练期典型工况分布

一次项的岭回归，以及本文的 $U_i^2 + 1/T_i$ 形式。岭回归的正则强度按训练期 3 个逐步加长的窗口滚动选择，本文形式则没有可调超参数。结果见表 8。

表8 功率模型设定比较与时间外推结果

设定	验证 R^2	验证 RMSE/kW	测试 R^2	测试 RMSE/kW	测试平均偏差/kW
均值基线	-5.3259	140.18	-0.3733	83.51	-43.54
电压一次+周期一次（岭）	0.9798	7.93	0.9180	20.41	-16.65
电压二次+周期一次（岭）	0.9837	7.12	0.9571	14.76	-10.78
本文 $U_i^2 + 1/T_i$	0.9885	5.97	0.9930	5.98	-0.02
本文设定+工况变量	0.9885	5.97	0.9929	5.98	+0.08

这张表有三点值得强调。

第一，**该系统性偏差来自模型设定误差，而非数据漂移**。线性 T 的设定在测试日留下 -10.78 kW 的平均偏差，而且偏差在高功率区还会扩大；改成 $1/T$ 后，偏差降到 -0.02 kW，测试 RMSE 从 14.76 kW 降到 5.98 kW， R^2 从 0.957 升到 0.993（图 9）。若沿用线性形式，该偏差很容易被误判为数据本身的问题而额外计入误差预算；实际上，它主要由错误的函数形式引入。

第二， T 是优化中的决策变量。线性形式意味着周期每延长单位时间所节省的功率恒定，而真实的边际节电按 c_i/T_i^2 衰减，即周期越长、再延长的收益越小。因此线性周期设定会使最优振打周期系统性地偏离真实值。

第三，再加入工况变量后各项指标不再改善，印证了 §4.1 的判断：总功率只由控制量决定。

验证集上绝对误差的 90% 分位数为 9.78 kW，本文把它作为做功率预算时的余量。它不改变各方案的优劣排序，也不保证在今后的日期里仍能覆盖 90% 的情形。

5.3 优化模型

设控制向量 $z = (U_1, \dots, U_4, T_1, \dots, T_4)$ 。对每类工况 k 与限值 C_{lim} 求解

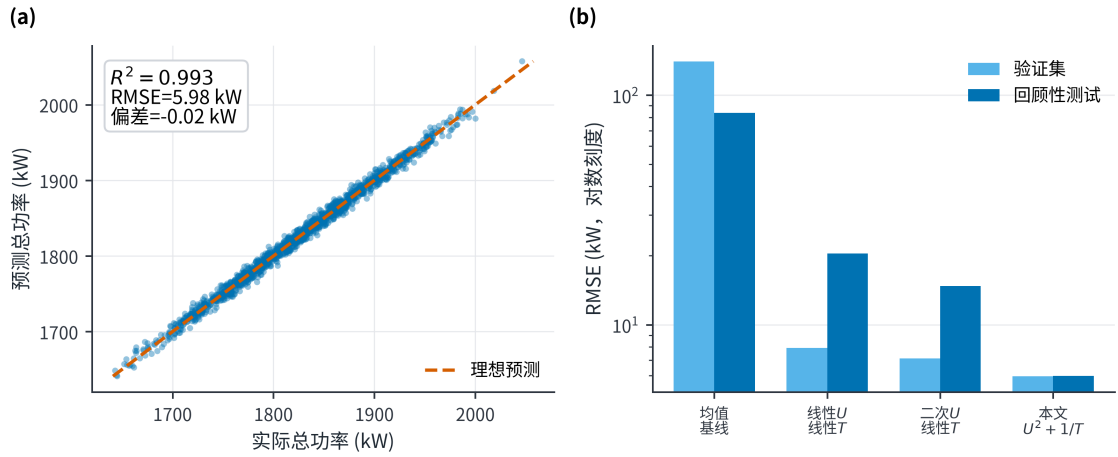


图8 功率模型预测结果

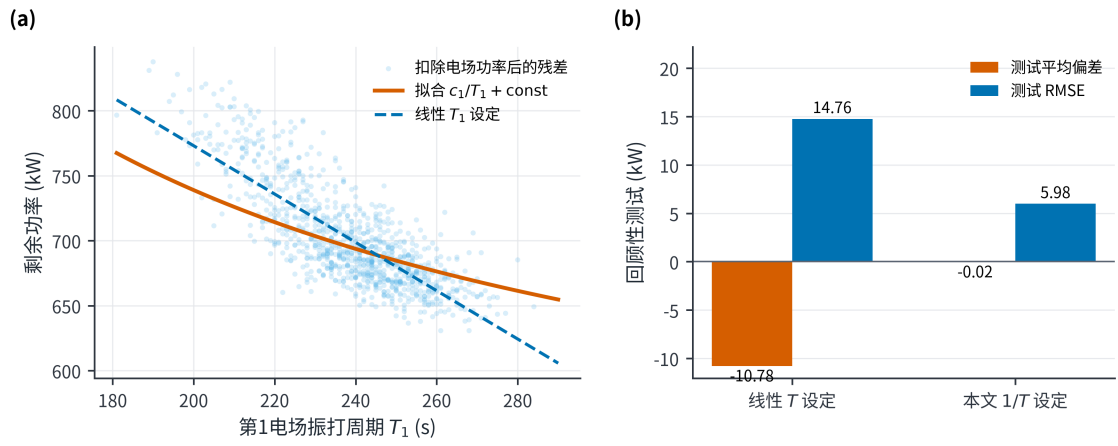


图9 振打周期的设定形式对比

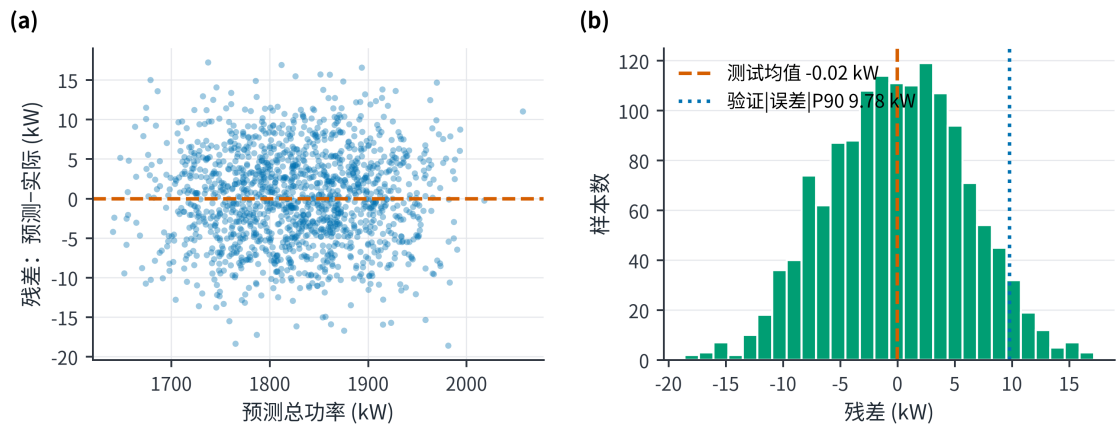


图10 回顾性测试残差诊断

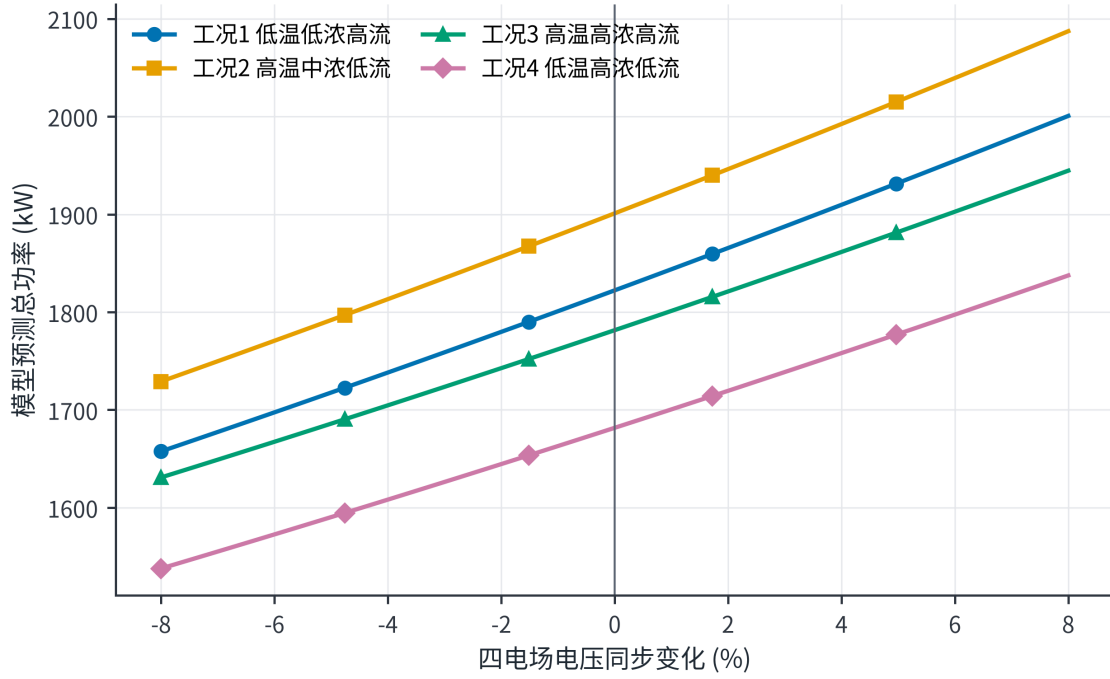


图 11 局部电压—功率响应

$$\min_z \hat{P}(z) = a_0 + \sum_i b_i U_i^2 + \sum_i \frac{c_i}{T_i} \quad (7)$$

$$\text{s.t. } C_{base,k}(z) + B_k(z) \leq C_{lim}, \quad U_1 \geq U_3 + 3, U_2 \geq U_4 + 3, T_3 \geq T_1 + 100, T_4 \geq T_2 + 100, \quad z \in \Omega_k \quad (8)$$

取值范围 Ω_k 取该工况训练期内各控制量实际出现过的最小值到最大值。5 mg/Nm³ 一档另外允许电压上界放宽到历史最大值的 1.05 倍，放宽了多少在表 11 中逐条列出。已有的多电场节能研究同样采用“排放受限、分场优化”的思路 [5,7]，但它们的排放模型都由现场实测标定，条件与本文不同，方法对照见附录 A。

关于求解方法，此处说明本文未采用凸优化求解器的原因。以电压 U 为变量时，目标函数 $\sum b_i U_i^2$ 本身是凸的，但排放约束要求这样一个凸函数取值**足够大**——这类约束把可行域挖成了非凸的形状，整体不是凸规划。若改用 $v_i = U_i^2$ 作变量，目标和排放约束都变成线性，可惜排序约束 $U_1 \geq U_3 + 3$ 在 v 下不再是凸集。即在任一坐标下均无法化为凸问题，故不宜宣称凸性。

本文因此采用**多起点 SLSQP 确定性求解**：每个“工况—限值”组合从 53 个起点分别求解（5 个有结构的起点：历史中位、上界、下界、中点、 T^* 点；再加 48 个随机起点），以检验各起点是否收敛至同一解。此外，在同一可行域内均匀随机抽取 4×10^5 个候选点作随机搜索，用作独立对照 (§8.1)。这一做法取代了早期版本的百万级随机采样：问题结构清楚之后，确定性求解既更快，也更容易复算。

5.4 电场优先级的解析判据

在给出数值解之前，先由模型结构推导各电场的调节优先级。把 $K_k = K_{0,k} \sum_i \alpha_i (U_i / U_{i,k}^{ref})^2 / \sum \alpha$ 与 $P = \sum_i b_i U_i^2$ 同时对 U_i^2 求导，得到**单位电功率的去污指数增益**

$$\left. \frac{dK}{dP} \right|_i = \frac{K_{0,k} \alpha_i}{\left(\sum_j \alpha_j \right) b_i \left(U_{i,k}^{ref} \right)^2} \quad (9)$$

该比值越大，表示该电场单位功率带来的除尘收益越高。由于四个电场的 b_i 几乎相同，排序完全取决于 $\alpha_i/(U_{i,k}^{ref})^2$ ，而不是取决于 α_i 。原因在于功率按电压平方增长，参考电压较低的电场按相同比例抬升时，所需电功率显著更少。代入中心权重与各工况的历史中位电压，得到表 9。可以看到，第三电场的去除权重 $\alpha_3 = 0.95$ 虽然低于前两级，但它的参考电压低约 10 kV，平方之后使它的单位功率收益在四类工况中都排第一。

表 9 电场优先级解析判据 ($\times 10^{-3} \text{ kW}^{-1}$)

工况	电场 1	电场 2	电场 3	电场 4	优先次序
1 低温低浓度高流量	5.768	5.504	6.668	5.573	E3 > E1 > E4 > E2
2 高温中浓度低流量	5.556	5.269	7.023	5.870	E3 > E4 > E1 > E2
3 高温高浓度高流量	6.630	6.327	8.024	6.706	E3 > E4 > E1 > E2
4 低温高浓度低流量	7.551	7.207	7.717	6.502	E3 > E1 > E2 > E4

这也解释了本文早期版本“前级优先”这一结论的真实来源：**它是排序约束 $U_1 \geq U_3 + 3$ 带出来的，而不是权重 α_i 决定的**。由于抬升 U_3 必须同时抬升 U_1 ，该排序约束使二者的调节之间存在耦合，故最优解中 U_1 与 U_3 总是同步走高。若排除该约束，单位功率收益最高的仍是第三电场。

振打侧同理： $\partial P / \partial T_i = -c_i / T_i^2$ 。前级周期短（中位 240 s），后级周期长（中位 441 s），故在相同的周期延长量下，前级的节电量约为后级的 **3.4 倍**。这与电压侧由前级承担主要捕集并不矛盾：电压用于提高捕集能力，周期用于降低电耗，二者的最优着力点分别位于不同电场。

5.5 10 mg/Nm³ 下的最优控制参数

四类工况的中心情景解如表 10（5 mg/Nm³ 行为第四问提供参数）。

表 10 四工况两档目标的最优控制参数

工况	目标/(mg/Nm ³)	U_1/kV	U_2/kV	U_3/kV	U_4/kV	T_1/s	T_2/s	T_3/s	T_4/s	峰值排放/(mg/Nm ³)	预测总功率/kW
1	10	70.2	68.6	57.6	57.4	290	289	492	493	9.948	2045.55
2	10	70.9	69.8	59.2	58.2	280	278	494	498	9.962	2104.30
3	10	68.0	60.3	56.3	57.3	275	280	506	510	9.971	1924.67
4	10	70.5	61.8	57.2	42.9	285	283	495	498	9.991	1851.15
1	5	73.7	71.6	60.5	60.3	290	289	492	493	4.993	2182.25
2	5	74.4	72.9	62.2	61.1	280	278	494	498	4.998	2245.51
3	5	71.4	63.4	59.1	60.4	275	280	506	510	4.991	2056.06
4	5	74.0	66.4	60.1	42.9	285	283	495	498	4.988	1973.10

第二问的答案：在 10 mg/Nm³ 下，四类工况的最低总功率为 1851.15~2104.30 kW，按训练期各工况占比加权为 **1973.0 kW**，折合日电耗 47.35 MWh。作为对照，历史实际加权功率是 1783.2 kW（日电耗 42.80 MWh）。也就是说，达到 10 mg/Nm³ 意味着**功率上升 10.6%、日电耗多用 4.55 MWh**。

这一方向是必然的：历史排放约 50 mg/Nm³，将出口浓度压至其五分之一必须提高捕集能力，而在线可调的手段只有电压，电压又按平方计入功率。因此，本文所称“节能”是指在同一排放目标下相对其他可行控制方案的功率降低，而非相对历史运行工况的降低。

解的结构与表 9 的判据一致。振打周期 T 在两档下都被推至历史区间上端，因为延长周期直接降低振打功率；电压则按单位功率收益由高到低分配——工况 3 与工况 4 中 U_2 明显低于 U_1 ，工况 4 的 U_4 更被压至历史下界 42.9 kV，正是由于该电场在此工况下单位功率收益最低（表 9 末行）。可见四场同步抬升并非最优。

表中数值为点预测值，编制功率预算时可另计 9.78 kW 的误差余量。电压无须按 0.1 kV 精度执行：多起点求解所得功率完全一致，但控制量本身存在若干功率近乎等价的组合 (§8.1)。

5.6 目标点已越出历史运行经验

上表的解能否直接下发给设备？答案是否定的，且理由可由数据定量给出。

本文引入马氏距离，以定量评价推荐控制组合偏离历史运行经验的程度。仅逐个检查各电压是否越限并不足够，因为八个控制量之间彼此相关：历史上从未出现过 U_1 很高而 U_3 很低的组合，即便二者各自都落在允许范围内。马氏距离平方 $d_M^2 = (z - \mu_k)^T \Sigma_k^{-1} (z - \mu_k)$ 正是把这种相关性一并计入的距离，其中 μ_k 和 Σ_k 是该工况训练期控制量的均值与协方差。将其与历史上 97.5% 的时间均不超过的水平 $q_{k,0.975}$ （四类工况依次为 20.55、18.72、16.85、19.09）相比，即可量化推荐解偏离历史经验的程度，结果见表 11。

表 11 推荐解相对历史运行经验的外推审计

工况与限值	d_M^2	$q_{0.975}$	倍数	最高电压超出历史最大值	判定
工况 1, 10 mg	49.39	20.55	2.40×	0.0%	外推
工况 2, 10 mg	46.46	18.72	2.48×	0.0%	外推
工况 3, 10 mg	77.69	16.85	4.61×	0.0%	外推
工况 4, 10 mg	102.79	19.09	5.38×	0.0%	外推
工况 1, 5 mg	65.84	20.55	3.20×	+5.05%	外推
工况 2, 5 mg	60.31	18.72	3.22×	+5.07%	外推
工况 3, 5 mg	106.26	16.85	6.31×	+5.00%	外推
工况 4, 5 mg	133.43	19.09	6.99×	+5.07%	外推

结论有三层。

其一，最低功率解全都落在历史经验之外， d_M^2 达到该水平的 2.40~6.99 倍。因此本文把 d_M^2 作为外推风险指标报告，而不把它当作必须满足的约束——否则求出来的就不是“最低功率”，而是“最低功率且最像以前的操作”，这是两个不同的问题。

其二，如果把 $d_M^2 \leq q_{k,0.975}$ 真的**加成硬约束**，八个“工况—限值”组合里仍有七个有解 (results/mahalanobis_constrained_feasibility.csv)，代价是功率上升：10 mg/Nm³ 档四类工况分别多耗 1.46%~4.38%（加权 2026.5 kW，比最低功率解高 2.7%），5 mg/Nm³ 档三个有解的工况多耗 2.22%~6.07%，只有工况 1 在 5 mg/Nm³ 下无解。这就给出了一个可选的**保守方案**：若现场要求控制组合严格落在历史经验支持的范围内，可以略增电耗为代价采用该方案。本文以最低功率解为主报告结果，同时保留该保守方案，供现场按风险偏好选用。

其三，从电源容量看，10 mg/Nm³ 的四个解**仍位于历史电压范围之内**：各电场电压均不超过该工况历史出现过的最大值，说明该目标在现有高压电源容量下原则上可达。而 5 mg/Nm³ 需要各电场电压比历史最大值再高，最多高出 5.07%；四类工况所需的最小裕度分别是 5%、4%、1%、0%。**这说明 5 mg/Nm³ 首先是容量问题，其次才是电耗问题**：若现场高压电源不具备该裕度，仅靠优化运行参数无法达标，须通过增设电场、改造电源或前端减尘等措施解决。

5.7 排放—功率权衡前沿

把排放限值由 10 逐档收紧到 5 mg/Nm³ 并重复求解，得到“排放—功率”权衡曲线（图 13）。四类工况的最优功率随限值收紧单调上升，工况加权功率由 1973.0 kW（10 mg/Nm³）升至 2105.5 kW（5 mg/Nm³），中间各档依次为 1991.3、2012.7、2046.3、2067.7 kW；所需电压裕度同步由 0% 增至 5%（每收紧 1 mg/Nm³ 约增 1 个百分点）。全部档位均高于历史实际加权功率 1783.2 kW。该图把限值与功率的关系整体呈现出来：**沿曲线向左（限值更严）必然增加电耗，向右（限值更松）才降低电耗**；历史运行点则位于该图右侧更远处。

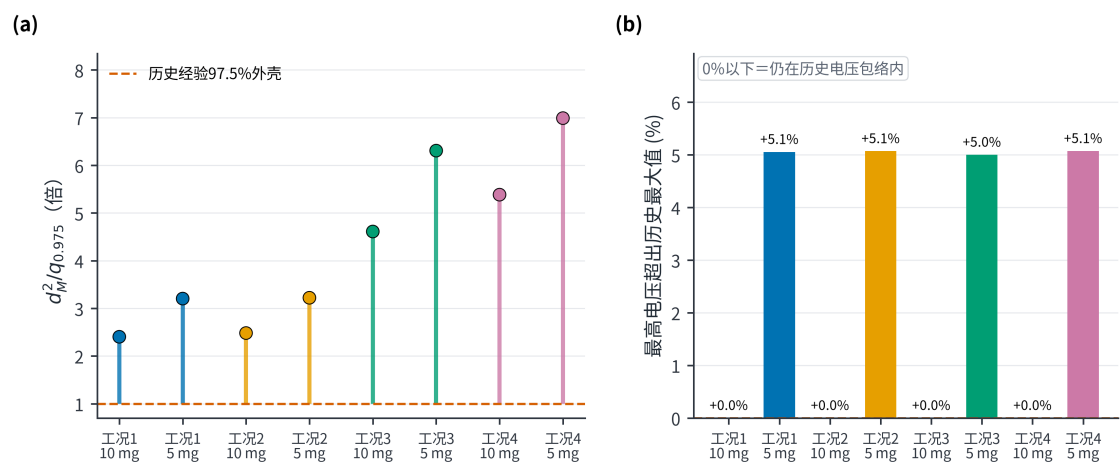


图 12 推荐解相对历史运行经验的外推审计

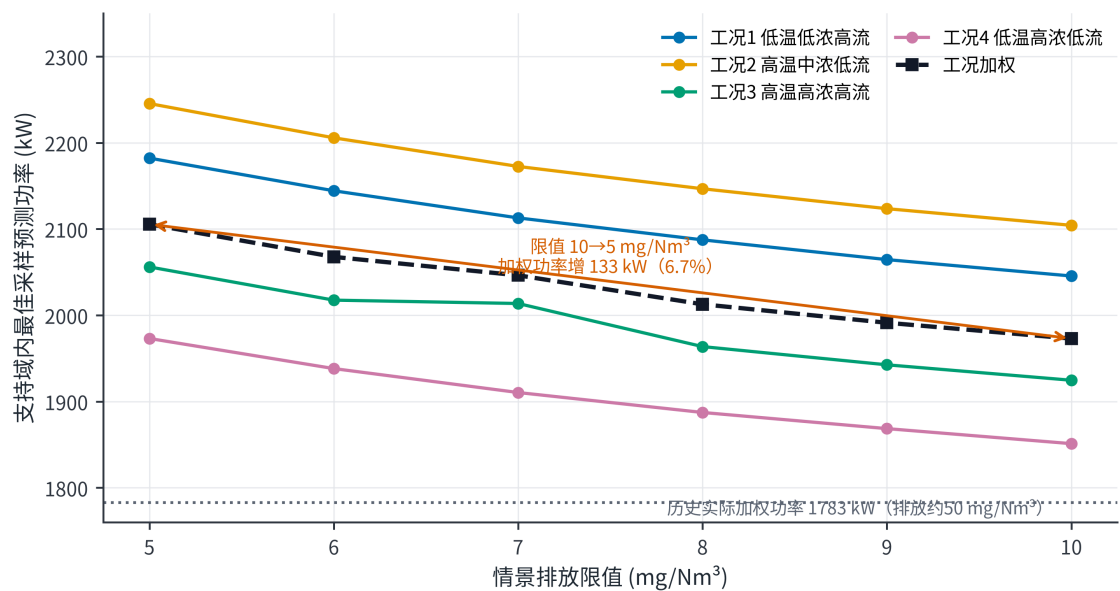


图 13 排放—功率权衡前沿

6 问题三：代表工况的控制策略比较

选取工况 1 和工况 3。二者流量都较高，但入口浓度和粉尘负荷差异明显：工况 1 平均负荷 12460 kg/h，工况 3 为 19560 kg/h，可用于比较低负荷与高负荷策略。

低负荷工况 1 采用四场同步升压策略（即四个电场按相近幅度同步抬升）：10 mg/Nm³ 下 $U = (70.2, 68.6, 57.6, 57.4)$ kV，四个电场相对各自参考值的抬升幅度接近。原因是该工况锚点 $K_0 = 6.257$ 最低，达标所需的强度因子 $S = 1.288$ 最大，四个电场均须出力，无从取舍。

高负荷工况 3 采用择优升压策略： $U = (68.0, 60.3, 56.3, 57.3)$ kV，其中 U_1 、 U_3 、 U_4 都接近历史最大值，唯独 U_2 停在 60.3 kV（它的历史最大值是 69.3 kV）。这正是表 9 判据的直接体现——工况 3 的优先次序为 $E3 > E4 > E1 > E2$ ，第二电场单位功率收益最低，故被留在低位。限值由 10 收紧到 5 mg/Nm³ 时，各场增量为 $\Delta U = (+3.4, +3.1, +2.8, +3.1)$ kV，这一格局保持不变。

因此，第三问不能简单归结为固定的“前级优先”，而应按单位功率的除尘收益确定调节顺序。以下三条判据可直接计算：

- **电压侧**：按 $\alpha_i/(b_i U_i^{ref2})$ 降序抬升，受排序约束 $U_1 \geq U_3 + 3$ 牵引时同步抬升 U_1 ；
- **振打侧**：按 c_i/T_i^2 降序延长周期，即优先延长周期较短的前级；
- **两者分工**：电压用于提高捕集能力，周期用于降低电耗，二者的最优着力点分别位于不同电场，不应混为一谈。

历史数据对上述判据只提供了部分支持。§4.2 的规律一（入口浓度升高时缩短各场周期）在全部时段方向一致，与高负荷下不宜为节电而延长周期相符；规律二（入口浓度升高时抬升前两级电压）仅在最后两天成立，故本文只将其列作参照。

为便于现场执行，按排放构成与容量状态把上述比较整理成一张决策表，见表 12。

表 12 控制优先级决策规则

状态特征	优先动作	原因
连续排放基值接近限值，振打峰值占比较低	按 $\alpha_i/(b_i U_i^{ref2})$ 降序小步提高电压	主要矛盾是连续捕集强度不足，应优先动用性价比最高的电场
峰值增量占比较高，周期明显长于参考周期	优先缩短过长周期并实施错相振打	主要矛盾是单次积灰脱落和峰值叠加
周期过短、振打频繁，且电耗压力大	优先延长前级周期	前级 c_i/T_i^2 最大，单位延长的节电最多
高粉尘负荷且基值和峰值均高	先错相、再调整周期，最后小步提升电压	先控制峰值，再补充连续捕集能力
某电场电压已达历史最大值	转向次优电场；若各场均达上限则停止优化	数学可行不等于电源容量可行
入口温度进入 140~160 °C 高比电阻带	退出优化，按现场反电晕处置预案运行	该带未被本文工况覆盖，模型外推无效
超出历史支持域或出现火花、联锁风险	停止优化，回退现场安全设定	数学可行不等于设备安全

在线运行设 **TARGET-10**、**TARGET-5**、**FALLBACK** 三种状态。排放目标由上级指令或管理规定选定，不设自动切换阈值——本文排放模型未经现场标定，尚不足以在线自动判定当前排放水平。TARGET-10 与 TARGET-5 分别读取表 10 中对应工况、对应目标的控制参数。控制量按斜坡逐步逼近目标值，试运行阶段暂定电压变化不超过 0.5 kV/min、振打周期不超过 5 s/min；若现场 DCS、联锁或设备制造商规定了更严的限制，以更严者为准。四场振打先按 $\rho = 1$ （同相振打、峰值完全叠加）这一最不利情形核验，再以 $\rho = 0.80$ 和 0.65 评估错相运行可带来的裕度。传感器失效、火花率超限、工况超出取值范围或联锁触发时，即切换至 FALLBACK。

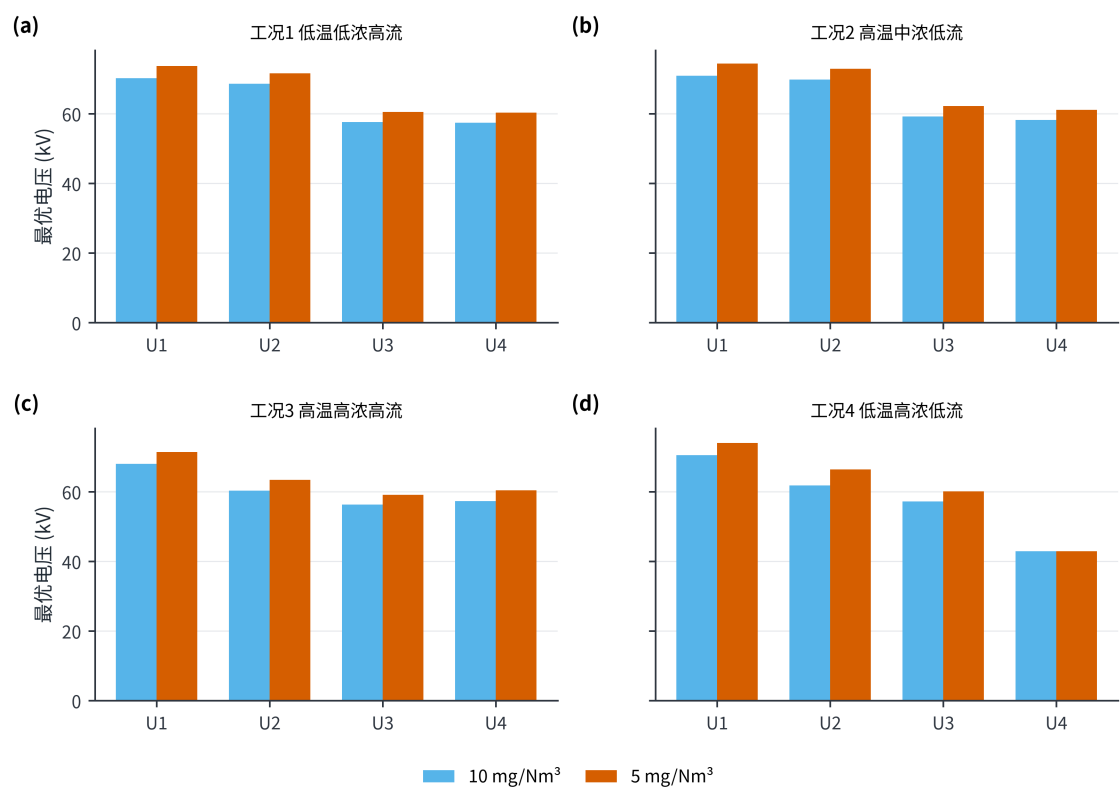


图 14 两档排放限值下的电压配置

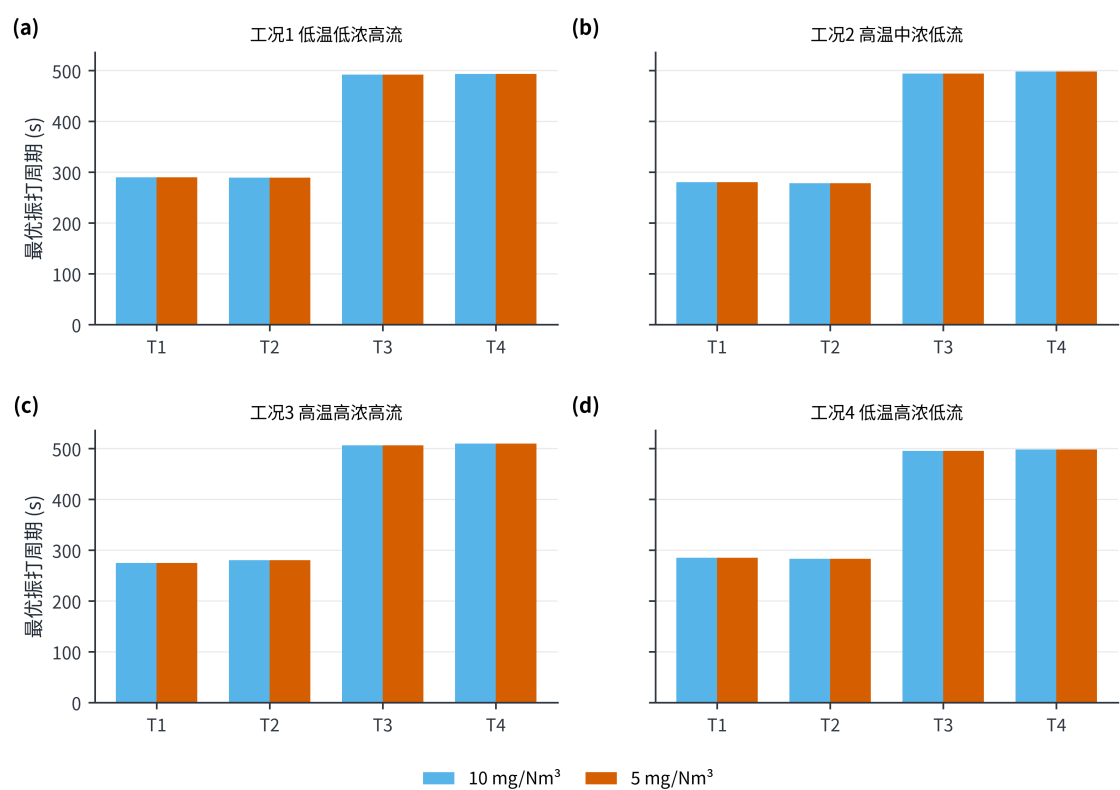


图 15 两档排放限值下的振打周期

7 问题四：10 收紧至 5 mg/Nm³ 的功率代价

7.1 闭式解

第四问可导出闭式解，其解释力强于纯数值优化结果。设在 10 mg/Nm³ 的最优点上，电场功率为 P_{field} 、去除指数为 K_{10} 。把四个电场的电压同时放大 f 倍，则去除指数变成 $K_{10}f^p$ ，电场功率变成 $P_{field}f^2$ 。要让排放减半（计入峰值项后，所需的去除指数增量是 $\Delta K = \ln \frac{10-B}{5-B}$ ），解得 $f = (1 + \Delta K/K_{10})^{1/p}$ ，于是

$$\Delta P = P_{field} \left[\left(1 + \frac{\Delta K}{K_{10}} \right)^{2/p} - 1 \right] \approx \frac{2}{p} \cdot \frac{P_{field}}{K_{10}} \Delta K \quad (10)$$

该式含义清晰：**限值每减半所增加的电耗，等于电场功率乘以所需的去除指数增量、除以当前去除指数，再乘以 $2/p$** 。由此可见，设备当前排放水平越低（ K 越大），进一步减半的相对代价越小。

7.2 分工况结果

表 13 中心情景下 10 收紧至 5 mg/Nm³ 的分工况功率代价

工况	10 mg 功率/kW	5 mg 功率/kW	K_{10}	ΔK	闭式 ΔP /kW	数值 ΔP /kW	功率增幅
1 低温低浓度高流量	2045.55	2182.25	8.059	0.804	137.7	136.7	6.68%
2 高温中浓度低流量	2104.30	2245.51	8.338	0.832	142.4	141.2	6.71%
3 高温高浓度高流量	1924.67	2056.06	8.594	0.903	131.3	131.4	6.83%
4 低温高浓度低流量	1851.15	1973.10	8.690	0.906	123.1	122.0	6.59%

闭式解与数值解的相对差异不超过 1%，互为验证。按训练期工况占比加权，

$$\bar{P}_{10} = 1973.0 \text{ kW}, \quad \bar{P}_5 = 2105.5 \text{ kW}, \quad \Delta P / \bar{P}_{10} = 6.72\% \quad (11)$$

按相同运行时长折算，两档日电耗分别约为 47.35 与 50.53 MWh/d，收紧限值的日增电耗约 3.18 MWh/d；按年运行 8760 h 计，年增电耗约 1.16×10^3 MWh。加上 9.78 kW 验证期误差附加量后，两档加权功率分别为 1982.8 与 2115.3 kW；由于两者使用同一附加量，该比例不作为严格上界，只用于功率预算。

值得注意的是，四类工况的增幅高度一致（6.59%~6.83%）。这不是巧合，而是上述公式的直接推论：增幅 $\approx (2/p)(P_{field}/P)(\Delta K/K)$ ，而四类工况的电场功率占比和当前去除指数都相差不大，所以增幅几乎不随工况变化。

7.3 真正的权衡：振打周期能买回多少

将第四问的结果与 §4.1 的振打功率结构并列，可得到本文最具操作价值的一组对照。按工况加权计算，限值由 10 收紧到 5 mg/Nm³ 要多耗约 132.5 kW；而把四个电场的振打周期从历史中位延长到历史 P95 可省下 56.1 kW，延长到历史最大值可省下 109.1 kW。**即振打侧可挖掘的全部潜力（约 109 kW），相当于排放收紧一档所增电耗（约 133 kW）的 82%，二者处于同一量级。**

但上述节电收益必须与振打峰值排放的代价一并权衡。按 §4.5 的推导，把周期由中位延长至历史最大值（平均延长 17.9%）时，单次振打的灰饼质量与瞬时峰值同比放大约 17.9%。而在表 10 的解中，峰值项本就已经占掉限值的 0.95~1.61 mg/Nm³，相当于 5 mg/Nm³ 这一档约五分之一到三分之

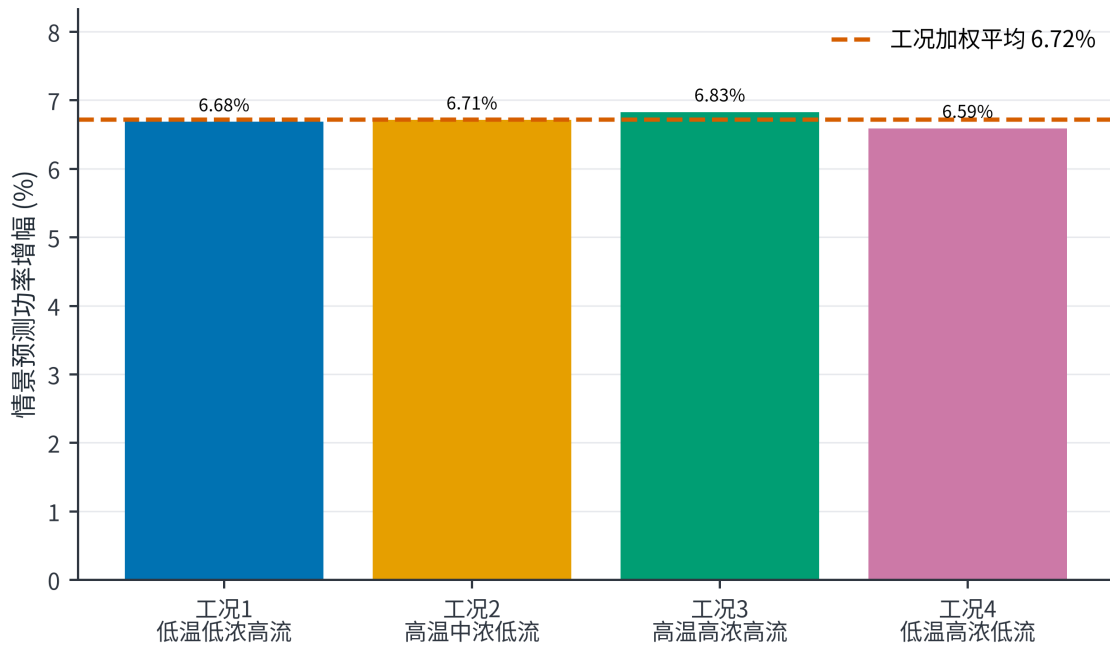


图 16 分工况功率增幅比较

一的余量。峰值再放大 17.9%，将进一步占用 0.17~0.29 mg/Nm³ 的排放裕度，该部分必须由电压侧补偿，而补偿电压本身又要付出功率代价。这正是振打周期不宜为节电而无限延长的定量表述：**振打周期是本设备上成本最低的节电手段，但其额度有限，且以瞬时排放峰值为代价。**

7.4 峰值约束口径的说明

在中心情景的解中，10 mg/Nm³ 方案的连续排放基础值为 8.38~9.00 mg/Nm³，振打带来的峰值增量为 0.95~1.61 mg/Nm³；5 mg/Nm³ 方案的基础值则降到 3.38~4.05 mg/Nm³。可见仅令平均值或基础值低于限值并不充分，否则振打时刻仍可能超限。

需要说明的是，本文把**瞬时峰值**直接卡到限值，而现行排放标准对颗粒物通常按小时均值考核。按 §4.5 的推导，一次持续数秒到数十秒的再飞扬峰值，对小时均值的贡献正比于 L 而与周期无关。因此本文口径系统性偏保守，会高估达标所需功率。若改用小时均值达标、峰值仅作软性惩罚的口径，所得功率增幅将低于本文报告值。在缺少逐次振打数据的条件下，本文选取保守口径，但不应将其视为唯一的合规解释。

8 模型检验、敏感性与稳健性

8.1 求解质量

八个“工况—限值”组合中，多起点 SLSQP 有 47~53 个起点成功收敛，**所有收敛起点给出的最优功率一致到 10^{-7} kW 以内**，表明各起点均收敛至同一解。作为独立对照，在同一可行域内均匀随机抽取 4×10^5 个候选点的随机搜索，在全部八个组合上均未找到更优解，其最优值反而高出 0.97%~5.17%；其中工况 1 的 10 mg/Nm³ 组合甚至未采到任何可行样本。原因在于可行域被排放约束与排序约束夹成一条窄带，均匀随机采样的命中效率很低。

该对照说明：在问题结构明确之后，确定性求解优于大规模随机搜索。但仍须指出，对非凸问题而言，多起点收敛至同一点是**很强的证据，而非数学证明**。因此本文只称“最优解”，不宣称“全局最优”。

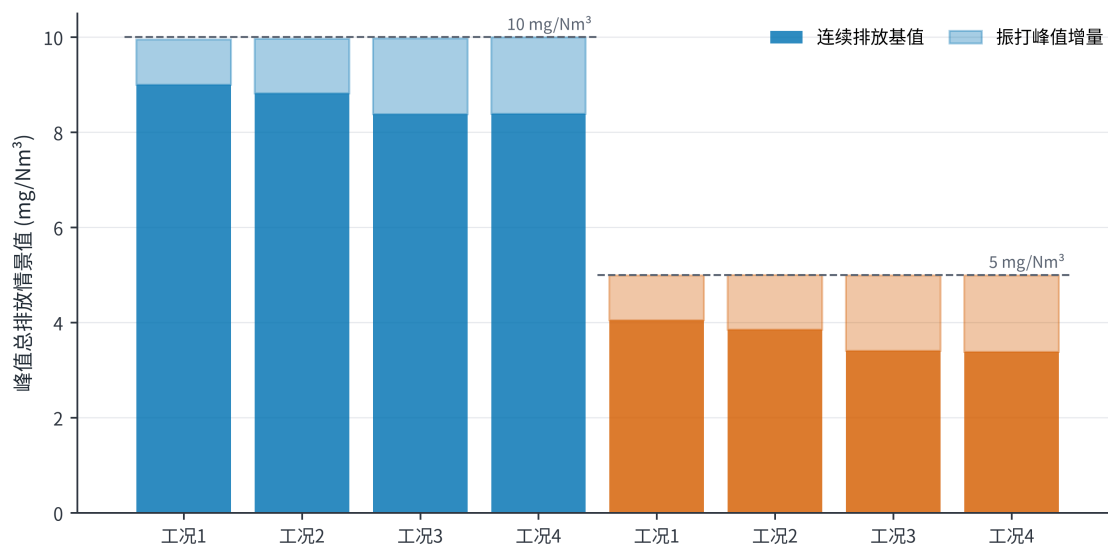


图 17 排放约束分解

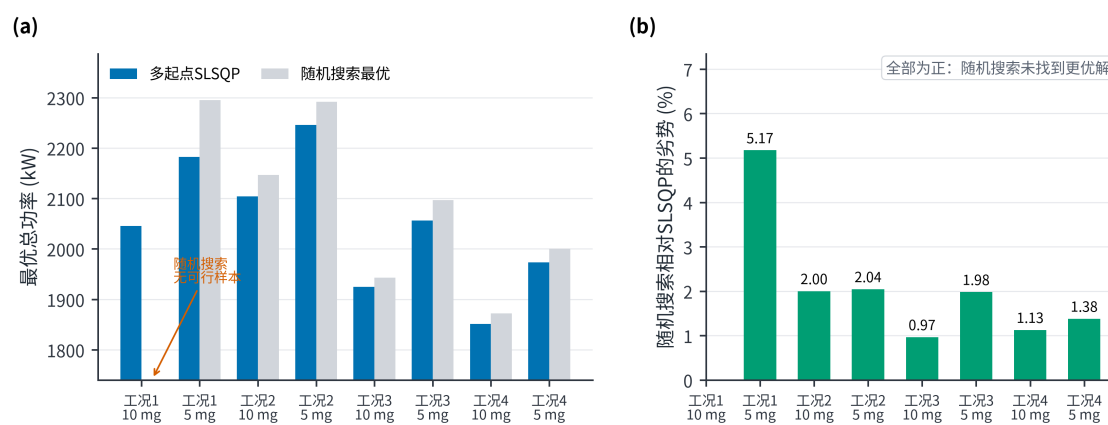


图 18 求解方法对照

8.2 先验敏感性

功率增幅结论对灰箱先验的依赖集中在单一标量 p 上。由 §7.1 的闭式解，增幅严格正比于 $2/p$ ；表 14 给出参考表。

表 14 功率增幅对幂次 p 的依赖

p	物理含义	10→5 mg/Nm ³ 加权增幅	5 mg/Nm ³ 在 5% 电压裕度内是否可达
1.0	去除指数正比于电压	14.2%	否，需更大电压裕度
1.5	中间情形	9.2%	否，需更大电压裕度
2.0	Deutsch 迁移速度形式（中心）	6.7%	是
2.5	强电压响应	5.3%	是
3.0	极强电压响应	4.4%	是

p 是本文结论中唯一需要现场标定的数，而且一次电压阶跃试验就能测出来：在工况稳定时把四个电场的电压同时抬高几个百分点，记录未截顶的出口浓度如何变化，再把 $\ln C_{out}$ 对 $\ln U$ 作回归，斜率就是 $-K_0 p$ 。对现场而言，给出一个可实测的参数，比给出一个宽区间更具指导意义。

表中增幅为闭式解值；对 $p \geq 2$ 的三档另作数值重解校验（图 20 叉号），与闭式值相差不超过 0.4 个百分点。

除 p 以外，本文还扫描了两组情景。

一是取值大小的组合： $p \in \{1.5, 2.0, 2.5\} \times$ 峰值尺度 $b_0 \in \{0.9, 1.2, 1.5\} \times$ 安全裕度 $s \in \{0, 0.08, 0.15\}$ ，共 27 组。其中 15 组在 5% 电压裕度内四类工况全部可行；不可行的 12 组全部来自 $p = 1.5$ ，这再次说明 p 首先决定目标的可达性，而不仅是代价大小。有解各组的加权增幅为 3.99%~7.04%，中位数 5.85%。

二是模型形状的组合：电场权重形状（3 种） $\times$ 峰值指数 $\gamma \in \{1.00, 1.30, 1.60\} \times$ 相位尺度 $\rho \in \{0.65, 0.80, 1.00\} \times$ 参考周期负荷指数（3 种），共 81 组，全部有解，加权增幅为 5.13%~7.55%，中位数 6.40%。峰值指数在推导值 1.00 附近变化时增幅几乎不变，这与 §4.5 的推导一致：峰值项占用的是限值余量，本身不是主导项。

需要强调，以上区间均为先验参数网格上的情景范围，而非统计意义上的置信区间。

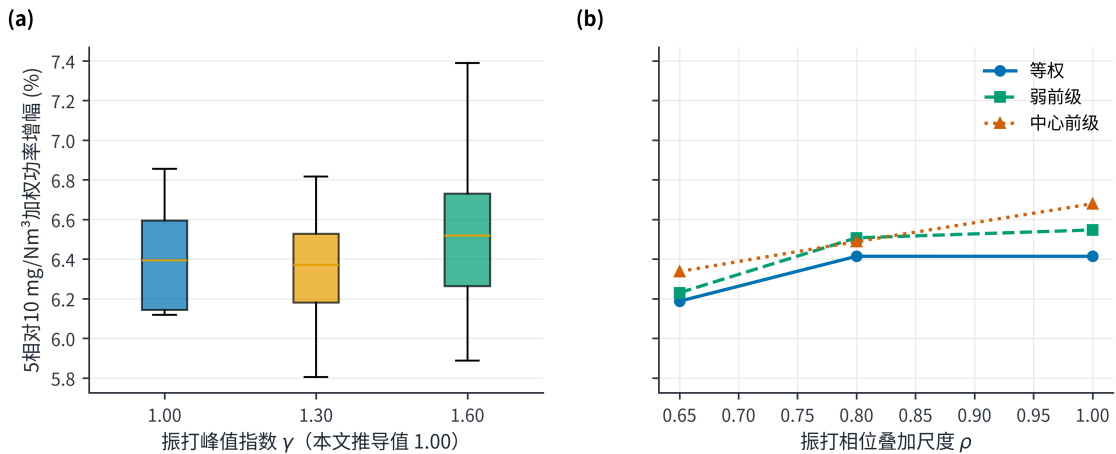


图 19 结构情景下的功率增幅分布

8.3 稳健性结论

稳健性证据分四层。第一，轮廓系数支持把工况分成四类。第二，四种设定的比较与时间外推支持所选的功率模型，其中 $1/T$ 形式在完全未参与拟合的测试日上表现最好，这是本文最强的实证

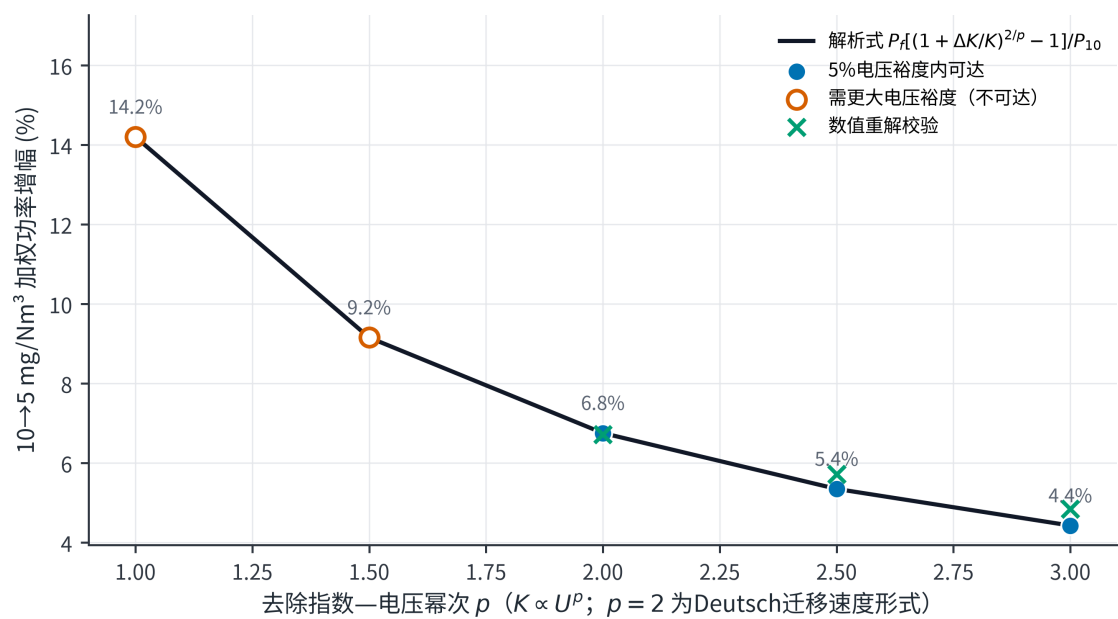


图 20 功率增幅对幂次 p 的依赖

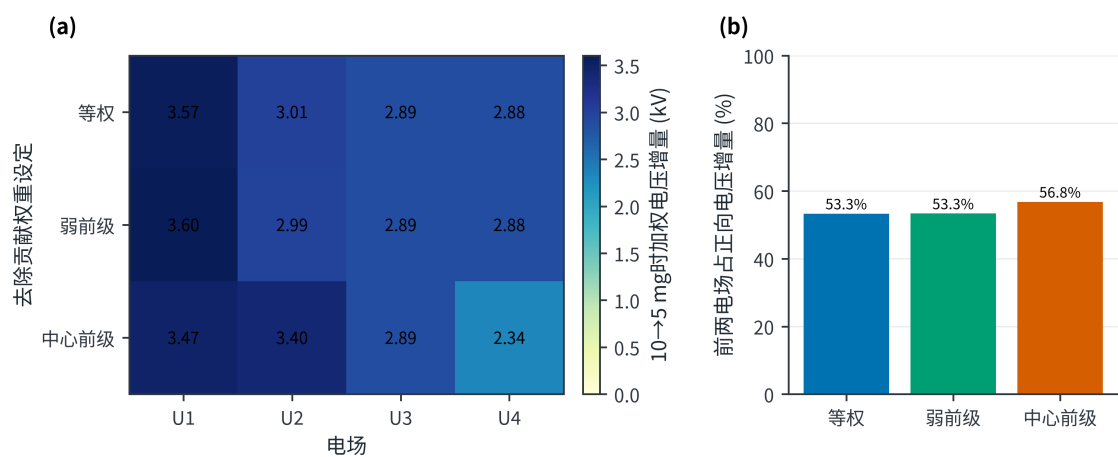


图 21 电场贡献权重消融结果

证据。第三，多起点收敛一致以及随机搜索的对照，支持数值解的质量。第四，闭式解与数值解互相验证，情景扫描与权重消融则检验了结论对建模选择的依赖程度。

但仍须强调：排放层的全部数值都是机理推演的结果。无论是误差余量还是情景范围，都不能把一个机理模型变成经过实测验证的模型。

9 模型评价、实施方案与局限

9.1 模型优点

第一，先逐列判定数据能够支持与不能支持的结论，既避免以复杂算法拟合截顶噪声，也未因一列不可用而放弃其余 13 列的信息。

第二，功率模型采用有物理含义的形式而非通用多项式，在未参与拟合的测试日上达到 $R^2 = 0.993$ 、偏差 -0.02 kW，并由此发现“振打占电耗近四成”这一可直接使用的结论。

第三，排放起算点直接取记录值，不引入未经证实的量纲假设，使第二问回答的是题目实际提出的问题。

第四，电场优先级由公式推出，而不是从某一次求解的现象里归纳。

第五，第四问给出闭式解，把全部不确定性收敛到一个可现场标定的数 p 。

第六，把推荐解偏离历史运行经验的程度作为结论的一部分报告，并给出严格落在历史经验之内的保守方案及其电耗代价，而非回避该问题。

9.2 模型局限

最主要的局限是，排放层的全部系数 ($p, \alpha_i, \beta_i, b_0, \gamma, \rho, s$) 都无法从附件中辨认，只能依据机理和文献设定；表 14 给出的区间是人为参数网格上的情景范围，不是统计置信区间。

其次，历史运行范围并不等于设备的安全边界， 5 mg/Nm^3 档所需的 5% 电压外推更须经高压电源容量与绝缘裕度核算。模型也没有二次电流、分场功率、火花率、极板积灰状态和联锁信息；分钟级数据不足以观察逐次振打峰值，因此相位叠加尺度 ρ 无法标定。四类工况的温度中心都没有覆盖 $140 \sim 160^\circ\text{C}$ 的高比电阻区间，本文结论在该区间无效。此外，测试日的数据曾被前期处理流程接触过，不能算作完全未触碰的盲测。

因此，本文给出的控制参数只能作为现场试验的起点，**不得直接下发生产设备**。

9.3 上线与补采方案

建议分四步部署。

第一阶段：影子运行。仅在后台计算、不改变设备设定，用以验证工况识别与 TARGET-10/TARGET-5 参数表的加载逻辑；同期完成高压电源容量核算，确认 5 mg/Nm^3 所需的电压裕度是否存在。

第二阶段：小步阶跃标定。在每类工况内执行单因素小步阶跃，重点是**标定幂次 p** (§8.2)，并由现场确定电压与周期的实际变化速率上限。

第三阶段：重新标定。以 $1 \sim 5 \text{ s}$ 的未截顶出口浓度、实际振打事件、二次电流、火花率与分场功率，重新标定排放层参数及相位尺度 ρ ，并单独补采 $140 \sim 160^\circ\text{C}$ 高比电阻区间的数据。

第四阶段：独立盲测与闭环投用。冻结模型，在全新日期上盲测，通过后方允许投入闭环。任何超出取值范围、传感器失效或联锁事件均应触发 FALLBACK。EPA 技术资料与相关实验都强调气流分布、收尘面积和再飞扬对性能评价的重要性 [2-3,6]，因此补采数据时应优先补齐这几项。

10 结论

1. **问题一：分层作答。**可由数据确定的部分：总功率几乎完全由控制量决定， $P = 73.1 + 0.0850 \sum U_i^2 + 5.39 \times 10^4 \sum T_i^{-1}$ （测试日 $R^2 = 0.993$ ），其中振打环节在中位工况下占总功率的 38.8%，而工况变量对总功率无额外解释力；历史操作规律中，入口浓度升高时缩短各场振打周期这一规律在四个时段方向一致，而入口浓度升高时抬升前级电压则仅在最后两天成立；附件还含两段持续异常工况，其中 05-06 的 158.2 °C 事件正落在粉尘比电阻的高值区。不可由数据确定的部分：出口浓度记录在 50 mg/Nm³ 附近被量程截顶，与全部 12 个变量的相关系数绝对值均小于 0.017，无法实证识别控制量对排放的作用。振打峰值则由积灰量守恒推得：单次峰值随周期近似线性放大，而其对小时均值的贡献与周期无关。
2. **问题二：4 类工况与 10 mg/Nm³ 方案。**基于入口三变量识别出 4 类典型工况（轮廓系数 0.4084）。以记录值 50 mg/Nm³ 为锚点，四类工况满足 10 mg/Nm³ 的最低功率为 1851.15~2104.30 kW，加权 1973.0 kW，日电耗 47.35 MWh，较历史实际的 1783.2 kW 上升 10.6%。收紧排放必然增加电耗，同一排放水平下不存在额外的节能空间。该方案电压仍位于历史范围之内，但已明显偏离历史常规操作（马氏距离达经验 97.5% 水平的 2.4~5.4 倍）；若要求严格落在历史经验之内，四类工况仍均可行，加权功率升至 2026.5 kW，多耗 2.7%。
3. **问题三：优先级可由解析判据确定。**单位电功率的除尘收益为 $K_0 \alpha_i / (\sum \alpha \cdot b_i U_i^{ref2})$ 。第三电场参考电压最低，该收益在四类工况中均居首；“前级优先”的结论实由排序约束 $U_1 \geq U_3 + 3$ 导出。振打侧按 c_i/T_i^2 排序，应优先延长周期较短的前级。低负荷工况须四场同步升压，高负荷工况则可择优升压，使单位功率收益最低的电场留在低位。
4. **问题四：功率代价与容量约束。**限值由 10 收紧至 5 mg/Nm³ 时，闭式解 $\Delta P = P_{field} \ln[(10 - B)/(5 - B)]/K_{10}$ 给出 123.1~142.4 kW，与数值解相差不足 1%；加权功率由 1973.0 升至 2105.5 kW，增幅 6.72%，日电耗由 47.35 增至 50.53 MWh。增幅与 $2/p$ 成正比，而 p 是影响功率增量与目标可达性的首要待标定参数。更关键的约束不是电耗而是电源容量：5 mg/Nm³ 要求各电场电压超出历史最大值最多 5.07%，四类工况所需的最小裕度分别为 5%、4%、1%、0%。若既有电源无法提供该裕度，仅靠优化运行参数无法达标，应考虑电源改造、增设电场或前端减尘等措施。以上排放数值均属机理推演，必须在补采未截顶的排放数据、振打事件与电气状态后重新标定。

参考文献

- [1] 中华人民共和国生态环境部. 水泥工业大气污染物排放标准：GB 4915—2013（含 2025 年修改单）[EB/OL]. (2013-12-27)[2026-08-06]. https://www.mee.gov.cn/ywgz/fgbz/bz/bzwb/dqhjbh/dqgdwrywrwpfbz/201312/t20131227_265765.htm.
- [2] OGLESBY S Jr, NICHOLS G B. *A Manual of Electrostatic Precipitator Technology: Part I—Fundamentals*[R/OL]. Birmingham: Southern Research Institute, 1970[2026-08-06]. <https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=9101FJFA.TXT>.
- [3] SZABO M F, SHAH Y M, SCHLIESSER S P. *Inspection Manual for Evaluation of Electrostatic Precipitator Performance*[R/OL]. Washington, DC: U.S. Environmental Protection Agency, 1981[2026-08-06]. <https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=9400034C.TXT>.
- [4] MIZUNO A. Electrostatic precipitation[J]. IEEE Transactions on Dielectrics and Electrical Insulation, 2000, 7(5): 615-624. DOI: 10.1109/94.879357.
- [5] LI Z, SHAO C, AN Y, et al. Energy-saving optimal control for a factual electrostatic precipitator with multiple electric-field stages based on GA[J]. Journal of Process Control, 2013, 23(8): 1041-1051. DOI:

10.1016/j.jprocont.2013.06.007.

[6] NAVARRETE B, VILCHES L F, RODRIGUEZ-GALAN M, et al. A pilot scale study of the rapping reentrainment and fouling in electrostatic precipitation[J]. Environmental Progress & Sustainable Energy, 2015, 34(1): 7-14. DOI: 10.1002/ep.11934.

[7] ZHENG C, ZHAO Z, GUO Y, et al. A real-time optimization method for economic and effective operation of electrostatic precipitators[J]. Journal of the Air & Waste Management Association, 2020, 70(7): 708-720. DOI: 10.1080/10962247.2020.1767227.

[8] FU Y, SU Z. Deep neural network based concentration prediction model of dry electrostatic precipitator with a modified PID optimizer[J]. Journal of Physics: Conference Series, 2022, 2189: 012018. DOI: 10.1088/1742-6596/2189/1/012018.

[9] MACQUEEN J. Some methods for classification and analysis of multivariate observations[C]//Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability. Berkeley: University of California Press, 1967, 1: 281-297.

[10] ROUSSEUW P J. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis[J]. Journal of Computational and Applied Mathematics, 1987, 20: 53-65. DOI: 10.1016/0377-0427(87)90125-7.

[11] HOERL A E, KENNARD R W. Ridge regression: Biased estimation for nonorthogonal problems[J]. Technometrics, 1970, 12(1): 55-67. DOI: 10.1080/00401706.1970.10488634.

[12] WHITE H J. *Industrial Electrostatic Precipitation*[M]. Reading, MA: Addison-Wesley, 1963.

[13] PARKER K R. *Electrical Operation of Electrostatic Precipitators*[M]. London: Institution of Electrical Engineers, 2003.

附录 A 与既有多电场节能优化研究的对照

本文与文献 [5]、[7] 同属“在排放约束下对多个电场分别优化”这一类工作，但两点不同：排放模型怎么标定，以及可行范围怎么划。对照见表 A1。

表 A1 与既有多电场节能优化研究的方法对照

维度	文献 [5] (GA 优化)	文献 [7] (实时优化)	本文
排放模型标定	由现场实测出口浓度回归	由在线仪表实时校正	出口浓度不可识别，改以记录值为锚点的 Deutsch 灰箱，全程标注为推演
功率模型	分场电源功率实测	分场功率与二次电流	仅有总功率，按 $U^2 + 1/T$ 物理设定辨识，测试 $R^2 = 0.993$
振打处理	未纳入优化	作为固定运行参数	纳入决策变量，并由积灰量守恒推导峰值标度
求解方法	遗传算法	滚动实时优化	多起点 SLSQP + 解析优先级判据 + 随机搜索对照
可行域	设备额定范围	在线安全约束	历史运行包络，越界量作为结论显式报告

上述工作的共同前提是排放模型可由现场数据标定。本文缺少这一条件，其处理方式是把不可标定的部分全部收敛到单一幂次 p ，并给出具体的标定方案 (§8.2) ——这是本文与上述工作的主要差别。

附录 B 以出口浓度为直接目标的对照建模

为便于与其他方案交叉比较，这里给出直接以 C_{out_mgNm3} 为目标建模的结果（训练/验证仍按时间划分）：

1. 以是否等于 50 为标签作分类，验证集 AUC=0.5098，与随机猜测无异；

2. 只用未截顶的样本做回归，验证集 $R^2 = -0.0190$ ；
3. 用全部记录值做回归，验证集 $R^2 = -0.0219$ 。

后两项的 R^2 为负，说明模型表现不及常数预测。因此，任何在该列上报告高 R^2 的模型，其性能几乎必然源于以下两种情形之一：随机划分数据导致的时间泄漏，或仅复现了 5491 条恒为 50 的截顶值。

附录 C 结果适用性声明

本文的功率模型、工况划分、历史操作规律、异常工况识别以及数据质量统计，均直接来自附件，可用 src/ 中的程序复算。

排放层的各项系数、振打峰值系数，以及由此得到的 5/10 mg/Nm³ 控制参数和功率增幅，则属于机理推演。若数据提供方给出新的未截顶排放数据，或现场标定得到的幂次与 $p = 2$ 不同，应按 §8.2 重新计算，不得沿用本文的数值。5 mg/Nm³ 档所需的电压外推，须先经高压电源容量与绝缘裕度核算。

附录 D 支撑材料与 AI 工具使用声明

求解流程：

输入：分钟级数据、排放限值 c_{lim}

1. 按时间划分训练/验证/回顾性测试；
2. 逐列可识别性审计；提取历史操作规律与异常工况；
3. 在训练期对 Temp、 C_{in} 、 Q 标准化，K-means 取 $k=4$ ；
4. 比较四种功率设定，选定 $P = a_0 + \sum b_i U_i^2 + \sum c_i / T_i$ ；
5. 以记录值 50 mg/Nm³ 为锚点建立灰箱排放层；由积灰量守恒定 $\gamma=1$ ；
6. 对每个工况与限值：
 - 6.1 取该工况控制量的历史最小—最大范围和支持域；
 - 6.2 多起点 SLSQP 求解（5 结构起点 + 48 随机起点）；
 - 6.3 舍入至 0.1 kV / 1 s 后按解析优先级修复可行性；
 - 6.4 以 4×10^5 点随机搜索独立对照；
7. 报告马氏距离外推审计与所需电压裕度；
8. 扫描 27 组尺度情景与 81 组结构情景，给出幂次 p 参考表；

输出：控制参数、功率代价、外推审计、敏感性、图表与运行摘要。

源程序清单： src/scenario_model.py（功率模型、灰箱排放情景、支持域优化与敏感性）、src/paper_figures.py（21 幅正文图与图注追踪表）、src/plot_utils.py（统一绘图样式）、src/integrity_check.py（确定性核验）、src/compute_frontier.py（排放—功率前沿）、src/build_paper_pdf.py（提交稿装配）。生成本文全部结果与图表的完整可运行源程序见附录 E，与电子支撑材料 src/ 目录逐字一致。

支撑材料清单： data/处理数据包；results/下功率基线、滚动验证、最优控制、外推审计、电压裕度需求、求解对照、结构情景、场权重消融、幂次参考表、动态控制表和运行摘要；figures_paper/下 21 幅正文图（20 幅 PNG/PDF 由绘图程序生成，图 1 为 TikZ 源文件独立编译）及 figure_manifest.csv；integrity/下确定性核验结果。

AI 工具使用声明（提交前由参赛队核验确认）： 本稿坚持参赛队人工主导。题意分析、模型构建、公式推导、参数设定、程序审阅、结果复算和论文定稿由参赛队承担；AI 编码助手仅用于代码排错、图表版式检查、文字校对与格式核验。AI 输出均经人工筛选和复核，未替代现场数据，也未把不可识别的排放关系改写为实证结论。

附录 E 完整源程序

本附录给出生成本文全部结果与图表的完整可运行源程序，与电子支撑材料 src/ 目录逐字一致。按依赖顺序排列：情景模型与优化、排放—功率前沿、绘图样式与图集生成、确定性核验、提交稿装配。

E.1 scenario_model.py

```
1  #!/usr/bin/env python3
2  """Reproducible scenario model for the cement ESP problem.
3
4  Two layers are kept strictly separate:
5
6  * the **power layer** is identified from the supplied data. Total power is a
7    near-deterministic function of the control settings alone, with the physically
8    motivated form  $P = a_0 + \sum_i b_i U_i^2 + \sum_i c_i / T_i$  (field power
9    proportional to voltage squared, rapping power proportional to rapping
10   frequency  $60/T_i$ );
11  * the **emission layer** is an explicitly labelled engineering scenario,
12    because the recorded outlet concentration is capped near 50 mg/Nm3 and
13    carries no learnable signal.
14
15  The historical emission anchor is the *recorded* outlet concentration; no
16  decimal-point rescaling is applied. The control problem solved here is
17  therefore the path from the historical ~50 mg/Nm3 level down to the 10 and
18  5 mg/Nm3 targets set by the problem statement.
19  """
20
21  from __future__ import annotations
22
23  import json
24  import math
25  from pathlib import Path
26
27  import numpy as np
28  import pandas as pd
29  from scipy.optimize import minimize
30
31
32  SEED = 20260805
33  ROOT = Path(__file__).resolve().parents[1]
34  DATA_FILE = ROOT / "data" / "full_timeseries_with_flags.csv"
35  CLUSTER_EVAL_FILE = ROOT / "data" / "condition_cluster_evaluation.csv"
36  OUT_DIR = ROOT / "results"
37
38  U_COLS = [f"U{i}_kV" for i in range(1, 5)]
39  T_COLS = [f"T{i}_s" for i in range(1, 5)]
40  COND_COLS = ["Temp_C", "C_in_gNm3", "Q_Nm3h"]
41
42  # ---- central scenario constants (engineering priors, not fitted to C_out) ----
43  ALPHA_CENTRAL = np.array([1.15, 1.10, 0.95, 0.80]) # per-field removal weights
44  BETA_CENTRAL = np.array([0.30, 0.27, 0.20, 0.17]) # rapping-period deviation loss
45  PEAK_SHARES = np.array([0.30, 0.30, 0.22, 0.18]) # per-field share of the peak proxy
46  P_EXPONENT_CENTRAL = 2.0 #  $K \sim U^p$ ;  $p = 2$  is the Deutsch drift-velocity form
47  PEAK_EXPONENT_CENTRAL = 1.0 # derived: cake mass per rap  $\sim L \cdot T \Rightarrow \text{peak} \sim T^1$ 
48  PEAK_SCALE_CENTRAL = 1.2
49  PHASE_SCALE_CENTRAL = 1.0
50  SAFETY_INDEX_CENTRAL = 0.08
51  # 5 mg/Nm3 needs voltage above every field's historical maximum; the margin is
52  # an explicitly declared extrapolation of the historical envelope, not a fact.
```

```

53 HEADROOM_10 = 0.00
54 HEADROOM_5 = 0.05
55
56 MULTISTART = 48
57 SWEEP_MULTISTART = 16 # sensitivity grids: fewer starts, cross-checked against the central solve
58
59
60 def _r2(y: np.ndarray, pred: np.ndarray) -> float:
61     return float(1.0 - np.sum((y - pred) ** 2) / np.sum((y - y.mean()) ** 2))
62
63
64 def _rmse(y: np.ndarray, pred: np.ndarray) -> float:
65     return float(np.sqrt(np.mean((y - pred) ** 2)))
66
67
68 # ----- power models
69
70 def voltage_polynomial(u: np.ndarray) -> np.ndarray:
71     """Equivalent to PolynomialFeatures(degree=2, include_bias=False) for 4 voltages."""
72     cols = [u[:, i] for i in range(4)]
73     for i in range(4):
74         for j in range(i, 4):
75             cols.append(u[:, i] * u[:, j])
76     return np.column_stack(cols)
77
78
79 class _ScaledOLS:
80     """Least squares with per-column scaling.
81
82     The scaling matters: ``1/T`` is of order 4e-3 while ``U^2`` is of order
83     3e3, and an unscaled ``lstsq`` truncates the rapping columns as though they
84     were rank-deficient, which makes the rapping term look irrelevant.
85     """
86
87     def __init__(self) -> None:
88         self.beta: np.ndarray | None = None
89
90     def _fit_design(self, x: np.ndarray, y: np.ndarray) -> None:
91         xx = np.column_stack([np.ones(len(x)), x])
92         scale = np.abs(xx).max(axis=0)
93         scale[scale == 0] = 1.0
94         beta, *_ = np.linalg.lstsq(xx / scale, y, rcond=None)
95         self.beta = beta / scale
96
97     def _predict_design(self, x: np.ndarray) -> np.ndarray:
98         if self.beta is None:
99             raise RuntimeError("model not fitted")
100         return self.beta[0] + x @ self.beta[1:]
101
102
103 class PhysicalPowerModel(_ScaledOLS):
104     """ $P = a_0 + \sum_i b_i U_i^2 + \sum_i c_i / T_i$  (optionally + condition terms)."""
105
106     name = "physical_voltage_square_plus_rapping_frequency"
107
108     def __init__(self, with_conditions: bool = False) -> None:
109         super().__init__()
110         self.with_conditions = with_conditions
111
112     def design(self, frame: pd.DataFrame) -> np.ndarray:
113         u = frame[U_COLS].to_numpy(float)
114         t = frame[T_COLS].to_numpy(float)

```

```

115     blocks = [u ** 2, 1.0 / t]
116     if self.with_conditions:
117         blocks.append(frame[COND_COLS].to_numpy(float))
118     return np.column_stack(blocks)
119
120 def fit(self, frame: pd.DataFrame, y: np.ndarray) -> "PhysicalPowerModel":
121     self._fit_design(self.design(frame), y)
122     return self
123
124 def predict(self, frame: pd.DataFrame) -> np.ndarray:
125     return self._predict_design(self.design(frame))
126
127 # -- fast path used inside the optimiser -----
128 @property
129 def intercept(self) -> float:
130     return float(self.beta[0])
131
132 @property
133 def b(self) -> np.ndarray:
134     return np.asarray(self.beta[1:5], float)
135
136 @property
137 def c(self) -> np.ndarray:
138     return np.asarray(self.beta[5:9], float)
139
140 def power(self, u: np.ndarray, t: np.ndarray) -> np.ndarray:
141     """Total power for raw control arrays (last axis = 4 fields)."""
142     return self.intercept + np.sum(self.b * u ** 2, axis=-1) + np.sum(self.c / t, axis=-1)
143
144 def field_power(self, u: np.ndarray) -> np.ndarray:
145     return np.sum(self.b * u ** 2, axis=-1)
146
147 def rapping_power(self, t: np.ndarray) -> np.ndarray:
148     return np.sum(self.c / t, axis=-1)
149
150
151 class RidgePowerModel:
152     """Standardised ridge baseline with the rapping period entered *linearly*.
153
154     Retained as the misspecified reference point: it is the specification the
155     first version of this paper used, and the comparison in
156     ``power_model_baselines.csv`` shows what treating ``T`` as a linear term
157     costs on the retrospective test window.
158     """
159
160     def __init__(self, alpha: float = 1e-3, quadratic: bool = True):
161         self.alpha = alpha
162         self.quadratic = quadratic
163         self.mean: np.ndarray | None = None
164         self.std: np.ndarray | None = None
165         self.beta: np.ndarray | None = None
166
167     def design(self, frame: pd.DataFrame) -> np.ndarray:
168         u = frame[U_COLS].to_numpy(float)
169         remainder = frame[T_COLS + COND_COLS].to_numpy(float)
170         voltage = voltage_polynomial(u) if self.quadratic else u
171         return np.column_stack([voltage, remainder])
172
173     def fit(self, frame: pd.DataFrame, y: np.ndarray) -> None:
174         x = self.design(frame)
175         self.mean = x.mean(axis=0)
176         self.std = x.std(axis=0)

```

```

177     self.std[self.std == 0] = 1.0
178     z = (x - self.mean) / self.std
179     zz = np.column_stack([np.ones(len(z)), z])
180     penalty = np.eye(zz.shape[1]) * math.sqrt(self.alpha)
181     penalty[0, 0] = 0.0
182     augmented_x = np.vstack([zz, penalty])
183     augmented_y = np.concatenate([y, np.zeros(zz.shape[1])])
184     self.beta = np.linalg.lstsq(augmented_x, augmented_y, rcond=None)[0]
185
186     def predict(self, frame: pd.DataFrame) -> np.ndarray:
187         if self.mean is None or self.std is None or self.beta is None:
188             raise RuntimeError("model not fitted")
189         x = self.design(frame)
190         z = (x - self.mean) / self.std
191         zz = np.column_stack([np.ones(len(z)), z])
192         return np.einsum("ij,j->i", zz, self.beta)
193
194
195     def evaluate_power_models(
196         train: pd.DataFrame,
197         validation: pd.DataFrame,
198         test: pd.DataFrame,
199     ) -> tuple[PhysicalPowerModel, dict[str, object], pd.DataFrame, pd.DataFrame]:
200         """Fit every candidate specification on the training window only."""
201         alpha_grid = (1e-4, 1e-3, 1e-2, 1e-1, 1.0, 10.0)
202         n = len(train)
203         rolling_bounds = ((0, int(0.40 * n), int(0.60 * n)),
204                          (0, int(0.60 * n), int(0.80 * n)),
205                          (0, int(0.80 * n), n))
206         rolling_rows: list[dict[str, object]] = []
207         selected: dict[str, float] = {}
208         for model_name, quadratic in (("Linear_ridge_T", False), ("quadratic_ridge_T", True)):
209             alpha_scores: list[tuple[float, float]] = []
210             for alpha in alpha_grid:
211                 fold_rmse = []
212                 for fold, (start, fit_end, eval_end) in enumerate(rolling_bounds, start=1):
213                     fit = train.iloc[start:fit_end]
214                     evaluate = train.iloc[fit_end:eval_end]
215                     candidate = RidgePowerModel(alpha=alpha, quadratic=quadratic)
216                     candidate.fit(fit, fit["P_total_kW"].to_numpy(float))
217                     pred = candidate.predict(evaluate)
218                     actual = evaluate["P_total_kW"].to_numpy(float)
219                     rmse = _rmse(actual, pred)
220                     fold_rmse.append(rmse)
221                     rolling_rows.append({
222                         "model": model_name, "alpha": alpha, "fold": fold,
223                         "fit_rows": len(fit), "evaluation_rows": len(evaluate),
224                         "r2": _r2(actual, pred), "rmse_kW": rmse,
225                         "bias_kW": float(np.mean(pred - actual)),
226                     })
227                 alpha_scores.append((float(np.mean(fold_rmse)), alpha))
228             selected[model_name] = min(alpha_scores)[1]
229
230         # the physical specification has no tuning parameter; still report it per fold
231         for fold, (start, fit_end, eval_end) in enumerate(rolling_bounds, start=1):
232             fit = train.iloc[start:fit_end]
233             evaluate = train.iloc[fit_end:eval_end]
234             candidate = PhysicalPowerModel().fit(fit, fit["P_total_kW"].to_numpy(float))
235             pred = candidate.predict(evaluate)
236             actual = evaluate["P_total_kW"].to_numpy(float)
237             rolling_rows.append({
238                 "model": "physical_invT", "alpha": np.nan, "fold": fold,

```

```

239         "fit_rows": len(fit), "evaluation_rows": len(evaluate),
240         "r2": _r2(actual, pred), "rmse_kW": _rmse(actual, pred),
241         "bias_kW": float(np.mean(pred - actual)),
242     })
243
244     y_train = train["P_total_kW"].to_numpy(float)
245     fitted: dict[str, object] = {}
246     for model_name, quadratic in (("Linear_ridge_T", False), ("quadratic_ridge_T", True)):
247         m = RidgePowerModel(alpha=selected[model_name], quadratic=quadratic)
248         m.fit(train, y_train)
249         fitted[model_name] = m
250     fitted["physical_invT"] = PhysicalPowerModel().fit(train, y_train)
251     fitted["physical_invT_plus_conditions"] = PhysicalPowerModel(with_conditions=True).fit(train, y_train)
252
253     mean_power = float(y_train.mean())
254     baseline_rows: list[dict[str, object]] = []
255     for split_name, frame in (("train", train), ("validation", validation), ("retrospective_test", test)):
256         actual = frame["P_total_kW"].to_numpy(float)
257         for model_name in ("mean_predictor", "Linear_ridge_T", "quadratic_ridge_T",
258                           "physical_invT", "physical_invT_plus_conditions"):
259             pred = (np.full(len(frame), mean_power) if model_name == "mean_predictor"
260                     else fitted[model_name].predict(frame))
261             baseline_rows.append({
262                 "model": model_name,
263                 "selected_alpha": selected.get(model_name, np.nan),
264                 "split": split_name,
265                 "r2": _r2(actual, pred),
266                 "rmse_kW": _rmse(actual, pred),
267                 "bias_kW": float(np.mean(pred - actual)),
268             })
269
270     chosen: PhysicalPowerModel = fitted["physical_invT"] # type: ignore[assignment]
271     val_actual = validation["P_total_kW"].to_numpy(float)
272     val_pred = chosen.predict(validation)
273     test_actual = test["P_total_kW"].to_numpy(float)
274     test_pred = chosen.predict(test)
275     guard = float(np.quantile(np.abs(val_pred - val_actual), 0.90))
276     metrics: dict[str, object] = {
277         "model": chosen.name,
278         "formula": "P = a0 + sum_i b_i U_i^2 + sum_i c_i / T_i",
279         "fit_scope": "train_only",
280         "intercept_kW": chosen.intercept,
281         "b_field_kW_per_kV2": chosen.b.tolist(),
282         "c_rapping_kW_s": chosen.c.tolist(),
283         "train_r2": _r2(y_train, chosen.predict(train)),
284         "validation_r2": _r2(val_actual, val_pred),
285         "validation_rmse_kW": _rmse(val_actual, val_pred),
286         "validation_bias_kW": float(np.mean(val_pred - val_actual)),
287         "validation_abs_error_q90_kW": guard,
288         "retrospective_test_r2": _r2(test_actual, test_pred),
289         "retrospective_test_rmse_kW": _rmse(test_actual, test_pred),
290         "retrospective_test_bias_kW": float(np.mean(test_pred - test_actual)),
291         "test_protocol": "retrospective_not_blind_due_to_prior_package_use",
292         "reporting_guard_rule": "point_prediction_plus_validation_absolute_error_q90",
293         "ridge_selected_alphas": selected,
294     }
295     return chosen, metrics, pd.DataFrame(rolling_rows), pd.DataFrame(baseline_rows)
296
297
298 # ----- condition layer
299
300 def condition_profiles(df: pd.DataFrame) -> pd.DataFrame:

```

```

301 train = df[df["split"] == "train"].copy()
302 overall_load = train["dust_load_kg_h"].median()
303 labels = {
304     1: "低温低浓度高流量",
305     2: "高温中浓度低流量",
306     3: "高温高浓度高流量",
307     4: "低温高浓度低流量",
308 }
309 rows: list[dict[str, float | int | str]] = []
310 for cluster, group in train.groupby("condition_cluster"):
311     row: dict[str, float | int | str] = {
312         "condition_cluster": int(cluster),
313         "condition_name": labels[int(cluster)],
314         "count": int(len(group)),
315         "share": float(len(group) / len(train)),
316         "Temp_C": float(group["Temp_C"].mean()),
317         "C_in_gNm3": float(group["C_in_gNm3"].mean()),
318         "Q_Nm3h": float(group["Q_Nm3h"].mean()),
319         "dust_load_kg_h": float(group["dust_load_kg_h"].mean()),
320         "C_out_recorded_median": float(group["C_out_mgNm3"].dropna().median()),
321         "historical_power_kW": float(group["P_total_kW"].mean()),
322     }
323     for col in U_COLS + T_COLS:
324         row[f"{col}_min"] = float(group[col].min())
325         row[f"{col}_p05"] = float(group[col].quantile(0.05))
326         row[f"{col}_median"] = float(group[col].median())
327         row[f"{col}_p95"] = float(group[col].quantile(0.95))
328         row[f"{col}_max"] = float(group[col].max())
329         row["load_ratio_to_train_median"] = float(row["dust_load_kg_h"] / overall_load)
330         row["K0_anchor"] = math.log(1000.0 * row["C_in_gNm3"] / row["C_out_recorded_median"])
331     rows.append(row)
332 return pd.DataFrame(rows).sort_values("condition_cluster").reset_index(drop=True)
333
334
335 def scenario_t_star(profile: pd.Series, global_t_median: np.ndarray,
336                    load_exponent: float = -0.18) -> np.ndarray:
337     load_ratio = float(profile["load_ratio_to_train_median"])
338     raw = global_t_median * load_ratio ** load_exponent
339     lower = np.array([profile[f"{c}_min"] for c in T_COLS], float)
340     upper = np.array([profile[f"{c}_max"] for c in T_COLS], float)
341     return np.clip(raw, lower, upper)
342
343
344 def support_statistics(train_group: pd.DataFrame) -> tuple[np.ndarray, np.ndarray, float]:
345     """Empirical control-support centre, inverse covariance and d^2 cut-off."""
346     hist = train_group[U_COLS + T_COLS].to_numpy(float)
347     mu = hist.mean(axis=0)
348     inv_cov = np.linalg.pinv(np.cov(hist, rowvar=False))
349     delta = hist - mu
350     historical_d2 = np.einsum("ij,jk,ik->i", delta, inv_cov, delta)
351     return mu, inv_cov, float(np.quantile(historical_d2, 0.975))
352
353
354 # ----- emission layer
355
356 def scenario_emission(
357     profile: pd.Series,
358     u: np.ndarray,
359     t: np.ndarray,
360     t_star: np.ndarray,
361     p_exponent: float = P_EXPONENT_CENTRAL,
362     peak_scale: float = PEAK_SCALE_CENTRAL,

```



```

363 safety_index: float = SAFETY_INDEX_CENTRAL,
364 alpha_weights: np.ndarray | None = None,
365 peak_exponent: float = PEAK_EXPONENT_CENTRAL,
366 phase_scale: float = PHASE_SCALE_CENTRAL,
367 ) -> tuple[np.ndarray, np.ndarray, np.ndarray]:
368     """Return continuous base emission, rapping peak excess, and their sum.
369
370     ``K = K0 * S(U) - [G(T) - G(T_ref)]`` with
371     ``S(U) = sum_i alpha_i (U_i/U_ref_i)^p / sum_i alpha_i``.
372
373     At the historical anchor ``U = U_ref`` and ``T = T_ref`` this returns
374     ``K = K0 = ln(1000 C_in / C_out_recorded)`` , i.e. the model reproduces the
375     recorded operating point by construction. Every non-control argument is a
376     scenario input; none is estimated from the recorded outlet series.
377     """
378     u_ref = np.array([profile[f"{c}_median"] for c in U_COLS], float)
379     t_ref = np.array([profile[f"{c}_median"] for c in T_COLS], float)
380     cin = float(profile["C_in_gNm3"])
381     k_anchor = float(profile["K0_anchor"])
382
383     if alpha_weights is None:
384         alpha_weights = ALPHA_CENTRAL
385     alpha = np.asarray(alpha_weights, float)
386     strength = np.sum(alpha * (u / u_ref) ** p_exponent, axis=-1) / alpha.sum()
387
388     load_ratio = float(profile["load_ratio_to_train_median"])
389     beta = BETA_CENTRAL * load_ratio ** 0.5
390
391     def loss(period: np.ndarray) -> np.ndarray:
392         ratio = period / t_star
393         return np.sum(beta * (ratio + 1.0 / ratio - 2.0), axis=-1)
394
395     ref_loss = float(loss(t_ref))
396     k = k_anchor * strength - (loss(t) - ref_loss)
397     base = cin * 1000.0 * np.exp(-k + safety_index)
398
399     # Derived form: cake mass per rap ~ L * T, so the per-event peak grows
400     # linearly with the period (peak_exponent = 1) while the contribution to
401     # the hourly mean, ~ L*T*(1/T) = L, does not depend on the period at all.
402     peak = (peak_scale * phase_scale * load_ratio
403             * np.sum(PEAK_SHARES * (t / t_star) ** peak_exponent, axis=-1))
404     return base, peak, base + peak
405
406
407 def field_priority(profile: pd.Series, model: PhysicalPowerModel,
408                  alpha_weights: np.ndarray | None = None) -> np.ndarray:
409     """Removal gain per unit of electrical power for each field.
410
411     Differentiating ``K = K0 * sum_i alpha_i (U_i/U_ref_i)^2 / sum(alpha)`` and
412     ``P = sum_i b_i U_i^2`` with respect to ``U_i^2`` gives
413
414         
$$dK/dP|_i = K0 * \alpha_i / (\sum(\alpha) * b_i * U_{ref\_i}^2),$$

415
416     so the field to raise first is the one with the largest
417     ``alpha_i / U_ref_i^2`` -- *not* the largest ``alpha_i``.
418     """
419     alpha = ALPHA_CENTRAL if alpha_weights is None else np.asarray(alpha_weights, float)
420     u_ref = np.array([profile[f"{c}_median"] for c in U_COLS], float)
421     return float(profile["K0_anchor"]) * alpha / (alpha.sum() * model.b * u_ref ** 2)
422
423
424 # ----- optimisation

```

```

425
426 def _bounds(profile: pd.Series, voltage_headroom: float) -> tuple[np.ndarray, np.ndarray]:
427     lo = np.concatenate([[profile[f"{c}_min"] for c in U_COLS],
428                          [profile[f"{c}_min"] for c in T_COLS]])
429     hi = np.concatenate([[profile[f"{c}_max"] * (1.0 + voltage_headroom) for c in U_COLS],
430                          [profile[f"{c}_max"] for c in T_COLS]])
431     return lo.astype(float), hi.astype(float)
432
433
434 def _ordering(z: np.ndarray) -> np.ndarray:
435     return np.array([z[0] - z[2] - 3.0, z[1] - z[3] - 3.0,
436                     z[6] - z[4] - 100.0, z[7] - z[5] - 100.0])
437
438
439 def _round_controls(z: np.ndarray) -> np.ndarray:
440     return np.concatenate([np.round(z[:4], 1), np.round(z[4:])])
441
442
443 def solve_condition(
444     profile: pd.Series,
445     model: PhysicalPowerModel,
446     t_star: np.ndarray,
447     limit: float,
448     voltage_headroom: float = 0.0,
449     mahalanobis: tuple[np.ndarray, np.ndarray, float] | None = None,
450     multistart: int = MULTISTART,
451     seed: int = SEED,
452     **emission_kwargs,
453 ) -> dict[str, object]:
454     """Deterministic multi-start SLSQP solve of one condition/limit pair.
455
456     The feasible set is *not* convex in ``U``: the objective ``sum b_i U_i^2``
457     is convex while the emission constraint requires a convex function of
458     ``U`` to be large enough, which is a reverse-convex constraint. A dense
459     multi-start is therefore used, and ``verify_with_random_search`` re-checks
460     the result against a large random sample of the same feasible set.
461     """
462     lo, hi = _bounds(profile, voltage_headroom)
463
464     def objective(z: np.ndarray) -> float:
465         return float(model.power(z[:4], z[4:]))
466
467     def emission_slack(z: np.ndarray) -> float:
468         return float(limit - scenario_emission(profile, z[:4], z[4:], t_star, **emission_kwargs)[2])
469
470     constraints: list[dict[str, object]] = [
471         {"type": "ineq", "fun": emission_slack},
472         {"type": "ineq", "fun": _ordering},
473     ]
474     if mahalanobis is not None:
475         mu, inv_cov, threshold = mahalanobis
476         constraints.append({
477             "type": "ineq",
478             "fun": lambda z: float(threshold - (z - mu) @ inv_cov @ (z - mu)),
479         })
480
481     u_ref = np.array([profile[f"{c}_median"] for c in U_COLS], float)
482     t_ref = np.array([profile[f"{c}_median"] for c in T_COLS], float)
483     rng = np.random.default_rng(seed)
484     starts = [np.concatenate([u_ref, t_ref]), hi.copy(), lo.copy(), 0.5 * (lo + hi),
485                    np.concatenate([hi[:4], t_star])]
486     starts += [lo + (hi - lo) * rng.random(8) for _ in range(multistart)]

```

```

487
488 best_z: np.ndarray | None = None
489 best_val = np.inf
490 converged = 0
491 values: list[float] = []
492 for z0 in starts:
493     result = minimize(objective, np.clip(z0, lo, hi), method="SLSQP",
494                       bounds=list(zip(lo, hi)), constraints=constraints,
495                       options={"maxiter": 600, "ftol": 1e-10})
496     if not result.success:
497         continue
498     z = np.clip(result.x, lo, hi)
499     if emission_slack(z) < -1e-7 or np.any(_ordering(z) < -1e-7):
500         continue
501     converged += 1
502     values.append(float(result.fun))
503     if result.fun < best_val:
504         best_val, best_z = float(result.fun), z
505
506 if best_z is None:
507     return {
508         "condition_cluster": int(profile["condition_cluster"]),
509         "condition_name": str(profile["condition_name"]),
510         "limit_mgNm3": float(limit),
511         "voltage_headroom": float(voltage_headroom),
512         "status": "INFEASIBLE_IN_DECLARED_SUPPORT",
513     }
514
515 # Round to the resolution a DCS can actually hold, then repair feasibility
516 # by nudging the most power-efficient field upward.
517 z = _round_controls(best_z)
518 order = np.argsort(-field_priority(profile, model, emission_kwargs.get("alpha_weights")))
519 for _ in range(400):
520     if emission_slack(z) >= 0 and np.all(_ordering(z) >= -1e-9):
521         break
522     for i in order:
523         if z[i] + 0.1 <= hi[i] + 1e-9:
524             z[i] = round(z[i] + 0.1, 1)
525             break
526     else:
527         break
528 if emission_slack(z) < 0:
529     z = best_z # keep the unrounded solution rather than report an infeasible one
530
531 base, peak, total = scenario_emission(profile, z[:4], z[4:], t_star, **emission_kwargs)
532 row: dict[str, object] = {
533     "condition_cluster": int(profile["condition_cluster"]),
534     "condition_name": str(profile["condition_name"]),
535     "limit_mgNm3": float(limit),
536     "voltage_headroom": float(voltage_headroom),
537     "status": "SCENARIO_FEASIBLE",
538     "scenario_base_emission_mgNm3": float(base),
539     "scenario_peak_excess_mgNm3": float(peak),
540     "scenario_peak_total_mgNm3": float(total),
541     "predicted_power_kw": float(model.power(z[:4], z[4:])),
542     "field_power_kw": float(model.field_power(z[:4])),
543     "rapping_power_kw": float(model.rapping_power(z[4:])),
544     "historical_power_kw": float(profile["historical_power_kw"]),
545     "multistart_converged": converged,
546     "multistart_power_spread_kw": float(np.max(values) - np.min(values)) if values else np.nan,
547 }
548 row["power_change_vs_historical_pct"] = 100.0 * (

```

```

549         row["predicted_power_kw"] / float(profile["historical_power_kw"]) - 1.0)
550     for i, col in enumerate(U_COLS + T_COLS):
551         row[col] = float(z[i])
552     return row
553
554
555 def verify_with_random_search(
556     profile: pd.Series,
557     model: PhysicalPowerModel,
558     t_star: np.ndarray,
559     limit: float,
560     voltage_headroom: float,
561     n: int = 400_000,
562     seed: int = SEED,
563     **emission_kwargs,
564 ) -> float:
565     """Best feasible power found by uniform random sampling of the same set."""
566     lo, hi = _bounds(profile, voltage_headroom)
567     rng = np.random.default_rng(seed)
568     z = lo + (hi - lo) * rng.random((n, 8))
569     u = np.round(z[:, :4], 1)
570     t = np.round(z[:, 4:], 1)
571     keep = ((u[:, 0] >= u[:, 2] + 3.0) & (u[:, 1] >= u[:, 3] + 3.0)
572            & (t[:, 2] >= t[:, 0] + 100.0) & (t[:, 3] >= t[:, 1] + 100.0))
573     u, t = u[keep], t[keep]
574     if not len(u):
575         return float("nan")
576     _, total = scenario_emission(profile, u, t, t_star, **emission_kwargs)
577     feasible = total <= limit
578     if not np.any(feasible):
579         return float("nan")
580     return float(np.min(model.power(u[feasible], t[feasible])))
581
582
583 def aggregate(rows: List[dict[str, object]], key: str = "predicted_power_kw") -> float:
584     weights = np.array([float(r["share"]) for r in rows])
585     return float(np.average([float(r[key]) for r in rows], weights=weights))
586
587
588 def solve_all(profiles: pd.DataFrame, model: PhysicalPowerModel,
589              t_stars: dict[int, np.ndarray], limit: float, headroom: float,
590              multistart: int = MULTISTART, **emission_kwargs) -> List[dict[str, object]]:
591     rows = []
592     for _, profile in profiles.iterrows():
593         cluster = int(profile["condition_cluster"])
594         r = solve_condition(profile, model, t_stars[cluster], limit,
595                             voltage_headroom=headroom, multistart=multistart,
596                             **emission_kwargs)
597         r["share"] = float(profile["share"])
598         rows.append(r)
599     return rows
600
601
602 # ----- data-side diagnostics
603
604 def historical_strategy_correlations(df: pd.DataFrame) -> pd.DataFrame:
605     """Operator response of the control settings to the exogenous conditions.
606
607     Unlike the outlet concentration, this relationship *is* identifiable, and
608     it gives Question 3 an empirical baseline. It is reported per window as
609     well as over the whole record, because the two are not the same: the
610     rapping-period rule holds throughout, while the voltage response to inlet

```

```

611 concentration reverses sign between the first five days and the last two.
612 """
613 windows = [("full_record", df), ("train", df[df["split"] == "train"]),
614            ("validation", df[df["split"] == "validation"]),
615            ("retrospective_test", df[df["split"] == "test"])]
616 rows = []
617 for name, frame in windows:
618     for control in U_COLS + T_COLS:
619         row: dict[str, object] = {"window": name, "rows": int(len(frame)), "control": control}
620         for cond in COND_COLS:
621             row[f"r_{cond}"] = float(np.corrcoef(frame[cond], frame[control])[0, 1])
622         rows.append(row)
623 return pd.DataFrame(rows)
624
625
626 def anomaly_events(df: pd.DataFrame) -> pd.DataFrame:
627     """The two sustained excursions planted in the attachment."""
628     specs = [
629         ("入口浓度尖峰", df["C_in_gNm3"] > 60.0),
630         ("入口温度尖峰", df["Temp_C"] > 150.0),
631     ]
632     baseline = {c: float(df[c].median()) for c in COND_COLS + U_COLS + T_COLS + ["P_total_kW"]}
633     rows = []
634     for name, mask in specs:
635         sub = df[mask]
636         row: dict[str, object] = {
637             "event": name,
638             "start": str(sub["timestamp"].min()),
639             "end": str(sub["timestamp"].max()),
640             "minutes": int(len(sub)),
641         }
642         for col in COND_COLS + U_COLS + T_COLS + ["P_total_kW"]:
643             row[f"{col}_event_mean"] = float(sub[col].mean())
644             row[f"{col}_event_max"] = float(sub[col].max())
645             row[f"{col}_event_min"] = float(sub[col].min())
646             row[f"{col}_baseline_median"] = baseline[col]
647         rows.append(row)
648     return pd.DataFrame(rows)
649
650
651 def rapping_power_tradeoff(train: pd.DataFrame, model: PhysicalPowerModel) -> pd.DataFrame:
652     """How much power the rapping term holds, and what extending it recovers."""
653     u_med = train[U_COLS].median().to_numpy(float)
654     t_med = train[T_COLS].median().to_numpy(float)
655     field = float(model.field_power(u_med))
656     base_rap = float(model.rapping_power(t_med))
657     total = model.intercept + field + base_rap
658     rows = [{
659         "scenario": "历史中位周期",
660         "T_setting": ";".join(f"{v:.0f}" for v in t_med),
661         "field_power_kW": field,
662         "rapping_power_kW": base_rap,
663         "total_power_kW": total,
664         "rapping_share_pct": 100.0 * base_rap / total,
665         "power_recovered_kW": 0.0,
666         "cake_mass_per_rap_change_pct": 0.0,
667     }]
668     for label, t_new in (("延长至历史P95", train[T_COLS].quantile(0.95).to_numpy(float)),
669                        ("延长至历史最大值", train[T_COLS].max().to_numpy(float))):
670         rap = float(model.rapping_power(t_new))
671         rows.append({
672             "scenario": label,

```

```

673         "T_setting": ";".join(f"{v:.0f}" for v in t_new),
674         "field_power_kW": field,
675         "rapping_power_kW": rap,
676         "total_power_kW": model.intercept + field + rap,
677         "rapping_share_pct": 100.0 * rap / (model.intercept + field + rap),
678         "power_recovered_kW": base_rap - rap,
679         "cake_mass_per_rap_change_pct": 100.0 * (float(np.mean(t_new / t_med)) - 1.0),
680     })
681     return pd.DataFrame(rows)
682
683
684 # ----- driver
685
686 def main() -> None:
687     OUT_DIR.mkdir(parents=True, exist_ok=True)
688     df = pd.read_csv(DATA_FILE, parse_dates=["timestamp"])
689     df["condition_cluster"] = df["condition_cluster"].astype(int)
690     profiles = condition_profiles(df)
691     profiles.to_csv(OUT_DIR / "condition_profiles.csv", index=False, encoding="utf-8-sig")
692
693     train = df[df["split"] == "train"].copy()
694     validation = df[df["split"] == "validation"].copy()
695     test = df[df["split"] == "test"].copy()
696
697     model, metrics, rolling_metrics, baseline_metrics = evaluate_power_models(train, validation, test)
698     rolling_metrics.to_csv(OUT_DIR / "power_model_rolling_validation.csv", index=False, encoding="utf-8-sig")
699     baseline_metrics.to_csv(OUT_DIR / "power_model_baselines.csv", index=False, encoding="utf-8-sig")
700     (OUT_DIR / "power_model_metrics.json").write_text(
701         json.dumps(metrics, ensure_ascii=False, indent=2), encoding="utf-8")
702     guard = float(metrics["validation_abs_error_q90_kW"])
703
704     pd.read_csv(CLUSTER_EVAL_FILE).to_csv(
705         OUT_DIR / "cluster_selection_metrics.csv", index=False, encoding="utf-8-sig")
706
707     historical_strategy_correlations(df).to_csv(
708         OUT_DIR / "historical_strategy_correlations.csv", index=False, encoding="utf-8-sig")
709     anomaly_events(df).to_csv(OUT_DIR / "anomaly_events.csv", index=False, encoding="utf-8-sig")
710     rapping_power_tradeoff(train, model).to_csv(
711         OUT_DIR / "rapping_power_tradeoff.csv", index=False, encoding="utf-8-sig")
712
713     global_t_median = train[T_COLS].median().to_numpy(float)
714     t_stars = {int(p["condition_cluster"]): scenario_t_star(p, global_t_median)
715                 for _, p in profiles.iterrows()}
716
717 # -- field priority criterion (analytic) -----
718 priority_rows = []
719 for _, profile in profiles.iterrows():
720     crit = field_priority(profile, model)
721     u_ref = np.array([profile[f"{c}_median"] for c in U_COLS], float)
722     row = {"condition_cluster": int(profile["condition_cluster"]),
723           "condition_name": profile["condition_name"]}
724     for i in range(4):
725         row[f"U_{i+1}_ref_kW"] = float(u_ref[i])
726         row[f"alpha_{i+1}"] = float(ALPHA_CENTRAL[i])
727         row[f"criteria_{i+1}_per_kW"] = float(crit[i])
728         row["rank_order"] = ">".join(f"E{i+1}" for i in np.argsort(-crit))
729     priority_rows.append(row)
730 pd.DataFrame(priority_rows).to_csv(
731     OUT_DIR / "field_priority_criterion.csv", index=False, encoding="utf-8-sig")
732
733 # -- central solutions -----
734 rows10 = solve_all(profiles, model, t_stars, 10.0, HEADROOM_10)

```

```

735 rows5 = solve_all(profiles, model, t_stars, 5.0, HEADROOM_5)
736 optimum = pd.DataFrame(rows10 + rows5)
737 optimum["validation_error_adjusted_power_kw"] = optimum["predicted_power_kw"] + guard
738 optimum["validation_error_guard_kw"] = guard
739 optimum.to_csv(OUT_DIR / "optimal_controls_central_scenario.csv",
740               index=False, encoding="utf-8-sig")
741
742 # -- support-domain audit (reported, not imposed) -----
743 audit_rows = []
744 for row in rows10 + rows5:
745     cluster = int(row["condition_cluster"])
746     group = train[train["condition_cluster"] == cluster]
747     mu, inv_cov, threshold = support_statistics(group)
748     z = np.array([float(row[c]) for c in U_COLS + T_COLS])
749     d2 = float((z - mu) @ inv_cov @ (z - mu))
750     u_max = np.array([group[c].max() for c in U_COLS], float)
751     audit_rows.append({
752         "condition_cluster": cluster,
753         "condition_name": row["condition_name"],
754         "limit_mgNm3": row["limit_mgNm3"],
755         "mahalanobis_d2": d2,
756         "empirical_q975": threshold,
757         "ratio_to_q975": d2 / threshold,
758         "max_voltage_over_historical_max_pct": float(
759             100.0 * max((float(row[c]) / m - 1.0) for c, m in zip(U_COLS, u_max))),
760         "verdict": "SUPPORTED" if d2 <= threshold else "EXTRAPOLATION",
761     })
762 pd.DataFrame(audit_rows).to_csv(OUT_DIR / "support_boundary_audit.csv",
763                               index=False, encoding="utf-8-sig")
764
765 # Does the historical support ellipsoid admit any feasible point at all?
766 restricted_rows = []
767 for _, profile in profiles.iterrows():
768     cluster = int(profile["condition_cluster"])
769     stats = support_statistics(train[train["condition_cluster"] == cluster])
770     for limit, headroom in ((10.0, HEADROOM_10), (5.0, HEADROOM_5)):
771         r = solve_condition(profile, model, t_stars[cluster], limit,
772                             voltage_headroom=headroom, mahalanobis=stats)
773         restricted_rows.append({
774             "condition_cluster": cluster, "limit_mgNm3": limit,
775             "status_with_mahalanobis_constraint": r["status"],
776             "predicted_power_kw": r.get("predicted_power_kw", np.nan),
777         })
778 pd.DataFrame(restricted_rows).to_csv(
779     OUT_DIR / "mahalanobis_constrained_feasibility.csv", index=False, encoding="utf-8-sig")
780
781 # -- minimum voltage headroom needed at each limit -----
782 headroom_rows = []
783 for _, profile in profiles.iterrows():
784     cluster = int(profile["condition_cluster"])
785     for limit in (10.0, 5.0):
786         need = np.nan
787         for m in np.arange(0.0, 0.31, 0.01):
788             if solve_condition(profile, model, t_stars[cluster], limit,
789                               voltage_headroom=float(m), multistart=16
790                               )["status"] == "SCENARIO_FEASIBLE":
791                 need = float(m)
792                 break
793         headroom_rows.append({
794             "condition_cluster": cluster,
795             "condition_name": profile["condition_name"],
796             "limit_mgNm3": limit,

```

```

797         "min_voltage_headroom_pct": 100.0 * need,
798     })
799 pd.DataFrame(headroom_rows).to_csv(
800     OUT_DIR / "voltage_headroom_requirement.csv", index=False, encoding="utf-8-sig")
801
802 # -- optimiser verification against dense random search -----
803 verify_rows = []
804 for _, profile in profiles.iterrows():
805     cluster = int(profile["condition_cluster"])
806     for limit, headroom, rows in ((10.0, HEADROOM_10, rows10), (5.0, HEADROOM_5, rows5)):
807         solved = next(r for r in rows if r["condition_cluster"] == cluster)
808         if solved["status"] != "SCENARIO_FEASIBLE":
809             continue
810         rnd = verify_with_random_search(profile, model, t_stars[cluster], limit, headroom)
811         verify_rows.append({
812             "condition_cluster": cluster,
813             "condition_name": profile["condition_name"],
814             "limit_mgNm3": limit,
815             "slsqp_power_kw": solved["predicted_power_kw"],
816             "random_search_power_kw": rnd,
817             "random_search_gap_pct": 100.0 * (rnd / float(solved["predicted_power_kw"]) - 1.0),
818             "multistart_converged": solved["multistart_converged"],
819             "multistart_power_spread_kw": solved["multistart_power_spread_kw"],
820         })
821 verification = pd.DataFrame(verify_rows)
822 verification.to_csv(OUT_DIR / "optimization_verification.csv", index=False, encoding="utf-8-sig")
823
824 # -- Question 4 -----
825 q4_rows = []
826 for _, profile in profiles.iterrows():
827     cluster = int(profile["condition_cluster"])
828     a = next(r for r in rows10 if r["condition_cluster"] == cluster)
829     b = next(r for r in rows5 if r["condition_cluster"] == cluster)
830     k10 = float(profile["K0_anchor"]) * float(
831         np.sum(ALPHA_CENTRAL * (np.array([a[c] for c in U_COLS], float)
832             / np.array([profile[f"{c}_median"] for c in U_COLS], float)) ** 2)
833         / ALPHA_CENTRAL.sum())
834     delta_k = math.log((10.0 - float(a["scenario_peak_excess_mgNm3"]))
835         / (5.0 - float(a["scenario_peak_excess_mgNm3"])))
836     q4_rows.append({
837         "condition_cluster": cluster,
838         "condition_name": profile["condition_name"],
839         "share": float(profile["share"]),
840         "power_10_kw": float(a["predicted_power_kw"]),
841         "power_5_kw": float(b["predicted_power_kw"]),
842         "field_power_10_kw": float(a["field_power_kw"]),
843         "K_at_10": k10,
844         "delta_K_10_to_5": delta_k,
845         "analytic_delta_power_kw": float(a["field_power_kw"]) * delta_k / k10,
846         "numeric_delta_power_kw": float(b["predicted_power_kw"]) - float(a["predicted_power_kw"]),
847         "increase_pct": 100.0 * (float(b["predicted_power_kw"]) / float(a["predicted_power_kw"]) - 1.0),
848     })
849 q4 = pd.DataFrame(q4_rows)
850 q4.to_csv(OUT_DIR / "question4_by_condition.csv", index=False, encoding="utf-8-sig")
851 weighted_p10 = float(np.average(q4["power_10_kw"], weights=q4["share"]))
852 weighted_p5 = float(np.average(q4["power_5_kw"], weights=q4["share"]))
853 historical_weighted = float(np.average(profiles["historical_power_kw"], weights=profiles["share"]))
854 q4_summary = {
855     "weighted_power_10_kw": weighted_p10,
856     "weighted_power_5_kw": weighted_p5,
857     "weighted_increase_pct": 100.0 * (weighted_p5 / weighted_p10 - 1.0),
858     "historical_weighted_actual_power_kw": historical_weighted,

```



```

859     "power_10_vs_historical_pct": 100.0 * (weighted_p10 / historical_weighted - 1.0),
860     "daily_energy_10_MWh": weighted_p10 * 24.0 / 1000.0,
861     "daily_energy_5_MWh": weighted_p5 * 24.0 / 1000.0,
862     "daily_energy_historical_MWh": historical_weighted * 24.0 / 1000.0,
863     "validation_error_guard_kW": guard,
864 }
865 (OUT_DIR / "question4_summary.json").write_text(
866     json.dumps(q4_summary, ensure_ascii=False, indent=2), encoding="utf-8")
867
868 # -- p reference table: the single scalar that needs site calibration ----
869 # Holding the 10 mg/Nm^3 operating point fixed, scaling every voltage by f
870 # gives K = K_10 f^p and P_field = P_field,10 f^2, so halving the limit
871 # (delta_K) costs exactly P_field,10 [(1 + delta_K/K_10)^(2/p) - 1]:
872 # the penalty is governed by 2/p and nothing else.
873 p_rows = []
874 for p_exp in (1.0, 1.5, 2.0, 2.5, 3.0):
875     analytic = []
876     for row in q4_rows:
877         ratio = 1.0 + row["delta_K_10_to_5"] / row["K_at_10"]
878         analytic.append({"share": row["share"],
879                         "delta": row["field_power_10_kW"] * (ratio ** (2.0 / p_exp) - 1.0),
880                         "p10": row["power_10_kW"]})
881     w = np.array([a["share"] for a in analytic])
882     delta = float(np.average([a["delta"] for a in analytic], weights=w))
883     p10 = float(np.average([a["p10"] for a in analytic], weights=w))
884     # required voltage headroom above the historical maximum, same reasoning
885     f = float(np.mean([(1.0 + r["delta_K_10_to_5"] / r["K_at_10"]) ** (1.0 / p_exp)
886                        for r in q4_rows]))
887     r10 = solve_all(profiles, model, t_stars, 10.0, HEADROOM_10,
888                     multistart=SWEET_MULTISTART, p_exponent=p_exp)
889     r5 = solve_all(profiles, model, t_stars, 5.0, HEADROOM_5,
890                    multistart=SWEET_MULTISTART, p_exponent=p_exp)
891     feasible = all(r["status"] == "SCENARIO_FEASIBLE" for r in r10 + r5)
892     p_rows.append({
893         "p_exponent": p_exp,
894         "analytic_delta_power_kW": delta,
895         "analytic_increase_pct": 100.0 * delta / p10,
896         "implied_voltage_scaling_10_to_5": f,
897         "feasible_at_5pct_headroom": feasible,
898         "numeric_weighted_power_10_kW": aggregate(r10) if feasible else np.nan,
899         "numeric_weighted_power_5_kW": aggregate(r5) if feasible else np.nan,
900         "numeric_increase_pct": (100.0 * (aggregate(r5) / aggregate(r10) - 1.0)) if feasible else np.nan,
901     })
902 pd.DataFrame(p_rows).to_csv(OUT_DIR / "p_exponent_reference.csv",
903                             index=False, encoding="utf-8-sig")
904
905 # -- scale sensitivity: p x peak scale x safety index (27 combinations) --
906 sensitivity_rows = []
907 for p_exp in (1.5, 2.0, 2.5):
908     for peak_scale in (0.9, 1.2, 1.5):
909         for safety in (0.0, 0.08, 0.15):
910             kw = dict(p_exponent=p_exp, peak_scale=peak_scale, safety_index=safety)
911             r10 = solve_all(profiles, model, t_stars, 10.0, HEADROOM_10,
912                             multistart=SWEET_MULTISTART, **kw)
913             r5 = solve_all(profiles, model, t_stars, 5.0, HEADROOM_5,
914                             multistart=SWEET_MULTISTART, **kw)
915             ok = all(r["status"] == "SCENARIO_FEASIBLE" for r in r10 + r5)
916             sensitivity_rows.append({
917                 "p_exponent": p_exp, "peak_scale": peak_scale, "safety_index": safety,
918                 "all_conditions_feasible": ok,
919                 "weighted_power_10_kW": aggregate(r10) if ok else np.nan,
920                 "weighted_power_5_kW": aggregate(r5) if ok else np.nan,

```

```

921         "weighted_power_increase_pct": (100.0 * (aggregate(r5) / aggregate(r10) - 1.0)) if ok else np.nan,
922     })
923 sensitivity = pd.DataFrame(sensitivity_rows)
924 sensitivity.to_csv(OUT_DIR / "question4_sensitivity.csv", index=False, encoding="utf-8-sig")
925
926 # -- structural sensitivity: 81 grey-box shape combinations -----
927 alpha_profiles = {
928     "equal": np.array([1.00, 1.00, 1.00, 1.00]),
929     "weak_front": np.array([1.08, 1.04, 0.98, 0.90]),
930     "central_front": ALPHA_CENTRAL,
931 }
932 structural_rows = []
933 for alpha_name, alpha_weights in alpha_profiles.items():
934     for peak_exponent in (1.00, 1.30, 1.60):
935         for phase_scale in (0.65, 0.80, 1.00):
936             for tstar_exponent in (-0.30, -0.18, -0.10):
937                 stars = {int(p["condition_cluster"]): scenario_t_star(p, global_t_median, tstar_exponent)
938                     for _, p in profiles.iterrows()}
939                 kw = dict(alpha_weights=alpha_weights, peak_exponent=peak_exponent,
940                     phase_scale=phase_scale)
941                 r10 = solve_all(profiles, model, stars, 10.0, HEADROOM_10,
942                     multistart=SWEET_MULTISTART, **kw)
943                 r5 = solve_all(profiles, model, stars, 5.0, HEADROOM_5,
944                     multistart=SWEET_MULTISTART, **kw)
945                 ok = all(r["status"] == "SCENARIO_FEASIBLE" for r in r10 + r5)
946                 structural_rows.append({
947                     "alpha_profile": alpha_name,
948                     "peak_exponent": peak_exponent,
949                     "phase_scale": phase_scale,
950                     "tstar_load_exponent": tstar_exponent,
951                     "all_conditions_feasible": ok,
952                     "weighted_power_10_kw": aggregate(r10) if ok else np.nan,
953                     "weighted_power_5_kw": aggregate(r5) if ok else np.nan,
954                     "weighted_power_increase_pct": (100.0 * (aggregate(r5) / aggregate(r10) - 1.0)) if ok else np.nan,
955                 })
956 structural = pd.DataFrame(structural_rows)
957 structural.to_csv(OUT_DIR / "structural_sensitivity.csv", index=False, encoding="utf-8-sig")
958
959 # -- field weight ablation -----
960 ablation_rows, ablation_summary_rows = [], []
961 for alpha_name, alpha_weights in alpha_profiles.items():
962     r10 = solve_all(profiles, model, t_stars, 10.0, HEADROOM_10,
963         multistart=SWEET_MULTISTART, alpha_weights=alpha_weights)
964     r5 = solve_all(profiles, model, t_stars, 5.0, HEADROOM_5,
965         multistart=SWEET_MULTISTART, alpha_weights=alpha_weights)
966     per_profile = []
967     for a, b in zip(r10, r5):
968         row = {"alpha_profile": alpha_name, "condition_cluster": a["condition_cluster"],
969             "condition_name": a["condition_name"], "share": a["share"]}
970         for i, col in enumerate(U_COLS, start=1):
971             row[f"delta_U{i}_kV"] = float(b[col]) - float(a[col])
972         per_profile.append(row)
973     ablation_rows.append(row)
974     weights = np.array([x["share"] for x in per_profile])
975     deltas = np.array([x[f"delta_U{i}_kV"] for i in range(1, 5)] for x in per_profile)
976     weighted = np.average(deltas, axis=0, weights=weights)
977     positive = np.maximum(weighted, 0)
978     ablation_summary_rows.append({
979         "alpha_profile": alpha_name,
980         **{f"weighted_delta_U{i}_kV": weighted[i - 1] for i in range(1, 5)},
981         "front_two_share_of_positive_adjustment": float(positive[:2].sum() / positive.sum())
982         if positive.sum() else np.nan,

```

```

983     })
984 pd.DataFrame(ablation_rows).to_csv(
985     OUT_DIR / "field_priority_ablation_by_condition.csv", index=False, encoding="utf-8-sig")
986 pd.DataFrame(ablation_summary_rows).to_csv(
987     OUT_DIR / "field_priority_ablation_summary.csv", index=False, encoding="utf-8-sig")
988
989 # -- historical comparison and online schedule -----
990 historical_rows, dynamic_rows = [], []
991 for _, profile in profiles.iterrows():
992     cluster = int(profile["condition_cluster"])
993     u_med = np.array([profile[f"{c}_median"] for c in U_COLS], float)
994     t_med = np.array([profile[f"{c}_median"] for c in T_COLS], float)
995     median_power = float(model.power(u_med, t_med))
996     a = next(r for r in rows10 if r["condition_cluster"] == cluster)
997     b = next(r for r in rows5 if r["condition_cluster"] == cluster)
998     historical_rows.append({
999         "condition_cluster": cluster,
1000         "condition_name": profile["condition_name"],
1001         "share": float(profile["share"]),
1002         "historical_mean_actual_power_kw": float(profile["historical_power_kw"]),
1003         "model_power_at_historical_median_controls_kw": median_power,
1004         "scenario_optimal_10_power_kw": float(a["predicted_power_kw"]),
1005         "power_change_vs_model_median_pct": 100.0 * (float(a["predicted_power_kw"]) / median_power - 1.0),
1006     })
1007     dynamic_rows.append({
1008         "condition_cluster": cluster,
1009         "condition_name": profile["condition_name"],
1010         "mode_selection_rule": "EXTERNAL_TARGET_SELECTION",
1011         "available_modes": "TARGET-10/TARGET-5/FALLBACK",
1012         "automatic_emission_threshold_switching": False,
1013         "provisional_voltage_ramp_kV_per_min": 0.5,
1014         "provisional_period_ramp_s_per_min": 5.0,
1015         "phase_scenario_for_online_trial": "0.65/0.80/1.00",
1016         "implementation_status": "PROVISIONAL_REQUIRES_T/R_CAPACITY_CHECK_AND_SITE_INTERLOCK_LIMITS",
1017         **{f"target10_{c}": a[c] for c in U_COLS + T_COLS},
1018         **{f"target5_{c}": b[c] for c in U_COLS + T_COLS},
1019     })
1020 pd.DataFrame(historical_rows).to_csv(
1021     OUT_DIR / "historical_median_power_comparison.csv", index=False, encoding="utf-8-sig")
1022 pd.DataFrame(dynamic_rows).to_csv(
1023     OUT_DIR / "dynamic_control_schedule.csv", index=False, encoding="utf-8-sig")
1024
1025 scenario_params = {
1026     "status": "EMISSION_LAYER_IS_SCENARIO_ONLY_C_OUT_NOT_IDENTIFIABLE",
1027     "emission_anchor": "recorded outlet median, no decimal rescaling",
1028     "p_exponent_central": P_EXPONENT_CENTRAL,
1029     "p_exponent_grid": [1.0, 1.5, 2.0, 2.5, 3.0],
1030     "peak_exponent_central": PEAK_EXPONENT_CENTRAL,
1031     "peak_exponent_grid": [1.00, 1.30, 1.60],
1032     "voltage_weight_profiles": {k: v.tolist() for k, v in alpha_profiles.items()},
1033     "rapping_phase_scales": [0.65, 0.80, 1.00],
1034     "tstar_load_exponents": [-0.30, -0.18, -0.10],
1035     "rapping_loss_coefficients": BETA_CENTRAL.tolist(),
1036     "peak_field_shares": PEAK_SHARES.tolist(),
1037     "support_rule": "cluster-specific full historical range plus engineering ordering; "
1038     "Mahalanobis d2 reported as extrapolation audit, not imposed",
1039     "voltage_headroom_10": HEADROOM_10,
1040     "voltage_headroom_5": HEADROOM_5,
1041     "optimiser": "multi-start SLSQP (%d starts) verified against 400k random samples" % MULTISTART,
1042 }
1043 (OUT_DIR / "scenario_parameters.json").write_text(
1044     json.dumps(scenario_params, ensure_ascii=False, indent=2), encoding="utf-8")

```

```

1045
1046 feasible_structural = structural[structural["all_conditions_feasible"]]
1047 feasible_scale = sensitivity[sensitivity["all_conditions_feasible"]]
1048 summary = {
1049     "power_model": metrics,
1050     "cluster_k": 4,
1051     "central_q4": q4_summary,
1052     "rapping_share_at_median_pct": float(
1053         rapping_power_tradeoff(train, model).iloc[0]["rapping_share_pct"]),
1054     "scale_sensitivity": {
1055         "feasible_combinations": int(len(feasible_scale)),
1056         "total_combinations": int(len(sensitivity)),
1057         "increase_pct_min": float(feasible_scale["weighted_power_increase_pct"].min()),
1058         "increase_pct_median": float(feasible_scale["weighted_power_increase_pct"].median()),
1059         "increase_pct_max": float(feasible_scale["weighted_power_increase_pct"].max()),
1060     },
1061     "structural_sensitivity": {
1062         "feasible_combinations": int(len(feasible_structural)),
1063         "total_combinations": int(len(structural)),
1064         "increase_pct_min": float(feasible_structural["weighted_power_increase_pct"].min()),
1065         "increase_pct_median": float(feasible_structural["weighted_power_increase_pct"].median()),
1066         "increase_pct_max": float(feasible_structural["weighted_power_increase_pct"].max()),
1067     },
1068     "optimiser_verification": {
1069         "max_random_search_gap_pct": float(verification["random_search_gap_pct"].max()),
1070         "max_multistart_spread_kw": float(verification["multistart_power_spread_kw"].max()),
1071     },
1072     "central_all_conditions_feasible": bool((optimum["status"] == "SCENARIO_FEASIBLE").all()),
1073 }
1074 (OUT_DIR / "model_run_summary.json").write_text(
1075     json.dumps(summary, ensure_ascii=False, indent=2), encoding="utf-8")
1076 print(json.dumps(summary, ensure_ascii=False, indent=2))
1077
1078
1079 if __name__ == "__main__":
1080     main()

```

E.2 paper_figures.py

```

1  #!/usr/bin/env python3
2  """Generate the complete publication figure set for the cement ESP paper.
3
4  All empirical plots are regenerated from the supplied processed dataset and
5  model result tables. Scenario-only plots are explicitly labelled in captions.
6  Both PNG previews and vector PDF files are exported.
7  """
8
9  from __future__ import annotations
10
11  import sys
12  import hashlib
13  from pathlib import Path
14
15  import matplotlib.pyplot as plt
16  from matplotlib.patches import FancyArrowPatch, FancyBboxPatch
17  import numpy as np
18  import pandas as pd
19
20  SRC_DIR = Path(__file__).resolve().parent
21  if str(SRC_DIR) not in sys.path:
22      sys.path.insert(0, str(SRC_DIR))

```

```

23
24 from plot_utils import (GRAY, LIGHT_GRAY, NEUTRAL_FILL, PALETTE, add_subpanel_label,
25                          figure_legend, finish_axes, headroom, note, save_figure, set_paper_style)
26 from scenario_model import (ALPHA_CENTRAL, PhysicalPowerModel, RidgePowerModel,
27                             T_COLS, U_COLS)
28
29
30 ROOT = SRC_DIR.parent
31 DATA_FILE = ROOT / "data" / "full_timeseries_with_flags.csv"
32 RESULTS = ROOT / "results"
33 OUT = ROOT / "figures_paper"
34
35 COND_SHORT = {
36     1: "工况1\n低温低浓高流",
37     2: "工况2\n高温中浓低流",
38     3: "工况3\n高温高浓高流",
39     4: "工况4\n低温高浓低流",
40 }
41
42 CAPTIONS = {
43     "01_outlet_data_diagnosis": "出口浓度记录存在显著截顶且不覆盖目标区间。有效记录全部位于48.74~50.00 mg/Nm³，其中5491条恰好
44     ⇨ 等于50 mg/Nm³，因而不能从该列辨识5或10 mg/Nm³附近的控制—排放关系。",
45     "02_process_timeseries": "原始过量具有连续时序变化和工况切换特征。图中展示首24小时入口温度、入口浓度、烟气流量及总
46     ⇨ 功率的分钟级变化。",
47     "04b_operating_evidence": "附件中可识别的另一半信息。(a)(b)为两段持续异常工况及控制端的同步响应：入口浓度尖峰时抬高
48     ⇨  $\$U_{1\$}$ 、功率随之上冲，入口温度尖峰（比电阻高值区）时反而降压降功率；(c)为训练期工况变量与控制设定的相关矩阵，显
49     ⇨ 示“高浓度→抬升前两电场电压、缩短各场振打周期”的历史操作规律。",
50     "14_rapping_peak_mechanism": "(a)由“每周期积灰量 $\propto \$L \cdot T\$$ ”推得单次再飞扬峰值随周期近似线性放大，而对小时均值的贡献
51     ⇨  $\propto \$L \cdot T \cdot (1/T) = L\$$ 与周期无关（水平点线）；(b)振打环节在中位工况下占总功率约四成，把周期由中位延长到历史极
52     ⇨ 值可回收约109 kW，代价是单次积灰量增加约18%。",
53     "03_cluster_selection": "四类工况在轮廓系数与解释复杂度之间取得较好平衡。k=4的轮廓系数达到0.408，且避免继续增加聚类数带
54     ⇨ 来的解释负担。",
55     "04_condition_map": "四类工况在入口浓度—温度平面上形成清晰分区，并由烟气流量大小补充刻画负荷差异。",
56     "07_power_prediction": "总功率由控制量近乎确定地决定。物理设定 $\$P = a_0 + \sum b_i U_i^2 + \sum c_i / T_i\$$ 在回顾性测试段的 $\$R^2\$$ 与RMSE
57     ⇨ 均显著优于均值基线及把振打周期作为线性项的岭回归设定。",
58     "08b_feature_importance": "振打周期必须以 $\$1/T\$$ 进入功率模型。(a)扣除电场功率后的剩余功率对 $\$1/T\$$ 呈明显双曲形状，线性设定
59     ⇨ （虚线）在两端系统性偏离；(b)线性 $\$T\$$ 设定在回顾性测试段留下约-10.8 kW的平均偏差，改为 $\$1/T\$$ 后偏差降至约0。",
60     "08_power_residuals": "改用 $\$1/T\$$ 设定后，回顾性测试残差围绕零对称分布，原先“高功率区负偏差扩大”的现象消失，说明该偏差
61     ⇨ 源于设定误差而非数据漂移。",
62     "09_voltage_power_response": "在各工况典型状态下，模型给出的总功率随四电场电压同步提高而上升。曲线仅在各工况历史电压支
63     ⇨ 持区间附近绘制。",
64     "05b_boundary_proximity": "把排放由历史约50 mg/Nm³压到10与5 mg/Nm³，必然离开历史联合运行经验：八组解的马氏距离平方为经验97
65     ⇨ 0.5%外壳的2.4~7.0倍；10 mg解仍位于历史电压包络之内，5 mg解则需要各场电压超出历史最大值最多5%。",
66     "19_emission_power_frontier": "排放—功率权衡前沿。限值由10逐档收紧至5 mg/Nm³时四类工况的最优功率单调上升，工况加权功率
67     ⇨ 由约1972升至约2105 kW；两档均高于历史实际加权功率，说明超低排放改造是以增电耗换减排，而非节能。",
68     "11_optimal_voltage": "5 mg/Nm³方案在各工况均需进一步抬升电压，且抬升集中在单位功率去除收益最高的电场上；柱高已接近或
69     ⇨ 超出历史电压包络，必须先核验高压电源容量。",
70     "12_optimal_rapping_period": "两档限值下的最优振打周期均被推向历史区间上端——延长周期直接降低振打功率；前级短、后级长
71     ⇨ 的工程顺序约束始终保持。",
72     "13_emission_decomposition": "最优方案的约束值由连续排放基值和振打峰值增量共同构成。柱顶虚线分别表示10和5 mg/Nm³情景限
73     ⇨ 值。",
74     "18_condition_energy_penalty": "四类工况由10收紧至5 mg/Nm³的功率增幅高度一致（约6.7%），这与解析式 $\Delta P = P_{\text{field}} - P_{\text{K}} / K$ 
75     ⇨ 的预测相符：增幅主要由电场功率占比与当前去除指数决定，而与工况细节关系不大。",
76     "15_search_convergence": "多起点SLSQP与同一可行域内 $4 \times 10^5$ 点随机搜索的对照。八个工况—限值组合中，47~53个起点全部收敛到
77     ⇨ 同一解（功率一致到 $10^{-7}$  kW以内）；随机搜索均未找到更优解，其最优值反而高出0.97%~5.17%，工况1的10 mg/Nm³组合更是一个可行样本都没采到。可行域是被排放与排序约束夹出的窄带，随机撒点效率极低。",
78     "16_sensitivity_distribution": "81组结构情景全部可行。峰值指数在本文推导值1.00附近变化几乎不改变功率增幅，相位叠加尺度
79     ⇨ 与场权重形状的影响也有限；加权增幅为5.13%~7.55%，中位数6.40%。",
80     "20_eta_sensitivity": "功率增幅由单一可标定的数 $\$p\$$ 决定。 $\$K \propto U^p\$$ 时增幅近似正比于 $\$2/p\$$ ： $\$p=1\$$ 为14.2%， $\$p=2\$$ （Deutsch迂
81     ⇨ 移速度形式）为6.8%， $\$p=3\$$ 为4.4%。其中 $\$p=1\$$ 与 $\$p=1.5\$$ 在5%电压裕度内无解，说明 $\$p\$$ 首先决定能否达标。现场一次电压阶跃试
82     ⇨ 验即可标定 $\$p\$$ 。",

```

```

62 "17_sensitivity_heatmap": "场权重消融结果。三种去除贡献权重形状下，10→5 mg/Nm³所需的加权电压增量分布与按 $\alpha_i/(b_i U_{i-1})$ 排序的解析优先级一致。",
63 }
64
65 TRACE_META = {
66     "01_outlet_data_diagnosis": (str(DATA_FILE), "fig01_outlet_diagnosis", "§ 2.1—2.2；图2", "出口记录为量程截顶；不代表真实排放分布"),
67     "02_process_timeseries": (str(DATA_FILE), "fig02_timeseries", "§ 2.2；图3", "只展示首24 h，不代表全周期分布"),
68     "04b_operating_evidence": (str(RESULTS / "historical_strategy_correlations.csv"), "fig04b_operating_evidence", "§ 4.2—4.3；图4", "相关系数为同期线性关联，不等于因果；两段事件为个案，样本量有限"),
69     "14_rapping_peak_mechanism": (str(RESULTS / "rapping_power_tradeoff.csv"), "fig14_peak", "§ 4.4；图5", "峰值尺度 $b_0$ 仍为工程先验；功率侧数值由附件识别"),
70     "03_cluster_selection": (str(RESULTS / "cluster_selection_metrics.csv"), "fig03_cluster_selection", "§ 5.1；图6", "聚类指标只评价统计分离度，不等于物理工况唯一性"),
71     "04_condition_map": (str(DATA_FILE), "fig04_condition_map", "§ 5.1；图7", "为可视化抽样点；聚类实际使用全部训练样本"),
72     "07_power_prediction": (str(RESULTS / "power_model_baselines.csv"), "fig07_power_prediction + PhysicalPowerModel", "§ 5.2；图8", "回顾性测试并非真正未触碰盲测"),
73     "08b_feature_importance": (str(RESULTS / "power_model_baselines.csv"), "fig08b_specification", "§ 5.2；图9", "残差图按 $T_1$ 展开，其余三场周期固定在中位数"),
74     "08_power_residuals": (str(DATA_FILE), "fig08_residuals + PhysicalPowerModel", "§ 5.2；图10", "残差接近零均值，但仍非全新日期盲测"),
75     "09_voltage_power_response": (str(DATA_FILE), "fig09_voltage_response + PhysicalPowerModel", "§ 5.2；图11", "局部情景曲线，仅在历史电压附近解释"),
76     "05b_boundary_proximity": (str(RESULTS / "support_boundary_audit.csv"), "fig05b_boundary_proximity", "§ 5.4；图12", "经验支持域不是设备安全边界；超出部分必须由现场阶跃试验确认"),
77     "19_emission_power_frontier": (str(RESULTS / "emission_power_frontier.csv"), "fig19_frontier", "§ 5.5；图13", "各限值下均为灰箱情景解，前沿形状依赖排放先验"),
78     "11_optimal_voltage": (str(RESULTS / "optimal_controls_central_scenario.csv"), "grouped_controls(U_COLS)", "§ 6；图14", "最优电压是情景试验起点，不是设备安全设定"),
79     "12_optimal_rapping_period": (str(RESULTS / "optimal_controls_central_scenario.csv"), "grouped_controls(T_COLS)", "§ 6；图15", "无实际振打事件，周期结果依赖峰值先验"),
80     "13_emission_decomposition": (str(RESULTS / "optimal_controls_central_scenario.csv"), "fig13_emission", "§ 7.2；图17", "全部排放值为情景推演，不是实测"),
81     "18_condition_energy_penalty": (str(RESULTS / "question4_by_condition.csv"), "fig18_penalty", "§ 7.1；图16", "中心情景点估计，需与 $\rho$ 参考表同时解读"),
82     "15_search_convergence": (str(RESULTS / "optimization_verification.csv"), "fig15_optimizer_verification", "§ 8.1；图18", "随机搜索对照只能证伪，不构成全局最优的数学证明"),
83     "16_sensitivity_distribution": (str(RESULTS / "structural_sensitivity.csv"), "fig16_structural_sensitivity", "§ 8.2；图19", "扫描范围仍由显式情景网格决定"),
84     "20_eta_sensitivity": (str(RESULTS / "p_exponent_reference.csv"), "fig20_p_sensitivity", "§ 8.2；图20", " $\rho$ 的取值区间为先验网格；需由现场阶跃试验标定"),
85     "17_sensitivity_heatmap": (str(RESULTS / "field_priority_ablation_summary.csv"), "fig17_field_ablation", "§ 8.2；图21", "消融仍依赖同一灰箱结构与支持域"),
86 }
87
88 MANUSCRIPT_FIGURE = {
89     "01_outlet_data_diagnosis": 2,
90     "02_process_timeseries": 3,
91     "04b_operating_evidence": 4,
92     "14_rapping_peak_mechanism": 5,
93     "03_cluster_selection": 6,
94     "04_condition_map": 7,
95     "07_power_prediction": 8,
96     "08b_feature_importance": 9,
97     "08_power_residuals": 10,
98     "09_voltage_power_response": 11,
99     "05b_boundary_proximity": 12,
100     "19_emission_power_frontier": 13,
101     "11_optimal_voltage": 14,
102     "12_optimal_rapping_period": 15,
103     "18_condition_energy_penalty": 16,
104     "13_emission_decomposition": 17,

```

```

105     "15_search_convergence": 18,
106     "16_sensitivity_distribution": 19,
107     "20_eta_sensitivity": 20,
108     "17_sensitivity_heatmap": 21,
109 }
110
111
112 def _label_bars(ax, bars, fmt="{:.0f}", dy=3.0):
113     for bar in bars:
114         h = bar.get_height()
115         ax.text(bar.get_x() + bar.get_width() / 2, h + dy, fmt.format(h), ha="center", va="bottom", fontsize=7)
116
117
118 def fig01_outlet_diagnosis(df: pd.DataFrame) -> None:
119     valid = df["C_out_mgNm3"].dropna()
120     fig, axes = plt.subplots(1, 2, figsize=(8.2, 3.2), gridspec_kw={"width_ratios": [1.15, 1]})
121     ax = axes[0]
122     add_subpanel_label(ax, "a")
123     bins = np.linspace(valid.min() - 0.03, valid.max() + 0.03, 28)
124     ax.hist(valid, bins=bins, color=PALETTE[0], edgecolor="white")
125     ax.axvline(50, color=PALETTE[4], linestyle="--", label="记录上限 50")
126     ax.set_xlim(valid.min() - 0.05, valid.max() + 0.05)
127     ax.text(
128         0.03,
129         0.78,
130         "5与10 mg/Nm³目标均在图示范围外",
131         transform=ax.transAxes,
132         color=PALETTE[2],
133         fontsize=8,
134         bbox={"boxstyle": "round", "facecolor": "white", "edgecolor": LIGHT_GRAY},
135     )
136     ax.set_xlabel("出口粉尘浓度记录 (mg/Nm³)")
137     ax.set_ylabel("样本数")
138     ax.legend(frameon=False, loc="upper left")
139     finish_axes(ax)
140
141     ax = axes[1]
142     add_subpanel_label(ax, "b")
143     total = len(df)
144     categories = ["精确值", "50截顶", "缺失"]
145     values = [int(df["C_out_exact_flag"].sum()), int(df["C_out_cap50_flag"].sum()), int(df["C_out_missing_flag"].sum())]
146     bars = ax.bar(categories, values, color=[PALETTE[0], PALETTE[4], GRAY])
147     for bar, value in zip(bars, values):
148         ax.text(bar.get_x() + bar.get_width()/2, value + total*0.012, f"{value}\n({100*value/total:.1f}%)", ha="center", fontsize=8)
149     ax.set_ylabel("样本数")
150     ax.set_ylim(0, max(values) * 1.18)
151     note(ax, "百分比以全部10080条样本为分母", loc="upper right")
152     finish_axes(ax)
153     fig.tight_layout(w_pad=2.0)
154     save_figure(fig, OUT, "01_outlet_data_diagnosis")
155
156
157 def fig02_timeseries(df: pd.DataFrame) -> None:
158     d = df.iloc[:1440].copy()
159     hours = np.arange(len(d)) / 60
160     fields = [
161         ("Temp_C", "入口温度 (°C)", PALETTE[4]),
162         ("C_in_gNm3", "入口浓度 (g/Nm³)", PALETTE[0]),
163         ("Q_Nm3h", "烟气流量 (Nm³/h)", PALETTE[2]),
164         ("P_total_kw", "总功率 (kW)", PALETTE[3]),
165     ]
166     fig, axes = plt.subplots(4, 1, figsize=(8.2, 5.5), sharex=True)

```

```

167 for ax, (col, label, color), tag in zip(axes, fields, ["a", "b", "c", "d"]):
168     add_subpanel_label(ax, tag, x=-0.08, y=1.02)
169     ax.plot(hours, d[col], color=color, linewidth=0.8)
170     ax.set_ylabel(label)
171     finish_axes(ax)
172 axes[-1].set_xlabel("首日运行时间 (h)")
173 axes[-1].set_xlim(0, 24)
174 fig.tight_layout(h_pad=0.5)
175 save_figure(fig, OUT, "02_process_timeseries")
176
177
178 def fig03_cluster_selection(cluster_eval: pd.DataFrame) -> None:
179     fig, axes = plt.subplots(1, 2, figsize=(7.7, 3.0))
180     ax = axes[0]
181     add_subpanel_label(ax, "a")
182     ax.plot(cluster_eval["k"], cluster_eval["silhouette"], "o-", color=PALETTE[0])
183     chosen = cluster_eval[cluster_eval["selected"]].iloc[0]
184     ax.scatter([chosen["k"]], [chosen["silhouette"]], s=65, color=PALETTE[4], zorder=3, label="选定 k=4")
185     ax.set_xlabel("聚类数 k")
186     ax.set_ylabel("轮廓系数")
187     ax.legend(frameon=False)
188     finish_axes(ax)
189     ax = axes[1]
190     add_subpanel_label(ax, "b")
191     ax.plot(cluster_eval["k"], cluster_eval["inertia"] / 1000, "s--", color=PALETTE[2], label="惯性/1000")
192     ax.set_xlabel("聚类数 k")
193     ax.set_ylabel("类内惯性 ( $\times 10^3$ )")
194     finish_axes(ax)
195     fig.tight_layout(w_pad=2.0)
196     save_figure(fig, OUT, "03_cluster_selection")
197
198
199 def fig04_condition_map(df: pd.DataFrame) -> None:
200     train = df[df["split"] == "train"]
201     fig, ax = plt.subplots(figsize=(6.7, 4.4))
202     for cluster, group in train.groupby("condition_cluster"):
203         g = group.iloc[:5]
204         sizes = 8 + 28 * (g["Q_Nm3h"] - train["Q_Nm3h"].min()) / (train["Q_Nm3h"].max() - train["Q_Nm3h"].min())
205         ax.scatter(g["C_in_gNm3"], g["Temp_C"], s=sizes, alpha=0.38, color=PALETTE[int(cluster)-1], label=COND_SHORT[int(cluster)].replace("\
↔ n", " "))
206     ax.set_xlabel("入口粉尘浓度 (g/Nm³)")
207     ax.set_ylabel("入口温度 (°C)")
208     ax.legend(frameon=False, ncol=2)
209     finish_axes(ax, grid_axis="both")
210     fig.tight_layout()
211     save_figure(fig, OUT, "04_condition_map")
212
213
214
215 def fig04b_operating_evidence(df: pd.DataFrame, events: pd.DataFrame,
216                             strategy: pd.DataFrame) -> None:
217     """The identifiable half of Question 1.
218
219     The outlet column carries no signal, but the operator's response to the
220     inlet conditions is plainly visible: two sustained excursions (panels a, b)
221     and a stable correlation structure between conditions and settings (c).
222     """
223     fig = plt.figure(figsize=(8.4, 5.6))
224     gs = fig.add_gridspec(2, 2, height_ratios=[1, 1.05], hspace=0.60, wspace=0.62)
225
226     windows = [
227         ("2024-05-02 13:30", "2024-05-02 16:20", "C_in_gNm3", "入口浓度 (g/Nm³)", "a"),

```



```

228     ("2024-05-06 12:30", "2024-05-06 15:30", "Temp_C", "入口温度 (°C)", "b"),
229 ]
230 for col_idx, (t0, t1, col, ylabel, tag) in enumerate(enumerate(windows):
231     ax = fig.add_subplot(gs[0, col_idx])
232     add_subpanel_label(ax, tag, x=-0.14)
233     win = df[(df["timestamp"] >= t0) & (df["timestamp"] <= t1)]
234     minutes = (win["timestamp"] - win["timestamp"].iloc[0]).dt.total_seconds() / 60
235     ax.plot(minutes, win[col], color=PALETTE[4], linewidth=1.4, label=ylabel.split(" ")[0])
236     ax.set_ylabel(ylabel, color=PALETTE[4], fontsize=9)
237     ax.tick_params(axis="y", labelcolor=PALETTE[4])
238     ax.set_xlabel("事件窗口内时间 (min)")
239     twin = ax.twinx()
240     twin.plot(minutes, win["U1_kV"], color=PALETTE[0], linewidth=1.4, label="$U_1$")
241     twin.plot(minutes, win["P_total_kW"] / 30.0, color=PALETTE[2],
242               linewidth=1.2, linestyle="--", label="总功率/30")
243     twin.set_ylabel("$U_1$ (kV) 总功率/30 (kW)", fontsize=8.0)
244     twin.spines["top"].set_visible(False)
245     finish_axes(ax, grid_axis="none")
246     handles = ax.get_lines() + twin.get_lines()
247     ax.legend(handles, [h.get_label() for h in handles],
248               frameon=False, fontsize=7.4, ncol=3, loc="upper center",
249               bbox_to_anchor=(0.5, 1.30))
250
251 ax = fig.add_subplot(gs[1, :])
252 add_subpanel_label(ax, "c", x=-0.055)
253 cond_labels = {"r_Temp_C": "入口温度 $H$", "r_C_in_gNm3": "入口浓度 ${C}_{in}$", "r_Q_Nm3h": "烟气流量 $Q$"}
254 full = strategy[strategy["window"] == "full_record"].set_index("control")
255 matrix = full.loc[U_COLS + T_COLS, list(cond_labels)].to_numpy(float).T
256 im = ax.imshow(matrix, cmap="RdBu_r", aspect="auto", vmin=-0.30, vmax=0.30)
257 ax.set_xticks(range(matrix.shape[1]),
258               [f"${U}_{i}$" for i in range(1, 5)] + [f"${T}_{i}$" for i in range(1, 5)])
259 ax.set_yticks(range(3), list(cond_labels.values()))
260 for i in range(matrix.shape[0]):
261     for j in range(matrix.shape[1]):
262         ax.text(j, i, f"{matrix[i, j]:+.3f}", ha="center", va="center", fontsize=8,
263               color="white" if abs(matrix[i, j]) > 0.20 else "#1F2933")
264 cbar = fig.colorbar(im, ax=ax, fraction=0.028, pad=0.015)
265 cbar.set_label("Pearson 相关系数", fontsize=8.4)
266 ax.set_title("全记录期工况变量与控制设定的历史响应 (可由附件识别)", fontsize=9.2, pad=8)
267 tr = strategy[strategy["window"] == "train"].set_index("control")
268 te = strategy[strategy["window"] == "retrospective_test"].set_index("control")
269 ax.text(1.0, -0.32,
270        "注: 振打周期规律 (${C}_{in} \rightarrow T)$ 在四个时窗内一致; "
271        "电压规律并不平稳——${r}({C}_{in}, U_1)$ 在前5天为 %+.2f, 在末日为 %+.2f。"
272        % (float(tr.loc["U1_kV", "r_C_in_gNm3"]), float(te.loc["U1_kV", "r_C_in_gNm3"])),
273        transform=ax.transAxes, ha="right", va="top", fontsize=7.8, color=GRAY)
274 save_figure(fig, OUT, "04b_operating_evidence")
275
276
277 def fig05b_boundary_proximity(audit: pd.DataFrame) -> None:
278     """How far outside historical joint operating experience each solution sits."""
279     ordered = audit.sort_values(["condition_cluster", "limit_mgNm3"],
280                                ascending=[True, False]).reset_index(drop=True)
281     labels = [f"工况{int(c)}\n{int(l)} mg"
282              for c, l in zip(ordered["condition_cluster"], ordered["limit_mgNm3"])]
283     colors = [PALETTE[int(c) - 1] for c in ordered["condition_cluster"]]
284
285     fig, axes = plt.subplots(1, 2, figsize=(8.4, 3.6), gridspec_kw={"width_ratios": [1.15, 1]})
286     ax = axes[0]
287     add_subpanel_label(ax, "a")
288     x = np.arange(len(ordered))
289     ax.vlines(x, 1.0, ordered["ratio_to_q975"], color=colors, linewidth=2.2, alpha=0.8)

```

```

290 ax.scatter(x, ordered["ratio_to_q975"], color=colors, s=45,
291            edgecolor="black", linewidth=0.5, zorder=3)
292 ax.axhline(1.0, color=PALETTE[4], linestyle="--", linewidth=1.2, label="历史经验97.5%外壳")
293 ax.set_xticks(x, labels, fontsize=7.6)
294 ax.set_ylabel(r"$d_M^2/q_{0.975}$ (倍) ")
295 ax.legend(frameon=False, loc="upper left", fontsize=8)
296 headroom(ax, 0.16)
297 finish_axes(ax)
298
299 ax = axes[1]
300 add_subpanel_label(ax, "b")
301 bars = ax.bar(x, ordered["max_voltage_over_historical_max_pct"], color=colors, width=0.62)
302 _label_bars(ax, bars, fmt="{:.1f}%", dy=0.12)
303 ax.axhline(0, color=PALETTE[4], linestyle="--", linewidth=1.2)
304 ax.set_xticks(x, labels, fontsize=7.6)
305 ax.set_ylabel("最高电压超出历史最大值 (%)")
306 note(ax, "0%以下=仍在历史电压包络内", loc="upper left")
307 headroom(ax, 0.30)
308 finish_axes(ax)
309 fig.tight_layout(w_pad=1.9)
310 save_figure(fig, OUT, "05b_boundary_proximity")
311
312
313 def fit_power_model(df: pd.DataFrame) -> PhysicalPowerModel:
314     train = df[df["split"] == "train"]
315     return PhysicalPowerModel().fit(train, train["P_total_kw"].to_numpy(float))
316
317
318 def fit_linear_period_model(df: pd.DataFrame) -> RidgePowerModel:
319     """The misspecified reference: rapping period entered as a linear term."""
320     train = df[df["split"] == "train"]
321     metrics = pd.read_json(RESULTS / "power_model_metrics.json", typ="series")
322     alpha = float(metrics["ridge_selected_alphas"]["quadratic_ridge_T"])
323     model = RidgePowerModel(alpha=alpha)
324     model.fit(train, train["P_total_kw"].to_numpy(float))
325     return model
326
327
328 def fig07_power_prediction(df: pd.DataFrame, model: RidgePowerModel, baselines: pd.DataFrame) -> None:
329     test = df[df["split"] == "test"]
330     actual = test["P_total_kw"].to_numpy(float)
331     pred = model.predict(test)
332     fig, axes = plt.subplots(1, 2, figsize=(8.2, 3.5))
333     ax = axes[0]
334     add_subpanel_label(ax, "a")
335     ax.scatter(actual, pred, s=10, alpha=0.42, color=PALETTE[0], edgecolors="none")
336     lo, hi = min(actual.min(), pred.min()), max(actual.max(), pred.max())
337     ax.plot([lo, hi], [lo, hi], "--", color=PALETTE[4], label="理想预测")
338     metrics = pd.read_json(RESULTS / "power_model_metrics.json", typ="series")
339     ax.text(0.04, 0.94,
340            "$R^2$=%.3f$\n$RMSE$=%.2f kW$\n$偏差$=%.2f kW" % (float(metrics["retrospective_test_r2"]),
341                                                             float(metrics["retrospective_test_rmse_kw"]),
342                                                             float(metrics["retrospective_test_bias_kw"])),
343            transform=ax.transAxes, va="top",
344            bbox={"boxstyle": "round", "facecolor": "white", "edgecolor": "LIGHT_GRAY"})
345     ax.set_xlabel("实际总功率 (kW)")
346     ax.set_ylabel("预测总功率 (kW)")
347     ax.legend(frameon=False, loc="lower right")
348     finish_axes(ax, grid_axis="both")
349
350     ax = axes[1]
351     add_subpanel_label(ax, "b")

```

```

352 labels = ["均值\n基线", "线性$U$\n线性$T$", "二次$U$\n线性$T$", "本文\n$U^2+1/T$"]
353 model_order = ["mean_predictor", "linear_ridge_T", "quadratic_ridge_T", "physical_invT"]
354 x = np.arange(len(model_order))
355 val = baselines[baselines["split"] == "validation"].set_index("model").loc[model_order, "rmse_kw"]
356 tst = baselines[baselines["split"] == "retrospective_test"].set_index("model").loc[model_order, "rmse_kw"]
357 ax.bar(x-0.18, val, 0.36, color=PALETTE[5], label="验证集")
358 ax.bar(x+0.18, tst, 0.36, color=PALETTE[0], label="回顾性测试")
359 ax.set_xticks(x, labels, fontsize=7.6)
360 ax.set_ylabel("RMSE (kW, 对数刻度)")
361 ax.set_yscale("log")
362 ax.legend(frameon=False, loc="upper right")
363 finish_axes(ax)
364 fig.tight_layout()
365 save_figure(fig, OUT, "07_power_prediction")
366
367
368 def fig08_residuals(df: pd.DataFrame, model: RidgePowerModel) -> None:
369     test = df[df["split"] == "test"]
370     actual = test["P_total_kw"].to_numpy(float)
371     pred = model.predict(test)
372     res = pred - actual
373     fig, axes = plt.subplots(1, 2, figsize=(7.7, 3.1))
374     add_subpanel_label(axes[0], "a")
375     axes[0].scatter(pred, res, s=9, alpha=0.38, color=PALETTE[0], edgecolors="none")
376     axes[0].axhline(0, color=PALETTE[4], linestyle="--")
377     axes[0].set_xlabel("预测总功率 (kW)")
378     axes[0].set_ylabel("残差: 预测-实际 (kW)")
379     finish_axes(axes[0], grid_axis="both")
380     add_subpanel_label(axes[1], "b")
381     axes[1].hist(res, bins=28, color=PALETTE[2], edgecolor="white")
382     metrics = pd.read_json(RESULTS / "power_model_metrics.json", typ="series")
383     guard = float(metrics["validation_abs_error_q90_kw"])
384     axes[1].axvline(res.mean(), color=PALETTE[4], linestyle="--", label=f"测试均值 {res.mean():.2f} kW")
385     axes[1].axvline(guard, color=PALETTE[0], linestyle=":", label=f"验证|误差|P90 {guard:.2f} kW")
386     axes[1].set_xlabel("残差 (kW)")
387     axes[1].set_ylabel("样本数")
388     axes[1].legend(frameon=False)
389     finish_axes(axes[1])
390     fig.tight_layout(w_pad=2.0)
391     save_figure(fig, OUT, "08_power_residuals")
392
393
394 def fig08b_specification(df: pd.DataFrame, model: PhysicalPowerModel,
395                         linear_model: RidgePowerModel, baselines: pd.DataFrame) -> None:
396     """Why the rapping period must enter as 1/T rather than T.
397
398     Panel (a) shows the residual of a voltage-only fit against the period:
399     the curvature is the signature of a 1/T term. Panel (b) contrasts the
400     retrospective-test bias of both specifications.
401     """
402     train = df[df["split"] == "train"]
403     test = df[df["split"] == "test"]
404     fig, axes = plt.subplots(1, 2, figsize=(8.2, 3.4), gridspec_kw={"width_ratios": [1.15, 1]})
405
406     ax = axes[0]
407     add_subpanel_label(ax, "a")
408     u = train[U_COLS].to_numpy(float)
409     resid = train["P_total_kw"].to_numpy(float) - model.field_power(u) - model.intercept
410     t1 = train["T1_s"].to_numpy(float)
411     order = np.argsort(t1)
412     ax.scatter(t1[order][:7], resid[order][:7], s=7, alpha=0.22,
413               color=PALETTE[5], edgecolors="none", label="扣除电场功率后的残差")

```

```

414 grid = np.linspace(t1.min(), t1.max(), 120)
415 rap_other = float(np.sum(model.c[1:] / train[T_COLS[1:]].median().to_numpy(float)))
416 ax.plot(grid, model.c[0] / grid + rap_other, color=PALETTE[4], linewidth=2.0,
417         label=r"拟合  $\$c_1/T_1+\mathrm{const}\$")
418 slope = np.polyfit(t1, resid, 1)
419 ax.plot(grid, np.polyval(slope, grid), "--", color=PALETTE[0], linewidth=1.6,
420         label=r"线性  $\$T_1\$$  设定")
421 ax.set_xlabel("第1电场振动周期  $\$T_1\$$  (s)")
422 ax.set_ylabel("剩余功率 (kW)")
423 ax.legend(frameon=False, fontsize=8, loc="upper right")
424 finish_axes(ax, grid_axis="both")
425
426 ax = axes[1]
427 add_subpanel_label(ax, "b")
428 names = {"quadratic_ridge_T": "线性  $\$T\$$  设定", "physical_invT": "本文  $\$1/T\$$  设定"}
429 tst = baselines[baselines["split"] == "retrospective_test"].set_index("model")
430 x = np.arange(2)
431 bias = [float(tst.loc[k, "bias_kw"]) for k in names]
432 rmse = [float(tst.loc[k, "rmse_kw"]) for k in names]
433 ax.bar(x - 0.18, bias, 0.36, color=PALETTE[4], label="测试平均偏差")
434 ax.bar(x + 0.18, rmse, 0.36, color=PALETTE[0], label="测试 RMSE")
435 for xi, (b, r) in enumerate(zip(bias, rmse)):
436     ax.text(xi - 0.18, b + (0.6 if b >= 0 else -1.6), f"{b:+.2f}", ha="center", fontsize=8)
437     ax.text(xi + 0.18, r + 0.6, f"{r:.2f}", ha="center", fontsize=8)
438 ax.axhline(0, color=GRAY, linewidth=0.8)
439 ax.set_xticks(x, list(names.values()))
440 ax.set_ylabel("回顾性测试 (kW)")
441 ax.legend(frameon=False, fontsize=8)
442 headroom(ax, 0.22)
443 finish_axes(ax)
444 fig.tight_layout(w_pad=2.0)
445 save_figure(fig, OUT, "08b_feature_importance")
446
447
448 def fig09_voltage_response(df: pd.DataFrame, profiles: pd.DataFrame, model: RidgePowerModel) -> None:
449     train = df[df["split"] == "train"]
450     fig, ax = plt.subplots(figsize=(6.4, 4.0))
451     for _, p in profiles.iterrows():
452         c = int(p["condition_cluster"])
453         group = train[train["condition_cluster"] == c]
454         scale = np.linspace(0.92, 1.08, 80)
455         rows = []
456         u0 = group[U_COLS].median().to_numpy(float)
457         t0 = group[T_COLS].median().to_numpy(float)
458         for s in scale:
459             row = {col: val for col, val in zip(U_COLS, u0*s)}
460             row.update({col: val for col, val in zip(T_COLS, t0)})
461             row.update({"Temp_C": p["Temp_C"], "C_in_gNm3": p["C_in_gNm3"], "Q_Nm3h": p["Q_Nm3h"]})
462             rows.append(row)
463         y = model.predict(pd.DataFrame(rows))
464         ax.plot(100*(scale-1), y, color=PALETTE[c-1], marker=["o", "s", "A", "D"][c-1], markevery=16,
465             label=COND_SHORT[c].replace("\n", " "))
466         ax.set_xlabel("四电场电压同步变化 (%)")
467         ax.set_ylabel("模型预测总功率 (kW)")
468         ax.axvline(0, color=GRAY, linewidth=0.8)
469         ax.legend(frameon=False, ncol=2)
470         finish_axes(ax, grid_axis="both")
471         fig.tight_layout()
472         save_figure(fig, OUT, "09_voltage_power_response")
473
474
475$ 
```

```

476 def grouped_controls(optimum: pd.DataFrame, cols: list[str], stem: str, ylabel: str) -> None:
477     fig, axes = plt.subplots(2, 2, figsize=(8.2, 5.8), sharey=True)
478     for ax, cluster, tag in zip(axes.flat, range(1,5), ["a", "b", "c", "d"]):
479         add_subpanel_label(ax, tag)
480         sub = optimum[optimum["condition_cluster"] == cluster].set_index("limit_mgNm3")
481         x = np.arange(4)
482         ax.bar(x-0.18, sub.loc[10.0, cols], 0.36, color=PALETTE[5], label="10 mg/Nm³")
483         ax.bar(x+0.18, sub.loc[5.0, cols], 0.36, color=PALETTE[4], label="5 mg/Nm³")
484         ax.set_xticks(x, [c.split("_")[0] for c in cols])
485         ax.set_title(COND_SHORT[cluster].replace("\n", " "), fontsize=9)
486         finish_axes(ax)
487     axes[0,0].set_ylabel(ylabel); axes[1,0].set_ylabel(ylabel)
488     handles, labels = axes[0,0].get_legend_handles_labels()
489     fig.tight_layout(h_pad=2.0, rect=(0, 0.06, 1, 1))
490     figure_legend(fig, handles, labels, ncol=2, bottom=0.0)
491     save_figure(fig, OUT, stem)
492
493
494
495 def fig13_emission(optimum: pd.DataFrame) -> None:
496     x = np.arange(8)
497     ordered = optimum.sort_values(["limit_mgNm3", "condition_cluster"], ascending=[False, True])
498     base = ordered["scenario_base_emission_mgNm3"].to_numpy()
499     peak = ordered["scenario_peak_excess_mgNm3"].to_numpy()
500     colors = [PALETTE[0]]*4 + [PALETTE[4]]*4
501     fig, ax = plt.subplots(figsize=(8.0, 4.0))
502     ax.bar(x, base, color=colors, alpha=0.82, label="连续排放基值")
503     ax.bar(x, peak, bottom=base, color=colors, alpha=0.35, edgecolor=colors, label="振打峰值增量")
504     ax.hlines(10, -0.5, 3.5, colors=GRAY, linestyles="--", linewidth=1.0)
505     ax.hlines(5, 3.5, 7.5, colors=GRAY, linestyles="--", linewidth=1.0)
506     ax.text(3.45, 10.15, "10 mg/Nm³", ha="right", fontsize=8, color=GRAY)
507     ax.text(7.45, 5.15, "5 mg/Nm³", ha="right", fontsize=8, color=GRAY)
508     ax.set_xticks(x, [f"工况{i}" for i in range(1,5)]*2)
509     ax.set_ylabel("峰值总排放情景值 (mg/Nm³)")
510     ax.legend(frameon=False, ncol=2)
511     finish_axes(ax)
512     fig.tight_layout()
513     save_figure(fig, OUT, "13_emission_decomposition")
514
515
516 def fig14_peak(profiles: pd.DataFrame) -> None:
517     """The rapping trade-off, derived rather than assumed.
518
519     Cake mass per rap grows like  $L \cdot T$ , so the per-event peak grows linearly with
520     the period while the contribution to the hourly mean,  $L \cdot T \cdot (1/T) = L$ , does
521     not depend on the period at all. Panel (b) prices the same lever in power.
522     """
523     trade = pd.read_csv(RESULTS / "rapping_power_tradeoff.csv")
524     ratio = np.linspace(0.75, 1.35, 200)
525     fig, axes = plt.subplots(1, 2, figsize=(8.3, 3.5), gridspec_kw={"width_ratios": [1.1, 1]})
526
527     ax = axes[0]
528     add_subpanel_label(ax, "a")
529     for cluster in (1, 3, 4):
530         pr = profiles[profiles["condition_cluster"] == cluster].iloc[0]
531         peak = 1.2 * float(pr["load_ratio_to_train_median"]) * ratio
532         ax.plot(100 * (ratio - 1), peak, color=PALETTE[cluster - 1],
533               marker=["o", "s", "A", "D"][cluster - 1], markevery=40,
534               label=COND_SHORT[cluster].replace("\n", " "))
535     ax.axhline(0, color=GRAY, linewidth=0.8)
536     ax.axvline(0, color=GRAY, linewidth=0.8)
537     ax.plot(100 * (ratio - 1), np.full_like(ratio, 1.2 * float(

```

```

538     profiles[profiles["condition_cluster"] == 3].iloc[0]["load_ratio_to_train_median"])),
539     ":", color="#111827", linewidth=1.5, label="对小时均值的贡献（工况3，与ST$无关）")
540 ax.set_xlabel("振打周期相对参考周期变化（%）")
541 ax.set_ylabel("单次再飞扬峰值代理（mg/Nm³）")
542 ax.legend(frameon=False, fontsize=7.8, loc="upper left")
543 headroom(ax, 0.18)
544 finish_axes(ax, grid_axis="both")
545
546 ax = axes[1]
547 add_subpanel_label(ax, "b")
548 x = np.arange(len(trade))
549 bars = ax.bar(x, trade["rapping_power_kw"], color=[PALETTE[5], PALETTE[2], PALETTE[4]], width=0.6)
550 _label_bars(ax, bars, fmt=":.0f", dy=6)
551 for xi, row in trade.iterrows():
552     if row["power_recovered_kw"] > 0:
553         ax.text(xi, row["rapping_power_kw"] / 2,
554                 "省%.0f kW\n积灰+%.0f%" % (row["power_recovered_kw"],
555                                             row["cake_mass_per_rap_change_pct"]),
556                 ha="center", va="center", fontsize=7.6, color="white")
557 ax.set_xticks(x, ["历史\n中位周期", "延长至\n历史P95", "延长至\n历史最大"], fontsize=8)
558 ax.set_ylabel("振打环节功率（kW）")
559 note(ax, "中位工况下振打占总功率 %.1f%" % float(trade.iloc[0]["rapping_share_pct"]),
560      loc="upper right")
561 headroom(ax, 0.24)
562 finish_axes(ax)
563 fig.tight_layout(w_pad=2.0)
564 save_figure(fig, OUT, "14_rapping_peak_mechanism")
565
566
567 def fig15_optimizer_verification(verification: pd.DataFrame) -> None:
568     """Multi-start SLSQP against a 4x10^5-point random search of the same set."""
569     ordered = verification.sort_values(["condition_cluster", "limit_mgNm3"],
570                                       ascending=[True, False]).reset_index(drop=True)
571     labels = [f"工况{int(c)}\n{int(l)} mg"
572              for c, l in zip(ordered["condition_cluster"], ordered["limit_mgNm3"])]
573     x = np.arange(len(ordered))
574     slsqp = ordered["slsqp_power_kw"].to_numpy(float)
575     rnd = ordered["random_search_power_kw"].to_numpy(float)
576     gaps = ordered["random_search_gap_pct"].to_numpy(float)
577
578     fig, axes = plt.subplots(1, 2, figsize=(8.4, 3.3), gridspec_kw={"width_ratios": [1.2, 1]})
579
580     ax = axes[0]
581     add_subpanel_label(ax, "a")
582     lo = float(np.nanmin(slsqp)) * 0.94
583     hi = float(np.nanmax([np.nanmax(slsqp), np.nanmax(rnd)])) * 1.04
584     ax.set_ylim(lo, hi)
585     ax.bar(x - 0.18, slsqp, 0.36, color=PALETTE[0], label="多起点SLSQP")
586     ax.bar(x + 0.18, np.where(np.isfinite(rnd), rnd, lo), 0.36,
587           color=LIGHT_GRAY, label="随机搜索最优")
588     for xi, value in enumerate(rnd):
589         if not np.isfinite(value):
590             ax.annotate("随机搜索\n无可行样本", xy=(xi + 0.18, lo),
591                       xytext=(xi + 0.55, lo + 0.30 * (hi - lo)),
592                       fontsize=7.2, color=PALETTE[4], ha="left",
593                       arrowprops={"arrowstyle": "->", "color": PALETTE[4], "linewidth": 0.9})
594     ax.set_xticks(x, labels, fontsize=7.4)
595     ax.set_ylabel("最优总功率（kW）")
596     ax.legend(frameon=False, fontsize=8, loc="upper left", ncol=2)
597     finish_axes(ax)
598
599     ax = axes[1]

```

```

600 add_subpanel_label(ax, "b")
601 finite = np.isfinite(gaps)
602 ax.bar(x[finite], gaps[finite], color=PALETTE[2], width=0.6)
603 for xi, value in zip(x[finite], gaps[finite]):
604     ax.text(xi, value + 0.12, f"{value:.2f}", ha="center", va="bottom", fontsize=7.4)
605 ax.set_ylim(0, float(np.nanmax(gaps)) * 1.42)
606 ax.set_xticks(x, labels, fontsize=7.4)
607 ax.set_ylabel("随机搜索相对SLSQP的劣势 (%)")
608 note(ax, "全部为正：随机搜索未找到更优解", loc="upper right")
609 finish_axes(ax)
610 fig.tight_layout(w_pad=2.2)
611 save_figure(fig, OUT, "15_search_convergence")
612
613
614 def fig16_structural_sensitivity(structural: pd.DataFrame) -> None:
615     valid = structural[structural["all_conditions_feasible"] == True]
616     fig, axes = plt.subplots(1, 2, figsize=(8.2, 3.5), sharey=True)
617     ax = axes[0]
618     add_subpanel_label(ax, "a")
619     expts = sorted(valid["peak_exponent"].unique())
620     data = [valid[valid["peak_exponent"] == x]["weighted_power_increase_pct"] for x in expts]
621     boxes = ax.boxplot(data, tick_labels=[f"x: {x:.2f}" for x in expts],
622                        patch_artist=True, showfliers=False)
623     for i, box in enumerate(boxes["boxes"]):
624         box.set_facecolor(PALETTE[i % len(PALETTE)]); box.set_alpha(0.72)
625     ax.set_xlabel(r"振打峰值指数  $\gamma$  (本文推导值 1.00) ")
626     ax.set_ylabel("5相对10 mg/Nm3加权功率增幅 (%)")
627     finish_axes(ax)
628
629     ax = axes[1]
630     add_subpanel_label(ax, "b")
631     phases = sorted(valid["phase_scale"].unique())
632     for profile, style, label, color in zip(
633         ["equal", "weak_front", "central_front"], ["o-", "s--", "A:"],
634         ["等权", "弱前级", "中心前级"], [PALETTE[0], PALETTE[2], PALETTE[4]]):
635         medians = [valid[(valid["phase_scale"] == ph) & (valid["alpha_profile"] == profile)]
636                    ["weighted_power_increase_pct"].median() for ph in phases]
637         ax.plot(phases, medians, style, color=color, label=label)
638     ax.set_xlabel(r"振打相位叠加尺度  $\rho$ ")
639     ax.legend(frameon=False)
640     finish_axes(ax, grid_axis="both")
641     fig.tight_layout(w_pad=2.0)
642     save_figure(fig, OUT, "16_sensitivity_distribution")
643
644
645 def fig17_field_ablation(ablation: pd.DataFrame) -> None:
646     profiles = ["equal", "weak_front", "central_front"]
647     labels = ["等权", "弱前级", "中心前级"]
648     cols = [f"weighted_delta_U{i}_kV" for i in range(1,5)]
649     d = ablation.set_index("alpha_profile").loc[profiles, cols]
650     fig, axes = plt.subplots(1, 2, figsize=(7.8, 3.3), gridspec_kw={"width_ratios": [1.45, 1]})
651     ax = axes[0]
652     add_subpanel_label(ax, "a")
653     im = ax.imshow(d.to_numpy(), cmap="YlGnBu", aspect="auto", vmin=0)
654     ax.set_xticks(range(4), [f"U{i}" for i in range(1,5)])
655     ax.set_yticks(range(3), labels)
656     ax.set_xlabel("电场")
657     ax.set_ylabel("去除贡献权重设定")
658     for i in range(3):
659         for j in range(4):
660             ax.text(j, i, f"{d.iloc[i,j]:.2f}", ha="center", va="center", color="white" if d.iloc[i,j]>5 else "black", fontsize=8)
661     cbar = fig.colorbar(im, ax=ax, fraction=0.05, pad=0.04)

```

```

662 cbar.set_label("10→5 mg时加权电压增量 (kV)")
663
664 ax = axes[1]
665 add_subpanel_label(ax, "b")
666 shares = 100*ablation.set_index("alpha_profile").loc[profiles,"front_two_share_of_positive_adjustment"]
667 bars = ax.bar(labels, shares, color=[PALETTE[0],PALETTE[2],PALETTE[4]])
668 _label_bars(ax, bars, fmt="{:.1f}%", dy=0.7)
669 ax.set_ylabel("前两电场占正向电压增量 (%)")
670 ax.set_ylim(0,100)
671 finish_axes(ax)
672 fig.tight_layout(w_pad=2.0)
673 save_figure(fig, OUT, "17_sensitivity_heatmap")
674
675
676 def fig18_penalty(q4: pd.DataFrame) -> None:
677     x = np.arange(4)
678     weighted_p10 = float(np.sum(q4["share"] * q4["power_10_kw"]) / q4["share"].sum())
679     weighted_p5 = float(np.sum(q4["share"] * q4["power_5_kw"]) / q4["share"].sum())
680     weighted = 100.0 * (weighted_p5 / weighted_p10 - 1.0)
681     fig, ax = plt.subplots(figsize=(6.4, 3.8))
682     bars = ax.bar(x, q4.sort_values("condition_cluster")["increase_pct"], color=PALETTE[:4])
683     _label_bars(ax, bars, fmt="{:.2f}%", dy=0.12)
684     ax.axhline(weighted, color=PALETTE[4], linestyle="--", label=f"工况加权平均 {weighted:.2f}%")
685     ax.set_xticks(x, [COND_SHORT[i] for i in range(1,5)])
686     ax.set_ylabel("情景预测功率增幅 (%)")
687     ax.set_ylim(0, max(q4["increase_pct"])*1.22)
688     ax.legend(frameon=False)
689     finish_axes(ax)
690     fig.tight_layout()
691     save_figure(fig, OUT, "18_condition_energy_penalty")
692
693
694 def _relpath(path: str) -> str:
695     """交付物中一律使用相对路径，避免写入本机绝对路径。"""
696     try:
697         return Path(path).resolve().relative_to(ROOT).as_posix()
698     except ValueError:
699         return Path(path).name
700
701
702 TIKZ_ROWS = [{
703     "figure_id": "01", "manuscript_figure": 1, "stem": "00_esp_schematic",
704     "caption": "四电场电除尘器结构与变量定义",
705     "png": "figures_paper/00_esp_schematic.pdf", "pdf": "figures_paper/00_esp_schematic.pdf",
706     "source_data": "figures_paper/00_esp_schematic.tex",
707     "transformation": "按赛题给定的设备结构示意图以TikZ矢量重绘, xelatex独立编译",
708     "supported_manuscript_claims": "§1.1; 图1",
709     "limitations": "结构示意图，不含实测数据；比例非工程真实尺寸",
710 }]
711
712
713 def fig19_frontier(frontier: pd.DataFrame, weighted: pd.DataFrame) -> None:
714     """排放—功率权衡前沿：说明放宽限值换来的功率下降不是等排放下的节能。"""
715     fig, ax = plt.subplots(figsize=(7.4, 4.0))
716     for idx, (cluster, grp) in enumerate(frontier.groupby("condition_cluster")):
717         grp = grp.sort_values("limit_mgNm3")
718         ax.plot(grp["limit_mgNm3"], grp["predicted_power_kw"], "o-",
719                 color=PALETTE[idx], markersize=4.5, linewidth=1.5,
720                 label=COND_SHORT[int(cluster)].replace("\n", " "))
721     w = weighted.sort_values("limit_mgNm3")
722     ax.plot(w["limit_mgNm3"], w["weighted_power_kw"], "s--", color="#111827",
723             markersize=5, linewidth=2.0, label="工况加权")

```



```

724 hist = float(frontier["historical_power_kw"].iloc[0]) if "historical_power_kw" in frontier else np.nan
725 p10 = float(w[w["limit_mgNm3"] == 10.0]["weighted_power_kw"].iloc[0])
726 p5 = float(w[w["limit_mgNm3"] == 5.0]["weighted_power_kw"].iloc[0])
727 if np.isfinite(hist):
728     ax.axhline(hist, color=GRAY, linestyle=":", linewidth=1.3)
729     ax.text(9.9, hist - 14, f"历史实际加权功率 {hist:.0f} kW (排放约50 mg/Nm³)",
730            fontsize=8, color=GRAY, ha="right")
731 ax.annotate("", xy=(10.0, p10), xytext=(5.0, p5),
732            arrowprops={"arrowstyle": "<->", "color": PALETTE[4], "linewidth": 1.3})
733 ax.text(7.5, (p5 + p10) / 2 + 10,
734        f"限值 10→5 mg/Nm³\n加权功率增 {p5 - p10:.0f} kW ({100*(p5-p10)/p10:.1f}%)",
735        fontsize=8.2, color=PALETTE[4], ha="center")
736 ax.set_xlabel("情景排放限值 (mg/Nm³)")
737 ax.set_ylabel("支持域内最佳采样预测功率 (kW)")
738 ax.set_xticks(sorted(frontier["limit_mgNm3"].unique()))
739 ax.legend(frameon=False, ncol=2, fontsize=8.4, loc="upper right")
740 headroom(ax, 0.16)
741 finish_axes(ax, grid_axis="both")
742 fig.tight_layout()
743 save_figure(fig, OUT, "19_emission_power_frontier")
744
745
746 def fig20_p_sensitivity(p_ref: pd.DataFrame, prior: pd.DataFrame) -> None:
747     """The increase is set by one calibratable scalar: the exponent p in  $K \sim U^p$ ."""
748     ref = p_ref.sort_values("p_exponent")
749     fig, ax = plt.subplots(figsize=(6.9, 3.9))
750     ax.plot(ref["p_exponent"], ref["analytic_increase_pct"], "-", color="#111827",
751            linewidth=1.8, zorder=2, label=r"解析式  $\$P_f[(1+\Delta K/K)^{2/p}-1]/P_{10}\$")
752     feas = ref[ref["feasible_at_5pct_headroom"] == True]
753     infeas = ref[ref["feasible_at_5pct_headroom"] != True]
754     ax.scatter(feas["p_exponent"], feas["analytic_increase_pct"], s=62, color=PALETTE[0],
755            edgecolor="white", linewidth=0.8, zorder=4, label="%电压裕度内可达")
756     ax.scatter(infeas["p_exponent"], infeas["analytic_increase_pct"], s=62,
757            facecolor="white", edgecolor=PALETTE[4], linewidth=1.6, zorder=4,
758            label="需更大电压裕度 (不可达)")
759     num = ref.dropna(subset=["numeric_increase_pct"])
760     ax.scatter(num["p_exponent"], num["numeric_increase_pct"], marker="x", s=52,
761            color=PALETTE[2], zorder=5, label="数值重解校验")
762     for _, row in ref.iterrows():
763         ax.annotate(f"{row['analytic_increase_pct']:.1f}%",
764                (row["p_exponent"], row["analytic_increase_pct"]),
765                textcoords="offset points", xytext=(0, 10), ha="center",
766                fontsize=8.2, color=GRAY)
767     ax.set_xlabel(r"去除指数—电压幂次  $\$p\$$  ( $\$K \propto U^p\$$ ;  $\$p=2\$$  为Deutsch迁移速度形式)")
768     ax.set_ylabel("10→5 mg/Nm³ 加权功率增幅 (%)")
769     ax.legend(frameon=False, fontsize=8, loc="upper right")
770     headroom(ax, 0.24)
771     finish_axes(ax, grid_axis="both")
772     fig.tight_layout()
773     save_figure(fig, OUT, "20_eta_sensitivity")
774
775
776 def write_manifest() -> None:
777     script_hash = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()
778     model_hash = hashlib.sha256((SRC_DIR / "scenario_model.py").read_bytes()).hexdigest()
779     rows = []
780     for stem, caption in CAPTIONS.items():
781         source_data, function_name, claims, limitations = TRACE_META[stem]
782         source_data = _relpath(source_data)
783         rows.append({"figure_id": stem.split("_")[0], "manuscript_figure": MANUSCRIPT_FIGURE[stem], "stem": stem, "caption": caption,
784                "png": f"figures_paper/{stem}.png", "pdf": f"figures_paper/{stem}.pdf",
785                "source_data": source_data,$ 
```

```

786         "transformation": f"paper_figures.py::{function_name}; plot_sha256={script_hash}; model_sha256={model_hash}",
787         "supported_manuscript_claims": claims,
788         "limitations": limitations))
789 rows = TIKZ_ROWS + rows
790 rows.sort(key=lambda r: r["manuscript_figure"])
791 pd.DataFrame(rows).to_csv(OUT / "figure_manifest.csv", index=False, encoding="utf-8-sig")
792 lines = ["# 论文图组与图注", "", "所有图均提供 PNG 预览和 PDF 矢量版本。", ""]
793 for row in sorted(rows, key=lambda x: x["manuscript_figure"]):
794     lines.extend([
795         f"## 正文图{row['manuscript_figure']} (文件编号{row['figure_id']}) {row['stem']}", "", row["caption"], "",
796         f"- 源数据: `{row['source_data']}`",
797         f"- 变换: `{row['transformation']}`",
798         f"- 支持的正文论断: {row['supported_manuscript_claims']}",
799         f"- 局限: {row['limitations']}",
800         f"- PNG: `{row['png']}`", f"- PDF: `{row['pdf']}`", ""])
801 (OUT / "图注清单.md").write_text("\n".join(lines), encoding="utf-8")
802
803
804 def main() -> None:
805     set_paper_style()
806     df = pd.read_csv(DATA_FILE, parse_dates=["timestamp"])
807     profiles = pd.read_csv(RESULTS / "condition_profiles.csv")
808     cluster_eval = pd.read_csv(RESULTS / "cluster_selection_metrics.csv")
809     optimum = pd.read_csv(RESULTS / "optimal_controls_central_scenario.csv")
810     audit = pd.read_csv(RESULTS / "support_boundary_audit.csv")
811     verification = pd.read_csv(RESULTS / "optimization_verification.csv")
812     structural = pd.read_csv(RESULTS / "structural_sensitivity.csv")
813     ablation = pd.read_csv(RESULTS / "field_priority_ablation_summary.csv")
814     baselines = pd.read_csv(RESULTS / "power_model_baselines.csv")
815     q4 = pd.read_csv(RESULTS / "question4_by_condition.csv")
816     frontier = pd.read_csv(RESULTS / "emission_power_frontier.csv")
817     frontier_w = pd.read_csv(RESULTS / "emission_power_frontier_weighted.csv")
818     prior = pd.read_csv(RESULTS / "question4_sensitivity.csv")
819     p_ref = pd.read_csv(RESULTS / "p_exponent_reference.csv")
820     events = pd.read_csv(RESULTS / "anomaly_events.csv")
821     strategy = pd.read_csv(RESULTS / "historical_strategy_correlations.csv")
822     model = fit_power_model(df)
823     linear_model = fit_linear_period_model(df)
824
825     fig01_outlet_diagnosis(df)
826     fig02_timeseries(df)
827     fig04b_operating_evidence(df, events, strategy)
828     fig14_peak(profiles)
829     fig03_cluster_selection(cluster_eval)
830     fig04_condition_map(df)
831     fig07_power_prediction(df, model, baselines)
832     fig08b_specification(df, model, linear_model, baselines)
833     fig08_residuals(df, model)
834     fig09_voltage_response(df, profiles, model)
835     fig05b_boundary_proximity(audit)
836     fig19_frontier(frontier, frontier_w)
837     grouped_controls(optimum, U_COLS, "11_optimal_voltage", "最优电压 (kV)")
838     grouped_controls(optimum, T_COLS, "12_optimal_rapping_period", "最优振打周期 (s)")
839     fig18_penalty(q4)
840     fig13_emission(optimum)
841     fig15_optimizer_verification(verification)
842     fig16_structural_sensitivity(structural)
843     fig20_p_sensitivity(p_ref, prior)
844     fig17_field_ablation(ablation)
845     write_manifest()
846     print(f"Generated {len(CAPTIONS)} figures in {OUT}")
847

```

```

848
849 if __name__ == "__main__":
850     main()

```

E.3 plot_utils.py

```

1  """Shared publication plotting style for the cement ESP paper."""
2
3  from __future__ import annotations
4
5  from pathlib import Path
6
7  import matplotlib.pyplot as plt
8  from cycler import cycler
9
10
11 # Okabe-Ito 色序: 色觉障碍友好, 灰度打印仍可区分
12 PALETTE = ["#0072B2", "#E69F00", "#009E73", "#CC79A7", "#D55E00", "#56B4E9"]
13 GRAY = "#5B6472"
14 LIGHT_GRAY = "#D1D5DB"
15 NEUTRAL_FILL = "#DFE3E8"
16
17
18 def set_paper_style() -> None:
19     plt.rcParams.update(
20         {
21             "font.family": "sans-serif",
22             # Linux 优先 Noto, macOS 回落到系统中文字体
23             "font.sans-serif": [
24                 "Noto Sans CJK SC",
25                 "Source Han Sans SC",
26                 "PingFang SC",
27                 "Arial Unicode MS",
28                 "WenQuanYi Zen Hei",
29                 "DejaVu Sans",
30             ],
31             "axes.unicode_minus": False,
32             "font.size": 10.0,
33             "axes.labelsize": 10.0,
34             "axes.titlesize": 10.0,
35             "xtick.labelsize": 9.0,
36             "ytick.labelsize": 9.0,
37             "legend.fontsize": 9.0,
38             "legend.frameon": False,
39             "legend.handlelength": 1.8,
40             "legend.columnspacing": 1.4,
41             "legend.borderaxespad": 0.4,
42             "axes.linewidth": 0.9,
43             "axes.edgecolor": "#2F3640",
44             "axes.labelcolor": "#1F2933",
45             "axes.prop_cycle": cycler(color=PALETTE),
46             "lines.linewidth": 1.7,
47             "lines.markersize": 5.0,
48             "xtick.direction": "out",
49             "ytick.direction": "out",
50             "xtick.major.width": 0.9,
51             "ytick.major.width": 0.9,
52             "xtick.color": "#2F3640",
53             "ytick.color": "#2F3640",
54             "grid.color": "#E5E7EB",
55             "grid.linewidth": 0.6,

```

```

56         "grid.alpha": 0.9,
57         "figure.dpi": 160,
58         "savefig.dpi": 400,
59         "savefig.bbox": "tight",
60         "savefig.pad_inches": 0.03,
61         "pdf.fonttype": 42,
62         "ps.fonttype": 42,
63     }
64 )
65
66
67 def finish_axes(ax, grid_axis: str = "y") -> None:
68     if grid_axis != "none":
69         ax.grid(True, axis=grid_axis)
70     ax.set_axisbelow(True)
71     ax.spines["top"].set_visible(False)
72     ax.spines["right"].set_visible(False)
73
74
75 def add_subpanel_label(ax, label: str, x: float = -0.12, y: float = 1.05) -> None:
76     """Add a compact (a), (b), ... identifier to a multi-panel figure."""
77     ax.text(
78         x,
79         y,
80         f"({label})",
81         transform=ax.transAxes,
82         fontsize=10.5,
83         fontweight="bold",
84         va="bottom",
85         ha="right",
86     )
87
88
89 def headroom(ax, frac: float = 0.18) -> None:
90     """在数据上方留出空白，避免图例或注释压住柱顶与折线。"""
91     lo, hi = ax.get_ylim()
92     ax.set_ylim(lo, lo + (hi - lo) * (1.0 + frac))
93
94
95 def figure_legend(fig, handles, labels, ncol: int, bottom: float = -0.02) -> None:
96     """把多面板共用图例统一放到画布下方，杜绝与数据重叠。"""
97     fig.legend(
98         handles,
99         labels,
100         loc="lower center",
101         ncol=ncol,
102         bbox_to_anchor=(0.5, bottom),
103         frameon=False,
104     )
105
106
107 def note(ax, text: str, loc: str = "upper left", color: str | None = None) -> None:
108     """角落说明文字，带白底以免压住网格与数据。"""
109     anchor = {
110         "upper left": (0.02, 0.98, "left", "top"),
111         "upper right": (0.98, 0.98, "right", "top"),
112         "lower left": (0.02, 0.03, "left", "bottom"),
113         "lower right": (0.98, 0.03, "right", "bottom"),
114     }[loc]
115     ax.text(
116         anchor[0],
117         anchor[1],

```

```

118     text,
119     transform=ax.transAxes,
120     ha=anchor[2],
121     va=anchor[3],
122     fontsize=8.0,
123     color=color or GRAY,
124     bbox={
125         "boxstyle": "round,pad=0.28",
126         "facecolor": "white",
127         "edgecolor": LIGHT_GRAY,
128         "linewidth": 0.6,
129         "alpha": 0.92,
130     },
131 )
132
133
134 def save_figure(fig, out_dir: Path, stem: str) -> None:
135     out_dir.mkdir(parents=True, exist_ok=True)
136     fig.savefig(out_dir / f"{stem}.png")
137     fig.savefig(out_dir / f"{stem}.pdf")
138     plt.close(fig)

```

E.4 compute_frontier.py

```

1  #!/usr/bin/env python3
2  """Compute the scenario emission-power trade-off frontier.
3
4  The support-domain search is repeated at a range of emission limits, so the
5  paper can show the whole trade-off curve rather than only the 10 and
6  5 mg/Nm^3 end points. Because the historical anchor is ~50 mg/Nm^3, every
7  point on this curve costs power relative to historical operation; the curve
8  also records how much voltage headroom above the historical maximum each
9  limit needs before it becomes reachable at all.
10
11  Usage: python3 src/compute_frontier.py
12  """
13
14  from __future__ import annotations
15
16  from pathlib import Path
17
18  import numpy as np
19  import pandas as pd
20
21  from scenario_model import (
22      DATA_FILE,
23      HEADROOM_5,
24      T_COLS,
25      condition_profiles,
26      evaluate_power_models,
27      scenario_t_star,
28      solve_condition,
29  )
30
31  ROOT = Path(__file__).resolve().parents[1]
32  OUT_DIR = ROOT / "results"
33  LIMITS = [5.0, 6.0, 7.0, 8.0, 9.0, 10.0]
34
35
36  def main() -> None:
37      df = pd.read_csv(DATA_FILE, parse_dates=["timestamp"])

```

```

38 df["condition_cluster"] = df["condition_cluster"].astype(int)
39 profiles = condition_profiles(df)
40
41 train = df[df["split"] == "train"].copy()
42 validation = df[df["split"] == "validation"].copy()
43 test = df[df["split"] == "test"].copy()
44 model, metrics, _, _ = evaluate_power_models(train, validation, test)
45 guard = float(metrics["validation_abs_error_q90_kw"])
46
47 global_t_median = train[T_COLS].median().to_numpy(float)
48 t_stars = {int(p["condition_cluster"]): scenario_t_star(p, global_t_median)
49             for _, p in profiles.iterrows()}
50 historical_weighted = float(np.average(profiles["historical_power_kw"],
51                                       weights=profiles["share"])))
52
53 rows: list[dict[str, object]] = []
54 weighted_rows: list[dict[str, object]] = []
55 for limit in LIMITS:
56     # smallest declared headroom that makes every condition reachable
57     headroom = 0.0
58     while headroom <= HEADROOM_5 + 1e-9:
59         trial = [solve_condition(p, model, t_stars[int(p["condition_cluster"])]), limit,
60                 voltage_headroom=headroom, multistart=12)
61         for _, p in profiles.iterrows()]
62         if all(r["status"] == "SCENARIO_FEASIBLE" for r in trial):
63             break
64         headroom = round(headroom + 0.01, 2)
65
66     solved = [solve_condition(p, model, t_stars[int(p["condition_cluster"])]), limit,
67              voltage_headroom=headroom)
68     for _, p in profiles.iterrows()]
69     for profile_row, r in zip(profiles.itertuples(), solved):
70         rows.append({
71             "condition_cluster": r["condition_cluster"],
72             "condition_name": r["condition_name"],
73             "limit_mgNm3": limit,
74             "voltage_headroom_pct": 100.0 * headroom,
75             "status": r["status"],
76             "predicted_power_kw": r.get("predicted_power_kw", np.nan),
77             "validation_error_adjusted_power_kw": r.get("predicted_power_kw", np.nan) + guard,
78             "share": float(profile_row.share),
79             "historical_power_kw": historical_weighted,
80         })
81     weights = np.array([float(p.share) for p in profiles.itertuples()])
82     powers = np.array([float(r.get("predicted_power_kw", np.nan)) for r in solved])
83     weighted_rows.append({
84         "limit_mgNm3": limit,
85         "voltage_headroom_pct": 100.0 * headroom,
86         "weighted_power_kw": float(np.average(powers, weights=weights)),
87         "weighted_daily_energy_MWh": float(np.average(powers, weights=weights)) * 24.0 / 1000.0,
88         "historical_power_kw": historical_weighted,
89     })
90
91 pd.DataFrame(rows).to_csv(OUT_DIR / "emission_power_frontier.csv",
92                          index=False, encoding="utf-8-sig")
93 pd.DataFrame(weighted_rows).to_csv(OUT_DIR / "emission_power_frontier_weighted.csv",
94                                   index=False, encoding="utf-8-sig")
95 print(pd.DataFrame(weighted_rows).to_string(index=False))
96
97
98 if __name__ == "__main__":
99     main()

```

E.5 integrity_check.py

```
1  #!/usr/bin/env python3
2  """Deterministic integrity checks for manuscript numbers and artifacts."""
3
4  from __future__ import annotations
5
6  import json
7  import re
8  import sys
9  from pathlib import Path
10
11  import numpy as np
12  import pandas as pd
13
14  SRC = Path(__file__).resolve().parent
15  if str(SRC) not in sys.path:
16      sys.path.insert(0, str(SRC))
17
18  from scenario_model import PhysicalPowerModel, _r2, _rmse
19
20
21  ROOT = SRC.parent
22  DATA = ROOT / "data" / "full_timeseries_with_flags.csv"
23  RESULTS = ROOT / "results"
24  FIGURES = ROOT / "figures_paper"
25  MANUSCRIPT = ROOT / "10_修订后完整论文_终稿.md"
26  OUT = ROOT / "integrity"
27
28
29  def close(a: float, b: float, tol: float = 1e-6) -> bool:
30      return bool(abs(a - b) <= tol)
31
32
33  def main() -> None:
34      checks: list[dict[str, object]] = []
35
36      def check(name: str, passed: bool, actual: object, expected: object) -> None:
37          checks.append({"check": name, "passed": bool(passed), "actual": actual, "expected": expected})
38
39      df = pd.read_csv(DATA)
40      check("raw_row_count", len(df) == 10080, len(df), 10080)
41      check("outlet_missing_count", int(df["C_out_missing_flag"].sum()) == 50, int(df["C_out_missing_flag"].sum()), 50)
42      valid = df["C_out_mgNm3"].dropna()
43      check("outlet_valid_count", len(valid) == 10030, len(valid), 10030)
44      check("outlet_min", close(float(valid.min()), 48.74), float(valid.min()), 48.74)
45      check("outlet_max", close(float(valid.max()), 50.0), float(valid.max()), 50.0)
46      check("outlet_cap50_count", int((valid == 50).sum()) == 5491, int((valid == 50).sum()), 5491)
47      check("outlet_le10_count", int((valid <= 10).sum()) == 0, int((valid <= 10).sum()), 0)
48      check("outlet_le5_count", int((valid <= 5).sum()) == 0, int((valid <= 5).sum()), 0)
49
50      metrics = json.loads((RESULTS / "power_model_metrics.json").read_text())
51      train, val, test = (df[df["split"] == s] for s in ("train", "validation", "test"))
52      model = PhysicalPowerModel().fit(train, train["P_total_kW"].to_numpy(float))
53      vp, tp = model.predict(val), model.predict(test)
54      recomputed = {
55          "validation_r2": _r2(val["P_total_kW"].to_numpy(float), vp),
56          "validation_rmse_kW": _rmse(val["P_total_kW"].to_numpy(float), vp),
57          "retrospective_test_r2": _r2(test["P_total_kW"].to_numpy(float), tp),
58          "retrospective_test_rmse_kW": _rmse(test["P_total_kW"].to_numpy(float), tp),
59          "retrospective_test_bias_kW": float(np.mean(tp - test["P_total_kW"].to_numpy(float))),
60      }
```

```

61 for key, value in recomputed.items():
62     check(f"power_metric_{key}", close(value, float(metrics[key]), 1e-9), value, metrics[key])
63 guard = float(np.quantile(np.abs(vp-val["P_total_kW"].to_numpy(float)), 0.90))
64 check("validation_error_guard_q90", close(guard, float(metrics["validation_abs_error_q90_kW"]), 1e-9), guard, metrics["
    ⇐ validation_abs_error_q90_kW"])
65 check(
66     "reporting_guard_rule_named_consistently",
67     metrics.get("reporting_guard_rule") == "point_prediction_plus_validation_absolute_error_q90"
68     and "conservative_reporting_rule" not in metrics,
69     {k: metrics[k] for k in metrics if "reporting" in k or "guard" in k},
70     {"reporting_guard_rule": "point_prediction_plus_validation_absolute_error_q90"},
71 )
72 check(
73     "power_model_is_physical_specification",
74     metrics["model"] == "physical_voltage_square_plus_rapping_frequency",
75     metrics["model"], "physical_voltage_square_plus_rapping_frequency",
76 )
77 # the four field coefficients must be near-identical: the fields share a design
78 b = np.asarray(metrics["b_field_kW_per_kV2"], float)
79 check("field_coefficients_consistent", float(b.std() / b.mean()) < 0.01,
80     float(b.std() / b.mean()), "<1% relative spread")
81
82 baselines = pd.read_csv(RESULTS / "power_model_baselines.csv")
83 tst_base = baselines[baselines["split"] == "retrospective_test"].set_index("model")
84 check(
85     "invT_beats_linear_T_on_test",
86     float(tst_base.loc["physical_invT", "rmse_kW"]) < float(tst_base.loc["quadratic_ridge_T", "rmse_kW"]),
87     tst_base["rmse_kW"].to_dict(), "physical_invT < quadratic_ridge_T",
88 )
89 check(
90     "invT_removes_test_bias",
91     abs(float(tst_base.loc["physical_invT", "bias_kW"])) < 1.0
92     and abs(float(tst_base.loc["quadratic_ridge_T", "bias_kW"])) > 5.0,
93     {"physical_invT": float(tst_base.loc["physical_invT", "bias_kW"]),
94     "quadratic_ridge_T": float(tst_base.loc["quadratic_ridge_T", "bias_kW"])},
95     "|bias| < 1 kW with 1/T, > 5 kW with linear T",
96 )
97 check(
98     "condition_variables_add_nothing",
99     abs(float(tst_base.loc["physical_invT_plus_conditions", "r2"])
100         - float(tst_base.loc["physical_invT", "r2"])) < 1e-3,
101     {"with": float(tst_base.loc["physical_invT_plus_conditions", "r2"]),
102     "without": float(tst_base.loc["physical_invT", "r2"])},
103     "delta R2 < 1e-3",
104 )
105 rolling = pd.read_csv(RESULTS / "power_model_rolling_validation.csv")
106 check("rolling_fold_count", rolling["fold"].nunique() == 3, int(rolling["fold"].nunique()), 3)
107
108 cluster = pd.read_csv(RESULTS / "cluster_selection_metrics.csv")
109 selected = cluster[cluster["selected"]].iloc[0]
110 check("silhouette_cluster_k", int(selected["k"]) == 4, int(selected["k"]), 4)
111 check("silhouette_k4", close(float(selected["silhouette"]), 0.4083689, 1e-6), float(selected["silhouette"]), 0.4083689)
112
113 optimum = pd.read_csv(RESULTS / "optimal_controls_central_scenario.csv")
114 check(
115     "central_parameter_table_complete",
116     len(optimum) == 8
117     and set(map(tuple, optimum[["condition_cluster", "limit_mgNm3"]].to_numpy()))
118     == {(cluster, limit) for cluster in range(1, 5) for limit in (10.0, 5.0)},
119     optimum[["condition_cluster", "limit_mgNm3"]].values.tolist(),
120     "4 conditions × 2 targets",
121 )

```



```

122 check(
123     "validation_error_columns_named_consistently",
124     {"validation_error_adjusted_power_kw", "validation_error_guard_kw"}.issubset(optimum.columns)
125     and {"conservative_power_kw", "power_guard_kw"}.isdisjoint(optimum.columns),
126     sorted(optimum.columns),
127     "new validation-error names present; old conservative names absent",
128 )
129 check("central_all_feasible", bool((optimum["status"] == "SCENARIO_FEASIBLE").all()), optimum["status"].unique().tolist(), ["
↳ SCENARIO_FEASIBLE"])
130 check("central_limits_respected", bool((optimum["scenario_peak_total_mgNm3"] <= optimum["limit_mgNm3"] + 1e-9).all()), float((optimum["
↳ scenario_peak_total_mgNm3"] - optimum["limit_mgNm3"]).max()), "<=0")
131 check(
132     "emission_anchor_not_rescaled",
133     bool((pd.read_csv(RESULTS / "condition_profiles.csv")["C_out_recorded_median"] == 50.0).all()),
134     pd.read_csv(RESULTS / "condition_profiles.csv")["C_out_recorded_median"].tolist(),
135     "all 50.0 mg/Nm3 (recorded, unscaled)",
136 )
137 scenario = json.loads((RESULTS / "scenario_parameters.json").read_text())
138 check("no_outlet_rescaling_parameter", "central_outlet_scale" not in scenario,
139     sorted(scenario), "no central_outlet_scale key")
140
141 audit = pd.read_csv(RESULTS / "support_boundary_audit.csv")
142 check("support_audit_rows", len(audit) == 8, len(audit), 8)
143 check(
144     "targets_leave_historical_experience",
145     bool((audit["ratio_to_q975"] > 1.0).all()),
146     float(audit["ratio_to_q975"].min()), ">1 for all eight solutions",
147 )
148 check(
149     "10mg_inside_voltage_envelope",
150     bool((audit[audit["limit_mgNm3"] == 10.0]["max_voltage_over_historical_max_pct"] <= 1e-6).all()),
151     float(audit[audit["limit_mgNm3"] == 10.0]["max_voltage_over_historical_max_pct"].max()),
152     "<=0% over the historical maximum",
153 )
154 restricted = pd.read_csv(RESULTS / "mahalanobis_constrained_feasibility.csv")
155 merged = restricted.merge(
156     optimum[["condition_cluster", "limit_mgNm3", "predicted_power_kw"]],
157     on=["condition_cluster", "limit_mgNm3"], suffixes=("_shell", "_free"))
158 feasible_shell = merged["status_with_mahalanobis_constraint"] == "SCENARIO_FEASIBLE"
159 check(
160     "mahalanobis_shell_variant_mostly_feasible",
161     int(feasible_shell.sum()) == 7,
162     int(feasible_shell.sum()), "7 of 8 combinations feasible inside the shell",
163 )
164 penalty = 100.0 * (merged.loc[feasible_shell, "predicted_power_kw_shell"]
165     / merged.loc[feasible_shell, "predicted_power_kw_free"] - 1.0)
166 check(
167     "shell_variant_costs_power",
168     bool((penalty > 0).all()),
169     [round(float(penalty.min()), 2), round(float(penalty.max()), 2)],
170     "staying inside the shell is never cheaper than the free optimum",
171 )
172 headroom = pd.read_csv(RESULTS / "voltage_headroom_requirement.csv")
173 check(
174     "headroom_10mg_is_zero",
175     bool((headroom[headroom["limit_mgNm3"] == 10.0]["min_voltage_headroom_pct"] == 0.0).all()),
176     headroom[headroom["limit_mgNm3"] == 10.0]["min_voltage_headroom_pct"].tolist(), "all 0%",
177 )
178 check(
179     "headroom_5mg_within_declared_margin",
180     float(headroom[headroom["limit_mgNm3"] == 5.0]["min_voltage_headroom_pct"].max()) <= 5.0,
181     float(headroom[headroom["limit_mgNm3"] == 5.0]["min_voltage_headroom_pct"].max()), "<=5%",

```

```

182 )
183
184 q4 = pd.read_csv(RESULTS / "question4_by_condition.csv")
185 p10 = float(np.sum(q4["share"] * q4["power_10_kW"]) / q4["share"].sum())
186 p5 = float(np.sum(q4["share"] * q4["power_5_kW"]) / q4["share"].sum())
187 increase = 100 * (p5 / p10 - 1)
188 summary = json.loads((RESULTS / "question4_summary.json").read_text())
189 check("weighted_power_10", close(p10, summary["weighted_power_10_kW"], 1e-9), p10, summary["weighted_power_10_kW"])
190 check("weighted_power_5", close(p5, summary["weighted_power_5_kW"], 1e-9), p5, summary["weighted_power_5_kW"])
191 check("weighted_increase", close(increase, summary["weighted_increase_pct"], 1e-9), increase, summary["weighted_increase_pct"])
192 check(
193     "tightening_costs_power_vs_history",
194     summary["power_10_vs_historical_pct"] > 0 and summary["weighted_increase_pct"] > 0,
195     {"10mg_vs_history_pct": summary["power_10_vs_historical_pct"],
196      "increase_pct": summary["weighted_increase_pct"]},
197     "both positive: tightening the limit costs power",
198 )
199 rel = np.abs(q4["analytic_delta_power_kW"] - q4["numeric_delta_power_kW"]) / q4["numeric_delta_power_kW"]
200 check(
201     "analytic_matches_numeric_delta_power",
202     float(rel.max()) < 0.02, float(rel.max()), "<2% relative difference",
203 )
204
205 verification = pd.read_csv(RESULTS / "optimization_verification.csv")
206 check("verification_rows", len(verification) == 8, len(verification), 8)
207 check(
208     "multistart_agrees_on_optimum",
209     float(verification["multistart_power_spread_kW"].max()) < 1.0,
210     float(verification["multistart_power_spread_kW"].max()), "<1 kW across converged starts",
211 )
212 gaps = verification["random_search_gap_pct"].dropna()
213 check(
214     "random_search_finds_nothing_better",
215     bool((gaps >= -1e-9).all()), float(gaps.min()), ">=0 for every combination",
216 )
217
218 sens = pd.read_csv(RESULTS / "question4_sensitivity.csv")
219 check("sensitivity_grid_size", len(sens) == 27, len(sens), 27)
220 p_ref = pd.read_csv(RESULTS / "p_exponent_reference.csv")
221 check("p_reference_rows", len(p_ref) == 5, len(p_ref), 5)
222 check(
223     "p_reference_monotone_decreasing",
224     bool((p_ref.sort_values("p_exponent")["analytic_increase_pct"].diff().dropna() < 0).all()),
225     p_ref.sort_values("p_exponent")["analytic_increase_pct"].round(3).tolist(),
226     "increase falls as p rises (proportional to 2/p)",
227 )
228
229 structural = pd.read_csv(RESULTS / "structural_sensitivity.csv")
230 structural_vals = structural.loc[structural["all_conditions_feasible"] == True, "weighted_power_increase_pct"]
231 check("structural_total_count", len(structural) == 81, len(structural), 81)
232 check("structural_all_feasible", len(structural_vals) == 81, len(structural_vals), 81)
233
234 rapping = pd.read_csv(RESULTS / "rapping_power_tradeoff.csv")
235 check(
236     "rapping_share_reported",
237     30.0 < float(rapping.iloc[0]["rapping_share_pct"]) < 45.0,
238     float(rapping.iloc[0]["rapping_share_pct"]), "between 30% and 45% at median settings",
239 )
240 strategy = pd.read_csv(RESULTS / "historical_strategy_correlations.csv")
241 check(
242     "strategy_windows_reported",
243     set(strategy["window"]) == {"full_record", "train", "validation", "retrospective_test"},

```

```

244     sorted(set(strategy["window"])),
245     "four evaluation windows",
246 )
247 period_rule = strategy[strategy["control"].str.startswith("T")]
248 check(
249     "rapping_period_rule_stable_across_windows",
250     bool((period_rule["r_C_in_gNm3"] < 0).all()),
251     float(period_rule["r_C_in_gNm3"].max()), "<0 in every window and field",
252 )
253 events = pd.read_csv(RESULTS / "anomaly_events.csv")
254 check("anomaly_events_documented", len(events) == 2, len(events), 2)
255
256 manuscript = MANUSCRIPT.read_text(encoding="utf-8")
257 # A final submission may print this verifier itself in Appendix D. Restrict
258 # manuscript-content checks to the paper and non-code appendices so string
259 # literals inside the printed source do not become false positives.
260 review_manuscript = manuscript.split("## 附录D 完整源程序", 1)[0]
261 ref_body = review_manuscript.split("## 参考文献", 1)[0]
262 cited = set()
263 for token in re.findall(r"\[[0-9,-]+\]", ref_body):
264     for part in token.split(","):
265         if "-" in part:
266             a, b = map(int, part.split("-"))
267             cited.update(range(a, b + 1))
268         else:
269             cited.add(int(part))
270 listed = set(map(int, re.findall(r"^[0-9]+\]", review_manuscript.split("## 参考文献", 1)[1], flags=re.M)))
271 check("all_references_cited", cited == set(range(1, 14)), sorted(cited), list(range(1, 14)))
272 check("reference_list_complete", listed == set(range(1, 14)), sorted(listed), list(range(1, 14)))
273 clean_manuscript = re.sub(r"<!--.*?-->\n?", "", review_manuscript)
274 table6_segment = clean_manuscript.split("表10\u3000四工况两档目标", 1)[1].split("表11", 1)[0]
275 table6_rows: dict[tuple[int, float], list[float]] = {}
276 for line in table6_segment.splitlines():
277     if not re.match(r"^\[ 1-4 \] (?:10|5) \]", line):
278         continue
279     cells = [cell.strip() for cell in line.strip().strip("|").split("|")]
280     table6_rows[(int(cells[0]), float(cells[1]))] = [float(x) for x in cells[2:]]
281 table6_matches = len(table6_rows) == 8
282 for key, displayed in table6_rows.items():
283     expected_row = optimum[
284         (optimum["condition_cluster"] == key[0]) & (optimum["limit_mgNm3"] == key[1])
285     ]
286     if len(expected_row) != 1:
287         table6_matches = False
288         continue
289     row = expected_row.iloc[0]
290     expected = [
291         *round(float(row[f"U{i}_kW"]), 1) for i in range(1, 5),
292         *round(float(row[f"T{i}_s"]), 0) for i in range(1, 5),
293         round(float(row["scenario_peak_total_mgNm3"]), 3),
294         round(float(row["predicted_power_kW"]), 2),
295     ]
296     table6_matches &= bool(np.allclose(displayed, expected, atol=5e-3))
297 check(
298     "table6_complete_and_matches_results",
299     table6_matches,
300     {f"condition_{key[0]}_target_{key[1]:g}": value for key, value in table6_rows.items()},
301     "8 rows matching optimal_controls_central_scenario.csv after displayed rounding",
302 )
303 abstract = clean_manuscript.split("## 摘要", 1)[1].split("***关键词", 1)[0]
304 abstract_chars = len(re.sub(r"\s+", "", abstract))
305 check("abstract_fills_one_page", 700 <= abstract_chars <= 1300, abstract_chars, "700-1300 chars (one page)")

```

```

306 check("anonymous_manuscript", "作者: " not in clean_manuscript and "学校: " not in clean_manuscript, ["作者: " in clean_manuscript, "
    ↳ 学校: " in clean_manuscript], [False, False])
307 check("no_english_abstract", "## Abstract" not in clean_manuscript, "## Abstract" in clean_manuscript, False)
308 check("outlet_rescaling_claim_removed", "乘0.10" not in clean_manuscript and "0.1缩放" not in clean_manuscript, ["乘0.10" in
    ↳ clean_manuscript, "0.1缩放" in clean_manuscript], [False, False])
309 check("table_caption_count", len(re.findall(r"^表\d+", clean_manuscript, flags=re.M)) == 14, len(re.findall(r"^表\d+", clean_manuscript,
    ↳ flags=re.M)), 14)
310 check("manuscript_figure_count", len(re.findall(r"^!\[图", clean_manuscript, flags=re.M)) == 21, len(re.findall(r"^!\[图",
    ↳ clean_manuscript, flags=re.M)), 21)
311 check("ai_disclosure_present", "AI工具使用声明" in clean_manuscript, "AI工具使用声明" in clean_manuscript, True)
312 check("support_list_exists", (ROOT / "支撑材料文件清单.md").exists(), (ROOT / "支撑材料文件清单.md").exists(), True)
313 ai_detail = next(ROOT.parent.glob("**/AI工具使用详情*.md"), None)
314 check("ai_detail_exists", ai_detail is not None, str(ai_detail), "AI工具使用详情*.md present in the process directory")
315
316 manifest = pd.read_csv(FIGURES / "figure_manifest.csv")
317 check("figure_manifest_count", len(manifest) == 21, len(manifest), 21)
318 required_trace = {"source_data", "transformation", "caption", "supported_manuscript_claims", "limitations"}
319 check("figure_trace_fields", required_trace.issubset(manifest.columns), sorted(manifest.columns), sorted(required_trace))
320 missing_artifacts = []
321 for _, row in manifest.iterrows():
322     for col in ("png", "pdf", "source_data"):
323         artifact = Path(row[col])
324         if not artifact.is_absolute():
325             artifact = ROOT / artifact
326         if not artifact.exists():
327             missing_artifacts.append(f"{row['stem']}:{col}")
328 check("figure_artifacts_and_sources_exist", not missing_artifacts, missing_artifacts, [])
329
330 OUT.mkdir(parents=True, exist_ok=True)
331 failed = [x for x in checks if not x["passed"]]
332 report = {"status": "PASS" if not failed else "FAIL", "total_checks": len(checks), "passed": len(checks)-len(failed), "failed": len(
    ↳ failed), "checks": checks}
333 (OUT / "deterministic_integrity_results.json").write_text(json.dumps(report, ensure_ascii=False, indent=2), encoding="utf-8")
334 print(json.dumps({k: report[k] for k in ("status", "total_checks", "passed", "failed")}, ensure_ascii=False, indent=2))
335 if failed:
336     for item in failed:
337         print("FAILED", item)
338     raise SystemExit(1)
339
340
341 if __name__ == "__main__":
342     main()

```

E.6 build_paper_pdf.py

```

1  #!/usr/bin/env python3
2  """Build the CUMCM submission PDF from the final Markdown manuscript.
3
4  Converts ``10_修订后完整论文_终稿.md`` into XeLaTeX and compiles it.
5  Layout follows the CUMCM paper format specification: A4, >=2.5 cm margins,
6  abstract alone on page 1, continuous arabic page numbers centred in the footer.
7
8  Usage: python3 src/build_paper_pdf.py [--no-code | --core-code]
9  """
10
11 from __future__ import annotations
12
13 import re
14 import subprocess
15 import sys
16 from pathlib import Path

```

```

17
18 ROOT = Path(__file__).resolve().parent.parent
19 MANUSCRIPT = ROOT / "10_修订后完整论文_终稿.md"
20 BUILD = ROOT / "build"
21 SRC = ROOT / "src"
22 CODE_MODULES = ["scenario_model.py", "paper_figures.py", "plot_utils.py",
23                 "compute_frontier.py", "integrity_check.py", "build_paper_pdf.py"]
24
25 # ----- inline text
26
27 _SPECIALS = {
28     "\\": r"\textbackslash{}", "&": r"&", "%": r"%", "$": r"$",
29     "#": r"#", "_": r"_", "{": r"\{", "(": r"\(",
30     "~": r"\textasciitilde{}", "^": r"\textasciicircum{}",
31     # symbols the CJK/serif fonts do not carry -> render them as maths
32     "\u2103": r"$^\circ C$", "\u221d": r"$\propto$", "\u2260": r"$\neq$",
33     "\u2264": r"$\leq$", "\u2265": r"$\geq$", "\u2248": r"$\approx$",
34     "\u207b": r"$^{-}$", "\u00b9": r"$^1$", "\u2070": r"$^0$",
35     "\u2192": r"$\rightarrow$", "\u2191": r"$\uparrow$", "\u2193": r"$\downarrow$",
36 }
37
38
39 def esc(text: str) -> str:
40     return "".join(_SPECIALS.get(ch, ch) for ch in text)
41
42
43 #: bare links and DOIs are single unbreakable words for TeX; \url can break them
44 _URL_RE = re.compile(r"(https?://[S+?](?=[\s\u3002\u201c\u201d]|$)|(?<DOI: )(10\.S+?)(?=[\s\u3002]|$)")
45
46
47 def _urlify(piece: str) -> str:
48     r"""Escape text, but wrap any bare URL or DOI in a breakable url command."""
49     out: list[str] = []
50     last = 0
51     for m in _URL_RE.finditer(piece):
52         out.append(esc(piece[last:m.start()]))
53         link = m.group(0).rstrip(".")
54         trailing = m.group(0)[len(link):]
55         out.append(r"\url{" + link + "}" + esc(trailing))
56         last = m.end()
57     out.append(esc(piece[last:]))
58     return "".join(out)
59
60
61 #: Markdown bold may wrap a maths span (***若把$x$作为约束***), so the markers
62 #: cannot be resolved inside a single plain-text piece. They are replaced by a
63 #: sentinel first and paired up after every segment has been rendered.
64 _BOLD = "\x00"
65
66
67 def _smart_quotes(piece: str, state: list[bool]) -> str:
68     """Straight double quotes in Chinese prose -> paired curly quotes.
69
70     ``state`` carries the open/close flag across the whole line, so a quoted
71     phrase that wraps a maths span still gets a closing quote at its end.
72     """
73     out = []
74     for ch in piece:
75         if ch == '"':
76             out.append("\u201c" if state[0] else "\u201d")
77             state[0] = not state[0]
78         else:

```

```

79         out.append(ch)
80     return "".join(out)
81
82
83 def inline(text: str) -> str:
84     """Convert inline Markdown to LaTeX, leaving  $...$  math untouched."""
85     out: List[str] = []
86     quote_open = [True]
87     for i, seg in enumerate(re.split(r"(\${^$}*\$)", text)):
88         if i % 2:
89             # math segment - pass through verbatim
90             out.append(seg)
91             continue
92         # code spans next: their content must be escaped but not styled
93         for j, piece in enumerate(re.split(r"(`[^\`]*`)", seg)):
94             if j % 2:
95                 out.append(r"\texttt{" + esc(piece[1:-1]) + "}")
96             else:
97                 piece = piece.replace("***", _BOLD)
98                 out.append(_urlify(_smart_quotes(piece, quote_open)))
99     rendered = "".join(out)
100
101     # pair the sentinels; an unmatched trailing marker is dropped
102     parts = rendered.split(_BOLD)
103     if len(parts) == 1:
104         return rendered
105     result = parts[0]
106     for k, chunk in enumerate(parts[1:], start=1):
107         result += (r"\textbf{" if k % 2 else "}") + chunk
108     if len(parts) % 2 == 0:
109         # odd number of markers
110         result += "}"
111     return result
112
113 # ----- block level
114
115 def _display_width(text: str) -> int:
116     """Rough typeset width in en-units (CJK glyphs are double width)."""
117     text = re.sub(r"(\${^$}*\$)", "xx", text) # math sets compactly
118     text = re.sub(r"[^"]", "", text)
119     return sum(2 if ord(ch) > 0x2E80 else 1 for ch in text)
120
121 # A row of ``\`` columns cannot wrap, so any table whose natural width exceeds
122 # the text block silently runs off the page. Text-heavy columns therefore
123 # become wrapping ``X`` columns and the table is set to exactly \textwidth.
124 _LINE_BUDGET = 84 # en-units that fit across the 16 cm text block
125 _WRAP_THRESHOLD = 16 # a column wider than this must be allowed to wrap
126
127
128 def convert_table(header: List[str], align: List[str], rows: List[List[str]],
129                 number: str, title: str) -> str:
130     ncol = len(header)
131     widths = [max([_display_width(header[j])
132                    + [_display_width(r[j]) for r in rows if j < len(r)]]
133                for j in range(ncol))]
134     natural = sum(widths) + 2 * ncol
135
136     wrappable = [j for j in range(ncol)
137                  if align[j] == "L" and widths[j] > _WRAP_THRESHOLD]
138     use_tabularx = natural > _LINE_BUDGET and bool(wrappable)
139
140     if use_tabularx:

```

```

141     # Share the wrapping width between the text columns in proportion to
142     # how much text each of them actually carries.
143     # Proportional-to-length shares starve short-but-not-short-enough
144     # columns, so damp the ratio with a square root and floor it.
145     raw = {j: max(widths[j], 1) ** 0.5 for j in wrappable}
146     total_wrap = sum(raw.values())
147     shares = {j: max(len(wrappable) * raw[j] / total_wrap, 0.55) for j in wrappable}
148     scale = len(wrappable) / sum(shares.values())
149     parts = []
150     for j in range(ncol):
151         if j in wrappable:
152             parts.append(r">{\hsize=3f\hsize\raggedright\arraybackslash}X"
153                          % (shares[j] * scale))
154         else:
155             parts.append({"r": "r", "c": "c"}.get(align[j], "l"))
156     open_env = r"\begin{tabularx}{\textwidth}{\textwidth}" + ".join(parts) + "]"
157     close_env = r"\end{tabularx}"
158     shrink = False
159 else:
160     spec = ".join({"r": "r", "c": "c"}.get(a, "l") for a in align)
161     open_env = r"\begin{tabular}" + spec + "]"
162     close_env = r"\end{tabular}"
163     shrink = natural > _LINE_BUDGET      # all-numeric wide table: scale it
164
165     lines = [r"\begin{table}[htbp]", r"\centering",
166             r"\caption*{\normalsize 表" + number + "\u3000" + inline(title) + "}",
167             r"\vspace{-2pt}", r"\zihao{-5}"]
168     if shrink:
169         lines.append(r"\resizebox{\textwidth}{!}{%")
170     lines += [open_env, r"\toprule",
171             " & ".join(inline(h) for h in header) + r" \\", r"\midrule"]
172     for row in rows:
173         lines.append(" & ".join(inline(c) for c in row) + r" \\\")
174     lines += [r"\bottomrule", close_env]
175     if shrink:
176         lines.append("%")
177     lines.append(r"\end{table}")
178     return "\n".join(lines)
179
180
181 def parse_align(sep_row: list[str]) -> list[str]:
182     out = []
183     for cell in sep_row:
184         cell = cell.strip()
185         if cell.startswith(":") and cell.endswith(":"):
186             out.append("c")
187         elif cell.endswith(":"):
188             out.append("r")
189         else:
190             out.append("l")
191     return out
192
193
194 def split_row(line: str) -> list[str]:
195     return [c.strip() for c in line.strip().strip("|").split("|")]
196
197
198 def convert(md: str) -> tuple[str, str, str]:
199     """Return (title, abstract_block, body)."""
200     md = re.sub(r"<!--.*?-->", "", md, flags=re.S)
201     lines = md.split("\n")
202 
```

```

203 title = ""
204 abstract: List[str] = []
205 body: List[str] = []
206 target = body
207
208 i = 0
209 pending_table: tuple[str, str] | None = None # (number, title)
210 while i < len(lines):
211     line = lines[i].rstrip()
212
213     # ---- headings
214     m = re.match(r"^# (.+)$", line)
215     if m:
216         title = m.group(1).strip()
217         i += 1
218         continue
219     m = re.match(r"^## (.+)$", line)
220     if m:
221         head = m.group(1).strip()
222         if head == "摘要":
223             target = abstract
224             i += 1
225             continue
226         target = body
227         hm = re.match(r"^(d+)\s+(.*)$", head)
228         if hm:
229             target.append(r"\section{" + inline(hm.group(2)) + "}")
230         else:
231             target.append(r"\section*" + inline(head) + "}")
232             target.append(r"\addcontentsline{toc}{section}" + inline(head) + "}")
233             i += 1
234             continue
235     m = re.match(r"^### (.+)$", line)
236     if m:
237         head = m.group(1).strip()
238         hm = re.match(r"^(d+|[A-Z]\.d+)\s+(.*)$", head)
239         target.append(r"\subsection*" + inline(head if not hm else head) + "}")
240         i += 1
241         continue
242
243     # ---- fenced code
244     if line.startswith("```"):
245         j = i + 1
246         buf = []
247         while j < len(lines) and not lines[j].startswith("```"):
248             buf.append(lines[j])
249             j += 1
250         target.append(r"\begin{lstlisting}[style=pseudo]")
251         target.extend(buf)
252         target.append(r"\end{lstlisting}")
253         i = j + 1
254         continue
255
256     # ---- display math
257     if line.strip() == "$$:
258         j = i + 1
259         buf = []
260         while j < len(lines) and lines[j].strip() != "$$:
261             buf.append(lines[j])
262             j += 1
263         expr = "\n".join(buf).strip().rstrip(",.")
264         target.append(r"\begin{equation}")

```



```

265     target.append(expr)
266     target.append(r"\end{equation}")
267     i = j + 1
268     continue
269
270 # ---- image -> figure float
271 m = re.match(r"^\[图(\d+)\s*([^\]]*)\]\s*(([^\s]+)\s*)$", line)
272 if m:
273     num, alt, path = m.group(1), m.group(2), m.group(3)
274     cap = alt
275     k = i + 1
276     while k < len(lines) and not lines[k].strip():
277         k += 1
278     cm = re.match(r"^\[图\s\u3000(.+)\s$" % num, lines[k].strip()) if k < len(lines) else None
279     if cm:
280         cap = cm.group(1)
281         i = k
282     width = "0.96" if path.endswith("00_esp_schematic.pdf") else "0.92"
283     target += [r"\begin{figure}[htbp]", r"\centering",
284               r"\includegraphics[width=\s\textwidth]{\s}" % (width, path),
285               r"\caption*\normalsize 图" + num + "\u3000" + inline(cap) + "]",
286               r"\end{figure}"]
287     i += 1
288     continue
289
290 # ---- standalone figure caption already consumed above; skip strays
291 if re.match(r"^\[图(\d+)\u3000", line.strip()):
292     i += 1
293     continue
294
295 # ---- table title line
296 m = re.match(r"^\[表(\d+|A\d+)\u3000(.+)\s$", line.strip())
297 if m:
298     pending_table = (m.group(1), m.group(2))
299     i += 1
300     continue
301
302 # ---- table body
303 if line.strip().startswith("|"):
304     header = split_row(line)
305     align = parse_align(split_row(lines[i + 1]))
306     rows = []
307     j = i + 2
308     while j < len(lines) and lines[j].strip().startswith("|"):
309         rows.append(split_row(lines[j]))
310         j += 1
311     num, ttl = pending_table if pending_table else ("", "")
312     target.append(convert_table(header, align, rows, num, ttl))
313     pending_table = None
314     i = j
315     continue
316
317 # ---- blockquote
318 if line.startswith(">"):
319     buf = [line.lstrip("> ").strip()]
320     j = i + 1
321     while j < len(lines) and lines[j].startswith(">"):
322         buf.append(lines[j].lstrip("> ").strip())
323         j += 1
324     target.append(r"\begin{quote}\small " + inline(" ".join(buf)) + r"\end{quote}")
325     i = j
326     continue

```

```

327
328 # ---- ordered / unordered list
329 if re.match(r"^\d+\.\s+", line) or re.match(r"^[*]\s+", line):
330     ordered = bool(re.match(r"^\d+\.\s+", line))
331     env = "enumerate" if ordered else "itemize"
332     items = []
333     j = i
334     while j < len(lines) and (re.match(r"^\d+\.\s+", lines[j]) or re.match(r"^[*]\s+", lines[j])):
335         items.append(re.sub(r"^\d+\.[-*]\s+", "", lines[j].rstrip()))
336         j += 1
337     target.append(r"\begin{%s}[itemsep=1pt,topsep=3pt]" % env)
338     target += [r"\item " + inline(it) for it in items]
339     target.append(r"\end{%s}" % env)
340     i = j
341     continue
342
343 # ---- blank
344 if not line.strip():
345     target.append("")
346     i += 1
347     continue
348
349 # ---- plain paragraph
350 target.append(inline(line.strip()))
351 i += 1
352
353 return title, "\n".join(abstract).strip(), "\n".join(body).strip()
354
355
356 # ----- preamble
357
358 PREAMBLE = r"""
359 \documentclass[UTF8,zihao=5,a4paper]{ctexart}
360 \usepackage[a4paper,top=2.5cm,bottom=2.5cm,left=2.5cm,right=2.5cm]{geometry}
361 \usepackage{amsmath,amssymb,bm}
362 \usepackage{graphicx}
363 \usepackage{booktabs}
364 \usepackage{caption}
365 \usepackage{enumitem}
366 \usepackage{listings}
367 \usepackage{xcolor}
368 \usepackage{setspace}
369 \usepackage{array}
370 \usepackage{tabularx}
371 \usepackage{fancyhdr}
372 \usepackage{xurl}
373
374 %%FONTS%%
375
376 \onehalfspacing
377 \setlength{\parindent}{2em}
378 \captionsetup{skip=4pt}
379 \emergencystretch=3em
380 \sloppy
381
382 \pagestyle{fancy}
383 \fancyhf{}
384 \renewcommand{\headrulewidth}{0pt}
385 \fancyfoot[C]{\thepage}
386
387 \ctexset{
388     section/format = {\centering\zihao{4}\bfseries},

```

```

389 subsection/format = {\raggedright\zihao{5}\bfseries},
390 section/beforeskip = 12pt plus 2pt minus 2pt,
391 section/afterskip = 8pt,
392 }
393
394 \lstdefinestyle{pseudo}{
395   basicstyle=\ttfamily\zihao{-5},
396   breaklines=true, breakatwhitespace=false,
397   columns=fullflexible, keepspaces=true,
398   frame=single, rulecolor=\color{gray!55},
399   framesep=4pt, xleftmargin=6pt, xrightmargin=2pt,
400   showstringspaces=false, extendedchars=true,
401 }
402 \lstdefinestyle{pycode}{
403   language=Python,
404   basicstyle=\ttfamily\zihao{6},
405   keywordstyle=\color{blue!65!black},
406   commentstyle=\color{green!42!black}\itshape,
407   stringstyle=\color{orange!75!black},
408   breaklines=true, breakatwhitespace=false,
409   breakindent=12pt, postbreak=\mbox{\textcolor{gray}{$\hookrightarrow$}}\space,
410   columns=fullflexible, keepspaces=true,
411   showstringspaces=false, extendedchars=true,
412   numbers=left, numberstyle=\tiny\color{gray}, numbersep=6pt,
413   frame=leftline, rulecolor=\color{gray!45}, framesep=5pt,
414   xleftmargin=17pt, tabsize=4, upquote=true,
415 }
416
417 % 手工编号的图表题注（与正文交叉引用一致）
418 \captionsetup[figure]{labelformat=empty,justification=centering}
419 \captionsetup[table]{labelformat=empty,justification=centering}
420 ""
421
422
423
424 # ----- fonts
425
426 #: Preferred first, portable fallback second. The submission machine normally
427 #: has TeX Gyre Termes and Noto CJK; a bare TeX Live may not.
428 FONT_CHOICES = {
429   "main": ["TeX Gyre Termes", "Liberation Serif", "DejaVu Serif"],
430   "mono": ["Noto Sans Mono CJK SC", "DejaVu Sans Mono"],
431   "CJKmain": ["Noto Serif CJK SC", "Fandol Song", "Noto Sans CJK SC"],
432   "CJksans": ["Noto Sans CJK SC", "Fandol Hei"],
433   "Cjkmono": ["Noto Sans Mono CJK SC", "Noto Sans CJK SC"],
434 }
435
436
437 def _installed_families() -> set[str]:
438     try:
439         out = subprocess.run(["fc-list", ":", "family"], capture_output=True,
440                               text=True, check=True).stdout
441     except (OSError, subprocess.CalledProcessError):
442         return set()
443     return {name.strip() for line in out.splitlines() for name in line.split(",")}
444
445
446 def font_block() -> str:
447     available = _installed_families()
448
449     def pick(key: str) -> str:
450         for name in FONT_CHOICES[key]:

```

```

451         if not available or name in available:
452             return name
453     return FONT_CHOICES[key][-1]
454
455     return "\n".join([
456         r"\setCJKmainfont{%s}" % pick("cjkmain"),
457         r"\setCJKsansfont{%s}" % pick("cjksans"),
458         r"\setCJKmonofont{%s}" % pick("cjkmmono"),
459         r"\setmainfont{%s}" % pick("main"),
460         r"\setmonofont{%s}[Scale=0.82]" % pick("mono"),
461     ])
462
463
464 def build_tex(title: str, abstract: str, body: str, code: str) -> str:
465     parts = [PREAMBLE.replace("%FONTS%", font_block()), r"\begin{document}", ""]
466     # --- page 1: title + abstract only
467     parts += [
468         r"\begin{center}",
469         r"{\zihao{3}\bfseries " + inline(title) + r"}",
470         r"\end{center}",
471         r"\vspace{6pt}",
472         r"\begin{center}{\zihao{4}\bfseries 摘\quad 要}\end{center}",
473         r"\vspace{2pt}",
474     ]
475     kw = ""
476     keep = []
477     for l in abstract.split("\n"):
478         # 保留空行以维持摘要的分段结构
479         if l.startswith(r"\textbf{关键词: }"):
480             kw = l
481         else:
482             keep.append(l)
483     while keep and not keep[-1].strip():
484         keep.pop()
485     parts += keep
486     if kw:
487         parts += ["", r"\vspace{8pt}", "", r"\noindent " + kw]
488     parts += [r"\clearpage", ""]
489     parts.append(body)
490     if code:
491         parts.append(code)
492     parts += ["", r"\end{document}", ""]
493     return "\n".join(parts)
494
495 def _extract(module: str, names: list[str]) -> str:
496     """Pull whole top-level definitions out of a source file, in order."""
497     text = (SRC / module).read_text(encoding="utf-8").split("\n")
498     out: list[str] = []
499     for name in names:
500         try:
501             i = next(k for k, l in enumerate(text)
502                     if l.startswith(("def " + name + "(", "class " + name)))
503         except StopIteration:
504             continue
505         j = i + 1
506         while j < len(text) and (not text[j].strip() or text[j][:1] in " )\t"):
507             j += 1
508         while j > i and not text[j - 1].strip():
509             j -= 1
510         out.extend(text[i:j] + [""])
511     return "\n".join(out).rstrip() + "\n"
512

```

```

513
514 #: Appendix E carries the full listing by default; ``--core-code`` prints only
515 #: the two definitions that determine every number in the paper.
516 CORE_EXTRACTS = [
517     ("scenario_model.py", ["PhysicalPowerModel"]),
518     ("scenario_model.py", ["scenario_emission", "field_priority"]),
519 ]
520
521
522 def code_appendix(full: bool = True) -> str:
523     title = "\u9644\u5f55E\quad " + ("\u5b8c\u6574\u6e90\u7a0b\u5e8f" if full else "\u6838\u5fc3\u6e90\u7a0b\u5e8f\u8282\u9009")
524     out = [r"\clearpage",
525           r"\section{%s}" % title,
526           r"\addcontentsline{toc}{section}{%s}" % title]
527     if full:
528         out.append("\u672c\u9644\u5f55\u7ed9\u51fa\u751f\u6210\u672c\u6587\u5168\u90e8\u7ed3\u679c\u4e0e\u56fe\u8868\u7684\u5b8c\u6574\u53ef\u2192
\u8fd0\u884c\u6e90\u7a0b\u5e8f\u5f0c\u4e0e\u7535\u5b50\u652f\u6491\u6750\u6599 \texttt{src/} \u76ee\u5f55\u9010\u5b57\u4e00\u81f4\u2192
u3002"
529         "\u6309\u4f9d\u8d56\u987a\u5e8f\u6392\u5217\u5f1a\u60c5\u666f\u6a21\u578b\u4e0e\u4f18\u5316\u3001\u6392\u653e\u2014\u529f\u2192
u7387\u524d\u6cbf\u3001\u7ed8\u56fe\u6837\u5f0f\u4e0e\u56fe\u96c6\u751f\u6210\u3001\u786e\u5b9a\u6027\u6838\u9a8c\u3001\u63d0\u4ea4\u2192
u7a3f\u88c5\u914d\u3002")
530         for n, name in enumerate(CODE_MODULES, start=1):
531             path = SRC / name
532             if not path.exists():
533                 continue
534             out += [r"\subsection*{E.%d\quad \texttt{%s}}" % (n, esc(name)),
535                   r"\lstinputlisting[style=pycode]{%s}" % (path.as_posix())]
536         return "\n".join(out)
537
538     out.append("\u4e3a\u63a7\u5236\u7bc7\u5e45\u5f0c\u672c\u9644\u5f55\u53ea\u7ed9\u51fa\u51b3\u5b9a\u5168\u6587\u6570\u503c\u7ed3\u679c\u2192
u7684\u4e24\u6bb5\u6838\u5fc3\u5b9a\u4e49\u5f1b"
539     "\u5b8c\u6574\u53ef\u8fd0\u884c\u6e90\u7801\u968f\u7535\u5b50\u652f\u6491\u6750\u6599\u63d0\u4ea4\u3002")
540     BUILD.mkdir(exist_ok=True)
541     for n, (module, names) in enumerate(CORE_EXTRACTS, start=1):
542         snippet = BUILD / ("code_excerpt_%d.py" % n)
543         snippet.write_text(_extract(module, names), encoding="utf-8")
544         out += [r"\subsection*{E.%d\quad \texttt{%s}\u2192\u2192}
% (n, esc(module), "\u3001".join(esc(x) for x in names)),
545               r"\lstinputlisting[style=pycode]{%s}" % snippet.as_posix()]
546     return "\n".join(out)
547
548
549
550 def main() -> None:
551     include_code = "--no-code" not in sys.argv
552     full_code = "--core-code" not in sys.argv
553     md = MANUSCRIPT.read_text(encoding="utf-8")
554     title, abstract, body = convert(md)
555     tex = build_tex(title, abstract, body, code_appendix(full_code) if include_code else "")
556
557     BUILD.mkdir(exist_ok=True)
558     tex_path = BUILD / "paper.tex"
559     tex_path.write_text(tex, encoding="utf-8")
560
561     for run in range(2):
562         proc = subprocess.run(
563             ["xelatex", "-interaction=nonstopmode", "-halt-on-error",
564              "-output-directory", str(BUILD), str(tex_path)],
565             cwd=ROOT, capture_output=True, text=True)
566         if proc.returncode != 0:
567             log = (BUILD / "paper.log").read_text(encoding="utf-8", errors="replace")
568             errs = [l for l in log.split("\n") if l.startswith("!")][:12]
569             print("XeLaTeX 失败: \n" + "\n".join(errs))

```

```
570         sys.exit(1)
571     print("已生成", BUILD / "paper.pdf")
572
573
574 if __name__ == "__main__":
575     main()
```